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Issued by **lilei11**	Approved by **ruiguang.wang**

Software Requirement Specification IGM

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0.2.2 Updated requirements in this issue

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0.2.3 Deleted requirements in this issue

1 LIST OF REQUIREMENTS WITH DEVIATIONS

2 TERMINOLOGY

2.1 Abbreviations

2.2 Definitions

2.3 Concepts

2.3.1 Identification of requirements

When a requirement is introduced it is given a tag including a requirement number. This tag follows the requirement throughout the development process. Requirements are identified using a tag as follows:

Req <Req.ID><Req.Revision_number> <Req.Name>

e.g.: 73456 v3 Seat Switch Stuck

This tag identifies the requirement via a unique number, variant (normally not used) and a revision number of the requirement.

A requirement tag shall be unique, i.e. there must not exist two different requirements with the same requirement tag. The same requirement may however be used more than once (i.e. the same requirement may exist several times, even within the same document).

2.3.2 Sub sections of a requirement

Each requirement expressed in this document consists of seven sub sections:

Requirement name and ID

The requirement is tagged by a unique id (see above) to be possible to refer to and is given a descriptive name.

Purpose

Give a brief overview of the requirement and explain its purpose.

ASIL

States the Automotive Safety Integrity Level, defined in the ISO26262 std, for the Requirements when applicable.

Verification method

How will the requirement be verified/tested?

Requirement Description

Describe the actual requirement using plain text, tables, diagrams etc.

3 INTRODUCTION

3.1 What is the purpose of this document?

This document - Software Requirement Specification (SWRS) - specifies the requirements for a specific software component and the methods to be used to ensure that each requirement has been met.

The SWRS possibly supplemented with referenced documents shall be used as the basis for design and qualification testing of the software product.

3.2 Who should read this document?

The document is mainly intended for the developer of the actual software. This may be an external supplier or a software department within Geely.

If the software product is a part of a larger system and interacts with other software products, this document also should be read and approved by the system designers and the designers responsible for each interacting software product.

3.3 How is this document organized?

The structure of this document is explained below:

Section 0 - Requirements/Objects Changelog. The Requirements/Objects in this document are tagged, and this section contains different types of tables listing the new, deleted or changed Requirements/Objects by their title/name and page no.

Section 1 - Introduction including revision history and requisite specifications.

Section 2 - Terminology. Gives a clarification of the definitions, concepts and abbreviations used in the document.

Section 3 - Software component description. States briefly the background and the purpose of the system and the software to which this document applies. Describes the general nature of the system and the software.

Section 4 - ECU Software requirements. This section lists the requirements put on the complete ECU SW. They are following the same concept as the requirements listed in Section 5.

Section 5 - Software component requirements. This section is divided into paragraphs in order to specify the software product requirements, that is, those characteristics of the software that are conditions for its acceptance. These requirements are generated in order to satisfy the system requirements (stated in SRD:s) allocated to this software product. Each requirement is tagged in order to support testing and traceability and is annotated with at least one associated qualification. The requirements cover both functionality as well as design aspects.

Section 6 - Computer Resource Requirements. This section lists the requirements put on the ECU Hardware, which have to be fulfilled in order to host the ECU Software in a proper way.

Section 7 - Software Quality Factors.

Section 8 - Delivery Requirements.

Section 9 - Appendix

3.4 How do you receive more information?

Additional information is available in the documents listed in section 1.6. Questions are answered by either the person responsible for this document (see page 1) or the person responsible for the revision (see section 1.5).

3.5 Document reference generated.

4 REVISION HISTORY

Revision	Date	Description	Logical Component	Responsible

5 SOFTWARE COMPONENT DESCRIPTION

6 SOFTWARE COMPONENTS REQUIREMENTS

6.1 Legal regulations

6.2 Required states and modes

6.3 Detailed functional requirements

6.3.1 IGM2

Configuration Summary:

ID	Name	Type	Version	Status	Last Changed
x04000000279757B9	IGM2	Composition software component	EEA4.026R3(18)	Released	2026-07-16
x04000000231892D3	ThermalManagementRequirements	Logical component	(8a4a3a3)	CS Released	2026-01-27
x04000000122A8FB6	GlobalSnapshotDataReceiver	Logical component	(15)	CS Released	2023-05-12
x0400000021EFE655	GeneralRequirements	Logical component	EEA3.525R3(8a1a6)	Released	2025-11-24
x0400000014C7D594	PreCdnforDiagComCtrlForNewEgy	Logical component	(2)	CS Released	2023-10-23

ID	Name	Type	Version	Status	Last Changed
x0400000021D8139E	EM2Obd2Interface	Logical component	EEA3.5 25R3_1 (3a3)	CS Released	2025-11-17
x04000000224AE8DC	EM2GeneralInterface	Logical component	EEA 25R3(5 b1a4)	CS Released	2025-12-09
x0400000007DBE257	EngineStartPriorityRequirements	Logical component	(2)	CS Released	2021-03-31
x0400000020A38138	EM2ThermalManagementInterface	Logical component	EEA3.5 _25R2(2b1a2)	CS Released	2025-09-04
x040000002358CFB3	EM2MotorControlInterface	Logical component	EEA3.5 26R1(7 a4)	CS Released	2026-01-27
x04000000218DF5C8	PreProgmCdnForNewEgy	Logical component	(9)	CS Released	2025-10-30
x0400000025F1CA27	MotorControlRequirements	Logical component	(10a4a1 1)	Released	2026-05-21
x0400000021F625C3	Obd2Requirements	Logical component	(6)	CS Released	2025-11-24
x040000002784CD3E	SecOCSecMngFVSIGM	Logical component	(3)	CS Released	2026-07-08
x0400000009543532	SnapshotDataUTC	Logical component	(1)	CS Released	2021-07-28
x0400000022E8F6B0	TerminatingECUsonCANorCANFD	Logical component	(3)	CS Released	2026-01-09
x040000002228C6E9	RmsComponentRequirements	Logical component	EEA3.5 25R3(5 b3a1)	Released	2025-11-28
x0400000024176850	CarConfigReceiver	Logical component	(11)	CS Released	2026-03-12
x040000002793CE1E	IGM2	SW-ECU	EEA4.0 26R3(1 9)	Released	2026-07-13

6.3.2 IGM2<Functional Safety>

6.3.3 Composition software component: IGM2

ID	Name	Version	Status	Last Changed
x04000000279757B9	IGM2	EEA4.0 26R3(1 8)	Released	2026-07-16

6.3.3.1 IGM2: Receiver Ports

Port	Interface	DataElement	DataType	E2E ID
AmbTRaw/	AmbTRaw/;5	AmbTRaw/;	AmbTWithQly/;2	
BkpOfDstTrvld/	BkpOfDstTrvld/;7	BkpOfDstTrvld/;	Len31/;2	
CarTiGlb/	CarTiGlb/;5	CarTiGlb/;	CarTiGlb/;2	
CooltFlowForDtElec/	CooltFlowForDtElec/;5	CooltFlowForDtElec/;	CooltFlow2/;2	
CooltTSigForDtElec/	CooltTSigForDtElec/;5	CooltTSigForDtElec/;	CooltTHiResl/;2	
CrashStsSafe/	CrashStsSafe/;2	CrashStsSafe/;	CrashStsSafe/;1	1035
DenoOfDiagcRateGen/	DenoOfDiagcRateGen/ ;1	DenoOfDiagcRateGen/;	Boolean/;2	
DrvgCyc/	DrvgCyc/;3	DrvgCyc/;	OnOff1/;2	
DrvgCycToClrPrmntDtc/	DrvgCycToClrPrmntDtc/	DrvgCycToClrPrmntDtc/;	Boolean/;2	

Port	Interface	DataElement	DataType	E2E ID
	;1			
EEMReqElecMot2OilSpd/	EEMReqElecMot2OilSpd/;1	EEMReqElecMot2OilSpd/;	EEMReqOilSpd/;1	
ElecMot2ActvDampgGain/	ElecMot2ActvDampgGain/;1	ElecMot2ActvDampgGain/;	GainTypActvDampg8/;2	
ElecMot2ActvDampgModReq/	ElecMot2ActvDampgModReq/;1	ElecMot2ActvDampgModReq/;	WhlMotSysActvDampgModCodSts/;2	
ElecMot2EngSpdOfActvDampg/	ElecMot2EngSpdOfActvDampg/;1	ElecMot2EngSpdOfActvDampg/;	N1/;2	
ElecMot2ModReq/	ElecMot2ModReq/;1	ElecMot2ModReq/;	IsgModTyp1/;4	
ElecMot2PwrLimMax/	ElecMot2PwrLimMax/;1	ElecMot2PwrLimMax/;	PrpsnPwrLimMax/;2	
ElecMot2PwrLimMin/	ElecMot2PwrLimMin/;1	ElecMot2PwrLimMin/;	PrpsnPwrLimMin/;2	
ElecMot2SpdReq/	ElecMot2SpdReq/;1	ElecMot2SpdReq/;	IsgSpdWSgnTyp/;2	
ElecMot2TqAllwd/	ElecMot2TqAllwd/;2	ElecMot2TqAllwd/;	IsgTqAllwd1/;3	91
ElecMot2TqReq/	ElecMot2TqReq/;1	ElecMot2TqReq/;	Tq34/;1	
EngCooltInhbFlow/	EngCooltInhbFlow/;4	EngCooltInhbFlow/;	CooltInhb/;2	
EngNSafe/	EngNSafe/;7	EngNSafe/;	EngNSafe/;2	68
EngSt1WdSts/	EngSt1WdSts/;4	EngSt1WdSts/;	EngSt1WdSts/;1	137
EngT/	EngT/;5	EngT/;	EngT/;2	
EngTiOn/	EngTiOn/;1	EngTiOn/;	TimeDesForSWDL/;2	
EngTqReqFast/	EngTqReqFast/;1	EngTqReqFast/;	PtTqType/;2	
HvBattSoc/	HvBattSoc/;3	HvBattSoc/;	NewBattSOC/;1	
HvBattUDc/	HvBattUDc/;2	HvBattUDc/;	BCLRqrdU/;1	
HvBattUDynMaxLim/	HvBattUDynMaxLim/;2	HvBattUDynMaxLim/;	NewBattVolt/;1	
HvBattUDynMinLim/	HvBattUDynMinLim/;2	HvBattUDynMinLim/;	NewBattVolt/;1	
HvSysRlySts/	HvSysRlySts/;8	HvSysRlySts/;	HvSysRlySts/;1	21
HvSysSts/	HvSysSts/;3	HvSysSts/;	HvSysSts/;2	
IuprDenominatorInhb/	IuprDenominatorInhb/;1	IuprDenominatorInhb/;	InhibitSts/;1	
LoadValCalcd/	LoadValCalcd/;3	LoadValCalcd/;	Perc5/;2	
MonrCycCmpl/	MonrCycCmpl/;3	MonrCycCmpl/;	NotCmpl1/;2	
OBMPrpsnActve/	OBMPrpsnActve/;1	OBMPrpsnActve/;	Boolean/;2	
PrpsnFbLimnIndcn/	PrpsnFbLimnIndcn/;2	PrpsnFbLimnIndcn/;	UInt4/;2	
PtEngTqAct/	PtEngTqAct/;1	PtEngTqAct/;	PtTqType/;2	
PtTqAtAxleMaxReq/	PtTqAtAxleMaxReq/;2	PtTqAtAxleMaxReq/;	PtTqAtAxleReq/;1	1109
ThrPos/	ThrPos/;3	ThrPos/;	Perc2/;2	
TiAndDateIndcn/	TiAndDateIndcn/;5	TiAndDateIndcn/;	DateTi30/;2	
TripResetSyncMesage/	TripResetSyncMesage/;4a1	TripResetSyncMesage/;	TripResetSync/;1	
VehBattU/	VehBattU/;8	VehBattU/;	SysU/;4	
VehCfgPrm/	VehCfgPrm/;5	VehCfgPrm/;	VehCfgPrm/;3	
VehCfgPrmExt/	VehCfgPrmExt/;2	VehCfgPrmExt/;	VehCfgPrm/;3	
VehCfgPrmExt2/	VehCfgPrmExt2/;1	VehCfgPrmExt2/;	VehCfgPrm/;3	
VehModMngtGlbSafe1/	VehModMngtGlbSafe1/;7	VehModMngtGlbSafe1/;	VehModMngtGlbSafe1/;2	116
VehMtnSt/	VehMtnSt/;10	VehMtnSt/;	VehMtnStSafe1/;2	54
VehSpdLgt/	VehSpdLgt/;15	VehSpdLgt/;	VehSpdLgtSafe1/;3	

6.3.3.2 IGM2: Send Ports

Port	Interface	DataElement	DataType	Generation Type	E2E ID
ElecMot2CoolgReq/	ElecMot2CoolgReq/;1	ElecMot2CoolgReq/;	TrCoolgReq/;2	Periodic	
ElecMot2CooltFlowMinReq/	ElecMot2CooltFlowMinReq/;1	ElecMot2CooltFlowMinReq/;	CooltFlow6/;1	Periodic	
ElecMot2CooltT/	ElecMot2CooltT/;1	ElecMot2CooltT/;	T3/;2	Periodic	
ElecMot2ErrIndcnReq/	ElecMot2ErrIndcnReq/;1	ElecMot2ErrIndcnReq/;	OnOff1/;2	Periodic	
ElecMot2IDc/	ElecMot2IDc/;1	ElecMot2IDc/;	Ampr1/;1	Periodic	

Port	Interface	DataElement	DataType	Generation Type	E2E ID
ElecMot2InvrT/	ElecMot2InvrT/;1	ElecMot2InvrT/;	T3/;2	Periodic	
ElecMot2LimnIndcn/	ElecMot2LimnIndcn/;1	ElecMot2LimnIndcn/;	UInt16/;2	Periodic	
ElecMot2ModSts/	ElecMot2ModSts/;2	ElecMot2ModSts/;	IsgModTyp2/;3	Periodic	
ElecMot2MotT/	ElecMot2MotT/;1	ElecMot2MotT/;	T3/;2	Periodic	
ElecMot2OilT/	ElecMot2OilT/;1	ElecMot2OilT/;	T3/;2	Periodic	
ElecMot2SpdActSgn/	ElecMot2SpdActSgn/;1	ElecMot2SpdActSgn/;	IsgSpdWSgnTyp/;2	Periodic	
ElecMot2TqAct/	ElecMot2TqAct/;2	ElecMot2TqAct/;	IsgTqActSafe2/;4	Periodic	1127
ElecMot2TqAvl/	ElecMot2TqAvl/;1	ElecMot2TqAvl/;	IsgTqAvl/;4	Periodic	
ElecMot2UDc/	ElecMot2UDc/;3	ElecMot2UDc/;	U52/;1	Periodic	
OBMElecMot2PendingSts/	OBMElecMot2PendingSts/;1	OBMElecMot2PendingSts/;	OnOff1/;2	Periodic	
TripResetSyncReqMsgIGM/	TripResetSyncReqMsgIGM/;1	TripResetSyncReqMsgIGM/;	NoYes1/;2	Sporadic	

6.3.3.3 Logical component: GlobalSnapshotDataReceiver

ID	Name	Version	Status	Last Changed
x04000000122A8FB6	GlobalSnapshotDataReceiver	(15)	CS Released	2023-05-12

Generic receiver LC providing all ports needed for reception of global snapshot data.

6.3.3.3.1 Included Virtual Function Clusters(VFCs)

6.3.3.3.2 GlobalSnapshotDataReceiver: Receiver Ports

Port	Interface	DataElement	DataType	E2E ID
BkpOfDstTrvld/	BkpOfDstTrvld/;7	BkpOfDstTrvld/;	Len31/;2	

Description

Total driven distance (odometer), in km

Port	Interface	DataElement	DataType	E2E ID
CarTiGlb/	CarTiGlb/;5	CarTiGlb/;	CarTiGlb/;2	

Description

Present a time value that shall be used by all ECUs to time-stamp diagnostic trouble codes(DTC) according to the requirements for Global DTC snapshot data.

Port	Interface	DataElement	DataType	E2E ID
VehBattU/	VehBattU/;8	VehBattU/;	SysU/;4	

Description

Vehicle System Voltage At 12V side

Port	Interface	DataElement	DataType	E2E ID
VehModMngtGlbSafe1/	VehModMngtGlbSafe1/;7	VehModMngtGlbSafe1/;	VehModMngtGlbSafe1/;2	

Description

VehModMngtGlbSafe1 is a system signal group sent globally in an end to end protected record that consists of CarModSts1, CarModSubtypWdCarModSubtyp, UsgModSts, EgyLvlElecMai, EgyLvlElecSubtyp, PwrLvlElecMai, PwrLvlElecSubtyp and FltEngyCnsWdSts

CarModSts1 : Factory, Transport, Dyno, Normal. Crash

UsgModSts : Abandoned, Inactive, Convenience, Active, Driving

Description
CarModSubtypWdCarModSubtyp: examples for CarModSubtyp: Factory Driving, Transport Driving

Port	Interface	DataElement	DataType	E2E ID
VehSpdLgt/	VehSpdLgt/;15	VehSpdLgt/;	VehSpdLgtSafe1/;3	

Description
Vehicle speed longitudinal based on wheel speed sensors and longitudinal acceleration Note: This signal is ASIL B And for ASIL D vehicle speed signal, please use signal 'WhlFastSpdSafe'.

6.3.3.3.3 GlobalSnapshotDataReceiver: Send Ports

6.3.3.3.4 Requirements

70416 v10 GlobalSnapshotDataReceiver [CS Released]

Purpose: To define how to obtain Global DTC snapshot information
Verification Method: Test
Legacy ID:

ECUs that need to receive Global Snapshot Data shall listen to the following signals, details about DTC snapshot data can be found in UDS Data section of the BaseTech SWRS, where:

- *Global real time* corresponds to the data element CarTiGlb,
- *Total distance* corresponds to the data elementBkpOfDstTrvld ,
- *Usage Mode* corresponds to the record item *UsgModSts* in the data element VehModMngtGlbSafe1
- *Vehicle battery voltage* corresponds to the data element VehBattU ,
- *ElPowerLevel* corresponds to the record items *PwrLvlElecMai* and *PwrLvlElecSubtyp* in the data element VehModMngtGlbSafe1 ,
- *Vehicle Speed corresponds to the data element* VehSpdLgt .

6.3.3.4 Logical component: SnapshotDataUTC

ID	Name	Version	Status	Last Changed
x0400000009543532	SnapshotDataUTC	(1)	CS Released	2021-07-28

UTC data information needed for reception of global snapshot data.

6.3.3.4.1 Included Virtual Function Clusters(VFCs)

6.3.3.4.2 SnapshotDataUTC: Receiver Ports

Port	Interface	DataElement	DataType	E2E ID
TiAndDateIndcn/	TiAndDateIndcn/;5	TiAndDateIndcn/;	DateTi30/;2	

Description
Current time (Hr, Min and Sec) and date (Year, Months and days) data for indication.

6.3.3.4.3 SnapshotDataUTC: Send Ports

6.3.3.4.4 Requirements

359907 v1 **SnapshotDataUTC [CS Released]**

Purpose: To define how to obtain Global DTC snapshot information
Verification Method: Test
Legacy ID:

ECUs that need to receive Global Snapshot Data shall listen to the UTC signals, details about DTC snapshot data can be found in UDS Data, where:

- *UTC time* corresponds to the data element TiAndDateIndcn .

6.3.3.5 Logical component: CarConfigReceiver

ID	Name	Version	Status	Last Changed
x0400000024176850	CarConfigReceiver	(11)	CS Released	2026-03-12

LC receives Car Configuration parameters in the vehicle.

6.3.3.5.1 Included Virtual Function Clusters(VFCs)

6.3.3.5.2 CarConfigReceiver: Receiver Ports

Port	Interface	DataElement	DataType	E2E ID
VehCfgPrm/	VehCfgPrm/;5	VehCfgPrm/;	VehCfgPrm/;3	
Description				
Container signal used to distribute the Car Configuration parameters in the vehicle.				

Port	Interface	DataElement	DataType	E2E ID
VehCfgPrmExt/	VehCfgPrmExt/;2	VehCfgPrmExt/;	VehCfgPrm/;3	
Description				
Container signal to be used for extentions of the Car Configuration parameter range beyond 504 parameters.				

Port	Interface	DataElement	DataType	E2E ID
VehCfgPrmExt2/	VehCfgPrmExt2/;1	VehCfgPrmExt2/;	VehCfgPrm/;3	
Description				
Container signal to be used for extentions of the Car Configuration parameter range beyond 1255 parameters.				

6.3.3.5.3 CarConfigReceiver: Send Ports

6.3.3.5.4 Requirements

70094 v6 **CarConfigReceiver [CS Released]**

Purpose: Receive the Car Configuration Parameter signals

Verification Method: Analysis

Legacy ID:

ECUs using CarConfig shall be recievers of signals VehCfgPrm ,VehCfgPrmExt and VehCfgPrmExt2used for the distribution of Vehicle Configuration Parameters (VCP), see BaseTech SWRS for more detailed requirements.

6.3.3.6 Logical component: PreProgmCdnForNewEgy

ID	Name	Version	Status	Last Changed
x04000000218DF5C8	PreProgmCdnForNewEgy	(9)	CS Released	2025-10-30

Entry conditions for service diagnosticSessionConrol (0x10), changing to programmingSession (0x02)

6.3.3.6.1 Included Virtual Function Clusters(VFCs)

6.3.3.6.2 PreProgmCdnForNewEgy: Receiver Ports

Port	Interface	DataElement	DataType	E2E ID
HvSysRlySts/	HvSysRlySts/;8	HvSysRlySts/;	HvSysRlySts/;1	

Description

Actual state of the main contactors.

Port	Interface	DataElement	DataType	E2E ID
VehModMngtGlbSafe1/	VehModMngtGlbSafe1/;7	VehModMngtGlbSafe1/;	VehModMngtGlbSafe1/;2	

Description

VehModMngtGlbSafe1 is a system signal group sent globally in an end to end protected record that consists of CarModSts1, CarModSubtypWdCarModSubtyp, UsgModSts, EgyLvlElecMai, EgyLvlElecSubtyp, PwrLvlElecMai, PwrLvlElecSubtyp and FltEngyCnsWdSts

CarModSts1 : Factory, Transport, Dyno, Normal. Crash

UsgModSts : Abandoned, Inactive, Convenience, Active, Driving

CarModSubtypWdCarModSubtyp: examples for CarModSubtyp: Factory Driving, Transport Driving

Port	Interface	DataElement	DataType	E2E ID
VehSpdLgt/	VehSpdLgt/;15	VehSpdLgt/;	VehSpdLgtSafe1/;3	

Description

Vehicle speed longitudinal based on wheel speed sensors and longitudinal acceleration

Note: This signal is ASIL B
And for ASIL D vehicle speed signal, please use signal 'WhlFastSpdSafe'.

6.3.3.6.3 PreProgmCdnForNewEgy: Send Ports

6.3.3.6.4 Requirement

Entry conditions for service diagnosticSessionConrol (0x10), changing to programmingSession

(0x02) (excluding programmingSession to programmingSession):
The implementer shall implement the ECU's condition for entering programmingSession based on the allocated functionality. The condition shall ensure a defined and safe vehicle state when entering programmingSession and shall at a minimum include vehicle speed < 3km/h if not otherwise approved by CEVT & GRI Electrical Architecture.
In an impaired vehicle or in a stand alone scenario if the vehicle signal(s) used in the evaluation of the condition e.g. speed and/or "main propulsion system not active" shall the safety mechanism not prevent the ECU to change to programmingSession to allow SWDL.

If the ECU implement safety requirements with an ASIL higher than QM (see *[Services_12] ISO 26262*) it shall, in all situations when diagnostic services may violate any of those safety requirements, reject the critical diagnostic service requests. Note that if the ECU rejects such critical diagnostic service requests, this requires an approval by Geely Electrical Architecture.

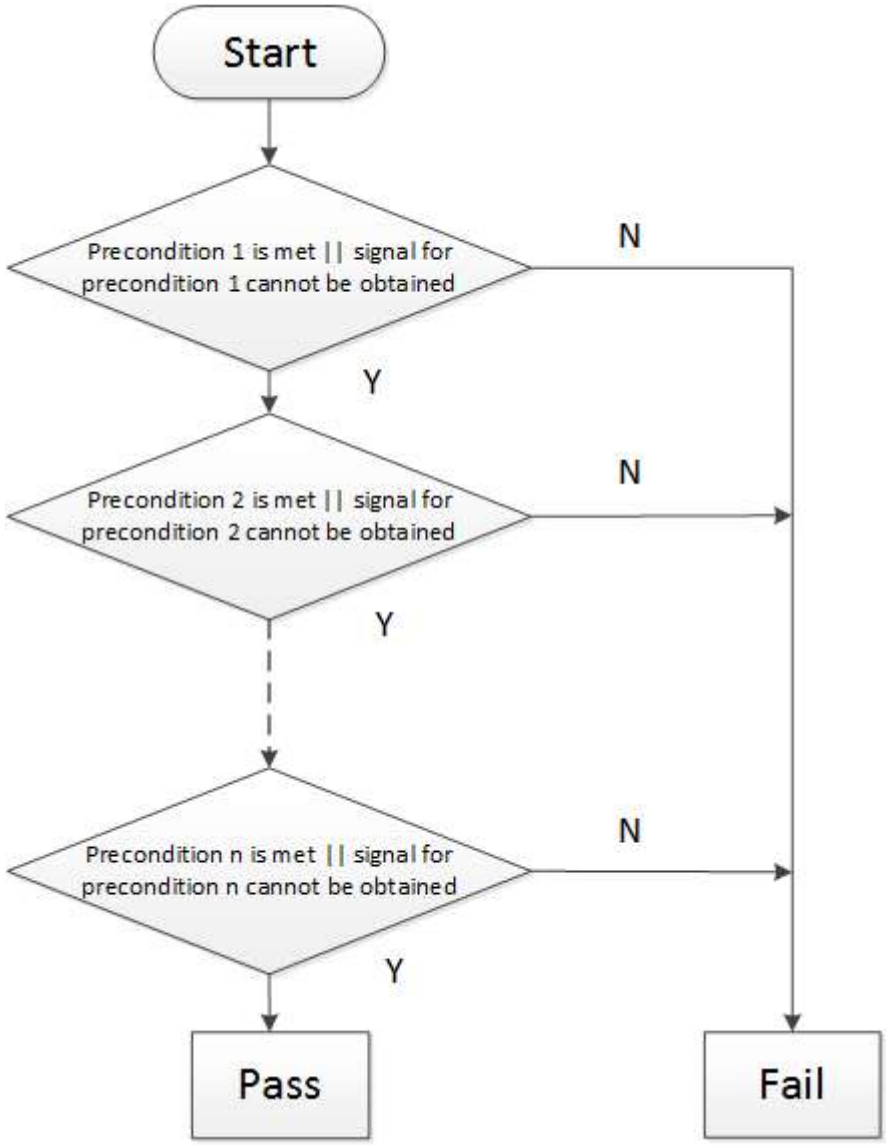
116702 v6 Pre-condition for New energy Domain MGM/IEM/IGM [CS Released]

Purpose: To ensure a defined and safe vehicle state upon entering programming session.
Verification Method: Test
Legacy ID:

The pre-conditions for plugin hybrid or electric vehicle to be met are as follows:
1) Vehicle speed VehSpdLgt< 3km/h and VehSpdLgtQf is equal to 2 or 3, where VehSpdLgt corresponds to the vehicle speed;
2) Signal UsgModSts in VehModMngtGlbSafe1is not equal to "Driving".
3) High voltage is less than 60V, the high voltage will be detected by MGM/IEM/IGM itself.
4) Main propulsion system is not active which means that the signal HvSysRlySts is equal to as "HvActvnSts_Open".

In an impaired vehicle or a stand alone scenario if the vehicle signal(s) used in the evaluation of the condition e.g. speed and/or "main propulsion system not active" shall the safety mechanism not prevent the ECU to change to programmingSession to allow SWDL. That is to say, for a specific pre-condition, it is considered to be met if the signal can't be obtained during the time to evaluate it. For example, vehicle speed condition is considered as met once the signal 'VehSpdLgt' cannot be obtained, but the other pre-condition (such as 'main propulsion') shall continue to evaluate in the next.

All of the pre-conditions shall be evaluated independently, only if all of the above pre-conditions are met can the ECU enter bootloader.



Pre-condition Evaluating Process

Other Pre-Conditions for entering bootloader(such as safety requirement) must be specified separately and confirmed by GRI & CEVT.

98793 v2 **Function safety related pre-condition [CS Released]**

Purpose: To define Pre-condition for function safety requiement
Verification Method: Analysis
Legacy ID:

If the ECU implement safety requirements with an ASIL higher than QM (see *[Services_12] ISO*

26262) it shall, in all situations when diagnostic services may violate any of those safety requirements, reject the critical diagnostic service requests. Note that if the ECU rejects such critical diagnostic service requests, this requires an approval by Geely Electrical Architecture.

If there is function safety requirement about pre-condition allocated to some ECUs, then the ECU should follow the function safety requirement.

6.3.3.7 Logical component: PreCdnforDiagComCtrlForNewEgy

ID	Name	Version	Status	Last Changed
x0400000014C7D594	PreCdnforDiagComCtrlForNewEgy	(2)	CS Released	2023-10-23

Entry conditions for service CommunicationControl (0x28) or ControlDTCSetting (0x85)

6.3.3.7.1 Included Virtual Function Clusters(VFCs)

6.3.3.7.2 PreCdnforDiagComCtrlForNewEgy: Receiver Ports

Port	Interface	DataElement	DataType	E2E ID
HvSysRlySts/	HvSysRlySts/;8	HvSysRlySts/;	HvSysRlySts/;1	

Description

Actual state of the main contactors.

Port	Interface	DataElement	DataType	E2E ID
VehModMngtGlbSafe1/	VehModMngtGlbSafe1/;7	VehModMngtGlbSafe1/;	VehModMngtGlbSafe1/;2	

Description

VehModMngtGlbSafe1 is a system signal group sent globally in an end to end protected record that consists of CarModSts1, CarModSubtypWdCarModSubtyp, UsgModSts, EgyLvlElecMai, EgyLvlElecSubtyp, PwrLvlElecMai, PwrLvlElecSubtyp and FltEngyCnsWdSts

CarModSts1 : Factory, Transport, Dyno, Normal. Crash

UsgModSts : Abandoned, Inactive, Convenience, Active, Driving

CarModSubtypWdCarModSubtyp: examples for CarModSubtyp: Factory Driving, Transport Driving

Port	Interface	DataElement	DataType	E2E ID
VehSpdLgt/	VehSpdLgt/;15	VehSpdLgt/;	VehSpdLgtSafe1/;3	

Description

Vehicle speed longitudinal based on wheel speed sensors and longitudinal acceleration

Note: This signal is ASIL B
And for ASIL D vehicle speed signal, please use signal 'WhlFastSpdSafe'.

6.3.3.7.3 PreCdnforDiagComCtrlForNewEgy: Send Ports

6.3.3.7.4 Requirement

676546 v2 **Pre-Condition for Entering diagnostic communication control [CS Released]**

Purpose: To define the entry conditions for service CommunicationControl (0x28) or ControlDTCSetting (0x85)

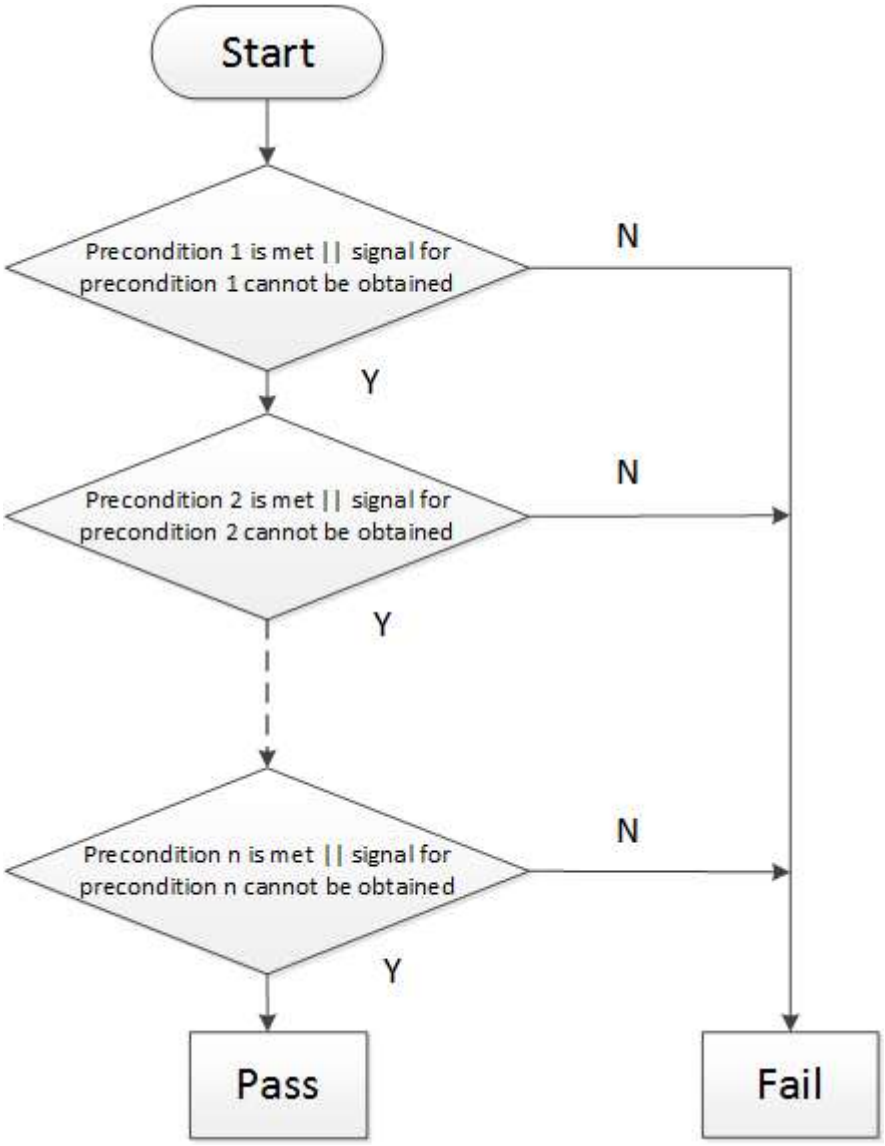
Verification Method: Test

Legacy ID:

While the pre-conditions for plugin hybrid or electric vehicle ECU to be met are as follows:
1) Vehicle speed VehSpdLgt< 3km/h and VehSpdLgtQf is equal to 2 or 3, where VehSpdLgt corresponds to the vehicle speed;
2) Signal UsgModSts in VehModMngtGlbSafe1 is not equal to “Driving”.
3) High voltage is less than 60V, the high voltage will be detected by MGM/IEM/IGM itself.
4) Main propulsion system is not active which means that the signal HvSysRlySts is equal to as “HvActvnSts_Open”.

In an impaired vehicle or a stand alone scenario if the vehicle signal(s) used in the evaluation of the condition e.g. speed and/or "main propulsion system not active" shall the safety mechanism not prevent the ECU to enter specific control. That is to say, for a specific pre-condition, it is considered to be met if the signal can't be obtained during the time to evaluate it. For example, vehicle speed condition is considered as met once the signal 'VehSpdLgt' cannot be obtained, but the other pre-condition (such as ‘main propulsion’) shall continue to evaluate in the next.

All of the pre-conditions shall be evaluated independently, only if all of the above pre-conditions are met can the ECU enter diagnostic communication control or DTCSetting control.



Pre-condition Evaluating Process

Other Pre-Conditions for entering diagnostic communication control or DTCSetting control (such as safety requirement) must be specified separately and confirmed by GRI.

6.3.3.8 Logical component: EM2ThermalManagementInterface

ID	Name	Version	Status	Last Changed
x0400000020A38138	EM2ThermalManagementInterface	EEA3.5_25R2(2b1a2)	CS Released	2025-09-04

High Voltage Electric Drive - Front Part
Auxillary Front Electric Drive System
Configuration From:

1. Front RIght Hub Motor
2. DHT(PCM)(P1)

6.3.3.8.1 Included Virtual Function Clusters(VFCs)

6.3.3.8.2 EM2ThermalManagementInterface: Receiver Ports

Port	Interface	DataElement	DataType	E2E ID
CooltFlowForDtElec/	CooltFlowForDtElec/;5	CooltFlowForDtElec/;	CooltFlow2/;2	

Description

Predicted coolant low for Electric Drivetrain (ED) coolant circuit.

Port	Interface	DataElement	DataType	E2E ID
CooltTSigForDtElec/	CooltTSigForDtElec/;5	CooltTSigForDtElec/;	CooltTHiResl/;2	

Description

Coolant temperature measured by sensors installed in emotor cooling citcuit.

Port	Interface	DataElement	DataType	E2E ID
EEMReqElecMot2OilSpd/	EEMReqElecMot2OilSpd/;1	EEMReqElecMot2OilSpd/;	EEMReqOilSpd/;1	

Description

Minimum oil speed required ISG by EEM.

Port	Interface	DataElement	DataType	E2E ID
EngCooltInhbFlow/	EngCooltInhbFlow/;4	EngCooltInhbFlow/;	CooltInhb/;2	

Description

Inhibitation of non-crucial coolant flow requests. Sent out to inform that the eninge is in heat up phase and that all additional coolant flow will postphone the time when the engine has reached normal operating temperature.

6.3.3.8.3 EM2ThermalManagementInterface: Send Ports

Port	Interface	DataElement	DataType	Generation Type	E2E ID
ElecMot2CoolgReq/	ElecMot2CoolgReq/;1	ElecMot2CoolgReq/;	TrCoolgReq/;2	Periodic	

Description

Rear motor cooling request

Port	Interface	DataElement	DataType	Generation Type	E2E ID
ElecMot2CooltFlowMinReq/	ElecMot2CooltFlowMinReq/;1	ElecMot2CooltFlowMinReq/;	CooltFlow6/;1	Periodic	

Description

P1 motor cooling flow request

l/min

Port	Interface	DataElement	DataType	Generation Type	E2E ID
ElecMot2CooltT/	ElecMot2CooltT/;1	ElecMot2CooltT/;	T3/;2	Periodic	

Description

Cooling temperature of rear motor

Port	Interface	DataElement	DataType	Generation Type	E2E ID
ElecMot2InvrT/	ElecMot2InvrT/;1	ElecMot2InvrT/;	T3/;2	Periodic	

Description

Highest or most relevant temperature measured in the inverter.

Port	Interface	DataElement	DataType	Generation Type	E2E ID
ElecMot2MotT/	ElecMot2MotT/;1	ElecMot2MotT/;	T3/;2	Periodic	

Description
Temperature of the auxiliary motor

Port	Interface	DataElement	DataType	Generation Type	E2E ID
ElecMot2OilT/	ElecMot2OilT/;1	ElecMot2OilT/;	T3/;2	Periodic	

Description
Rear electric oil temperature

6.3.3.8.4 Signal Map

port trace.

242222 v1a1 **Signal Map [CS Released]**

Purpose: To map SystemSignal to actual signals used on CAN
Verification Method: Documentation
Legacy ID:

Signal Map

System Signal	Actual Signal	Description
SysSigMotT	ElecMot2MotT	Measured Electric Motor Temperature
SysSigInvrT	ElecMot2InvrT	Measured Inverter Temperature
SysSigEngCooltInhbFlow	EngCooltInhbFlow	Coolant Flow Request inhibitation from Engine
SysSigCooltFlowElecDt	CooltFlowForDtElec	Predicted coolant flow from thermal manager
SysSigCooltTForElecDt	CooltTSigForDtElec	Coolant temperature from thermal manager
SysSigEEMReqElecMot2OilSpd	EEMReqElecMot2OilSpd	Minimum oil speed required ISG by EEM.

6.3.3.9 Logical component: EM2MotorControlInterface

ID	Name	Version	Status	Last Changed
x040000002358CFB3	EM2MotorControlInterface	EEA3.5 26R1(7 a4)	CS Released	2026-01-27

High Voltage Electric Drive - Front Part
Auxillary Front Electric Drive System
Configuration From:

1. Front RIght Hub Motor
2. DHT(PCM)(P1)

6.3.3.9.1 Included Virtual Function Clusters(VFCs)

6.3.3.9.2 EM2MotorControlInterface: Receiver Ports

Port	Interface	DataElement	DataType	E2E ID
CrashStsSafe/	CrashStsSafe/;2	CrashStsSafe/;	CrashStsSafe/;1	1035
Description				
Information if a crash has been detected by SRS. E2E protected.				

Port	Interface	DataElement	DataType	E2E ID
ElecMot2ActvDampgGain/	ElecMot2ActvDampgGain/;1	ElecMot2ActvDampgGain/;	GainTypActvDampg8/;2	
Description				
Generator motor active damping mode gain				

Port	Interface	DataElement	DataType	E2E ID
ElecMot2ActvDampgModReq/	ElecMot2ActvDampgModReq/;1	ElecMot2ActvDampgModReq/;	WhlMotSysActvDampgModCodSts/;2	
Description				
Generator motor active damping mode request 0 = No damping 1 = Low damping 2 = Medium damping 3 = High damping				

Port	Interface	DataElement	DataType	E2E ID
ElecMot2EngSpdOfActvDampg/	ElecMot2EngSpdOfActvDampg/;1	ElecMot2EngSpdOfActvDampg/;	N1/;2	
Description				
Generator motor reference speed of active damping				

Port	Interface	DataElement	DataType	E2E ID
ElecMot2ModReq/	ElecMot2ModReq/;1	ElecMot2ModReq/;	IsgModTyp1/;4	
Description				
Generator motor operation mode request 1: standby 2: torque control 3: speed control 4: UDC control 5: Speed Control Dead Zone				

Port	Interface	DataElement	DataType	E2E ID
ElecMot2PwrLimMax/	ElecMot2PwrLimMax/;1	ElecMot2PwrLimMax/;	PrpsnPwrLimMax/;2	
Description				
Maximum generator motor drive system powerlimit on DC side. Positive power means power flowing into inverter.				

Port	Interface	DataElement	DataType	E2E ID
ElecMot2PwrLimMin/	ElecMot2PwrLimMin/;1	ElecMot2PwrLimMin/;	PrpsnPwrLimMin/;2	
Description				
Minimum generator motor drive system powerlimit on DC side. Positive power means power flowing into inverter.				

Port	Interface	DataElement	DataType	E2E ID
ElecMot2SpdReq/	ElecMot2SpdReq/;1	ElecMot2SpdReq/;	IsgSpdWSgnTyp/;2	
Description				
Generator motor speed request				

Port	Interface	DataElement	DataType	E2E ID
ElecMot2TqAllwd/	ElecMot2TqAllwd/;2	ElecMot2TqAllwd/;	IsgTqAllwd1/;3	91
Description				
Indication if generator ISG is allowed to apply propulsive or regenerative torque on wheels. Coding: 0x0: Propulsive and regeneration torque not allowed; 0x1: Propulsive torque allowed; 0x2: Regenerative torque allowed; 0x3: Propulsive and regeneration torque allowed				

Port	Interface	DataElement	DataType	E2E ID
ElecMot2TqReq/	ElecMot2TqReq/;1	ElecMot2TqReq/;	Tq34/;1	
Description				
Generator motor torque request				

Port	Interface	DataElement	DataType	E2E ID
EngSt1WdSts/	EngSt1WdSts/;4	EngSt1WdSts/;	EngSt1WdSts/;1	137
Description				
Signaling the state of the powerpack				

Port	Interface	DataElement	DataType	E2E ID
EngTqReqFast/	EngTqReqFast/;1	EngTqReqFast/;	PtTqType/;2	
Description				
Engine request torque for fast path				

Port	Interface	DataElement	DataType	E2E ID
HvBattUDc/	HvBattUDc/;2	HvBattUDc/;	BCLRqrdU/;1	
Description				
Actual Hv Battery pack voltage (measuered before contactors, i.e. always giving pack voltage).				

Port	Interface	DataElement	DataType	E2E ID
HvSysRlySts/	HvSysRlySts/;8	HvSysRlySts/;	HvSysRlySts/;1	21
Description				
Actual state of the main contactors.				

Port	Interface	DataElement	DataType	E2E ID
PrpsnFbLimnIndcn/	PrpsnFbLimnIndcn/;2	PrpsnFbLimnIndcn/;	UInt4/;2	
Description				
Feedback signal that the present limitation has lead to a recofiguraton that has affected the driver.				

Port	Interface	DataElement	DataType	E2E ID
PtTqAtAxleMaxReq/	PtTqAtAxleMaxReq/;2	PtTqAtAxleMaxReq/;	PtTqAtAxleReq/;1	1109
Description				
Maximum allowed powertrain applied torque on the axle. Positive value indicates limitation of powertrain torque in the desired driving direction of the driver. The signal shall not be compensated for powertrain inertia.				

Port	Interface	DataElement	DataType	E2E ID
VehMtnSt/	VehMtnSt/;10	VehMtnSt/;	VehMtnStSafe1/;2	54
Description				
Vehicle motion state information.				

6.3.3.9.3 EM2MotorControlInterface: Send Ports

Port	Interface	DataElement	DataType	Generation Type	E2E ID
ElecMot2LimnIndcn/	ElecMot2LimnIndcn/;1	ElecMot2LimnIndcn/;	UInt16/;2	Periodic	

Description
Indicate auxiliary motor unavailable due to a fault

Port	Interface	DataElement	DataType	Generation Type	E2E ID
ElecMot2ModSts/	ElecMot2ModSts/;2	ElecMot2ModSts/;	IsgModTyp2/;3	Periodic	

Description
Auxiliary motor mode status

Port	Interface	DataElement	DataType	Generation Type	E2E ID
ElecMot2SpdActSgn/	ElecMot2SpdActSgn/;1	ElecMot2SpdActSgn/;	IsgSpdWSgnTyp/;2	Periodic	

Description
Actual auxiliary motor speed with sign (forward/backwards)

Port	Interface	DataElement	DataType	Generation Type	E2E ID
ElecMot2TqAct/	ElecMot2TqAct/;2	ElecMot2TqAct/;	IsgTqActSafe2/;4	Periodic	1127

Description
Estimated auxiliary motor torque on the crankshaft

Port	Interface	DataElement	DataType	Generation Type	E2E ID
ElecMot2TqAvl/	ElecMot2TqAvl/;1	ElecMot2TqAvl/;	IsgTqAvl/;4	Periodic	

Description
Maximum available auxiliary motor torque on the crankshaft. A function of motor drive system HW Limits, actual DC Voltage, motor temperature, etc. The torque is positive in the forward rotation direction. Minimum available auxiliary motor torque on the crankshaft. A function of motor drive system HW Limits, actual DC Voltage, motor temperature, etc. The torque is positive in the forward rotation direction

6.3.3.9.4 Signal Map

242482 v2a1 **Signal Map [CS Released]**

Purpose: To map SystemSignal to actual signals used on CAN
Verification Method: Documentation
Legacy ID:

Signal Map

System Signal	Actual Signal	Description
SysSigModReq	ElecMot2ModReq	Mode request to Electric Drive System
SysSigModSts	ElecMot2ModSts or ElecMot2ModstsEO11	Mode status of the Electric Drive System
SysSigTqAvl	ElecMot2TqAvl	Available torque from the Electric Drive System
SysSigSpdAct	ElecMot2SpdActSgn or ElecMot2SpdActSgnEO11	Rotational speed of the Electric Drive System
SysSigTqAct	ElecMot2TqAct	Actual Torque applied, positive sign is in the forward direction
SysSigLimnIndcn	ElecMot2LimnIndcn	Limitation Indication signal from Electric Drive System
SysSigPtTqAtAxleMaxReq	PtTqAtAxleMaxReq	Torque Limitations from Chassi on the wheel domain

SysSigTqAllwd	ElecMot2TqAllwd	Propulsive/regenerative torque allowance
SysSigHvRlySts	HvSysRlySts	Status & request signal from the HV-Battery
SysSigEngSt1WdSts	EngSt1WdSts	Actual status of the engine
SysSigVehMtnSts	VehMtnSt	Vehicle motion status
SysSigActvDampgGain	ElecMot2ActvDampgGain	Active damping (anti-shuffle) gain factor.
SysSigActvDampgModReq	ElecMot2ActvDampgModReq	Mode signal for Active damping
SysSigEngSpdOfActvDampg	ElecMot2EngSpdOfActvDampg	Engine speed target for Active damping
SysSigPwrLimMin	ElecMot2PwrLimMin	Electrical power limitation
SysSigPwrLimMax	ElecMot2PwrLimMax	Electrical power limitation
SysSigSpdReq	ElecMot2SpdReq	Rotational speed request to the Electric Drive System
SysSigTqReq	ElecMot2TqReq	Torque request to the Electric Drive System
SysSigFbLimnIndcn	PrpsnFbLimnIndcn	Feedback from Propulsion master of Limitations
SysSigCrashStsSafe	CrashStsSafe	Crash status signal
SysSigHvBattUDc	HvBattUDc	Measured voltage from the High Voltage Battery

6.3.3.10 Logical component: EM2Obd2Interface

ID	Name	Version	Status	Last Changed
x0400000021D8139E	EM2Obd2Interface	EEA3.525R3_1 (3a3)	CS Released	2025-11-17

High Voltage Electric Drive - Front Part
Auxillary Front Electric Drive System
Configuration From:

1. Front RIght Hub Motor
2. DHT(PCM)(P1)

6.3.3.10.1 Included Virtual Function Clusters(VFCs)

6.3.3.10.2 EM2Obd2Interface: Receiver Ports

Port	Interface	DataElement	DataType	E2E ID
AmbTRaw/	AmbTRaw/;5	AmbTRaw/;	AmbTWithQly/;2	
Description				
unfiltered AMB temperature				

Port	Interface	DataElement	DataType	E2E ID
DenoOfDiagcRateGen/	DenoOfDiagcRateGen/;1	DenoOfDiagcRateGen/;	Boolean/;2	
Description				
To indicate whether the driving cycle meets the condition as a diagnostics denominator cycle defined in China Six J3.4.3.2				

Port	Interface	DataElement	DataType	E2E ID
DrvgCyc/	DrvgCyc/;3	DrvgCyc/;	OnOff1/;2	
Description				
Transition from 0 to 1 indicates a start of a new Driving Cycle.				

Port	Interface	DataElement	DataType	E2E ID
DrvgCycToClrPrmntDtc/	DrvgCycToClrPrmntDtc/;1	DrvgCycToClrPrmntDtc/;	Boolean/;2	
Description				
China Six defines the cycle of clearing permanent fault codes. (J3.2.5.2 (B))				

Port	Interface	DataElement	DataType	E2E ID
EngNSafe/	EngNSafe/;7	EngNSafe/;	EngNSafe/;2	68
Description				
Engine Speed [rpm] and Engine Speed derivative [rpm/s]				

Port	Interface	DataElement	DataType	E2E ID
EngSt1WdSts/	EngSt1WdSts/;4	EngSt1WdSts/;	EngSt1WdSts/;1	137
Description				
Signaling the state of the powerpack				

Port	Interface	DataElement	DataType	E2E ID
EngT/	EngT/;5	EngT/;	EngT/;2	

Description
Engine coolant temperature signal group. Includes: - Engine coolant temperature. Resolution 1 degC. - Engine coolant temperature quality factor. Engine coolant temperature

Port	Interface	DataElement	DataType	E2E ID
EngTiOn/	EngTiOn/;1	EngTiOn/;	TimeDesForSWDL/;2	
Description				
Time since engine starts or propulsion on with movement for PID $1F.				

Port	Interface	DataElement	DataType	E2E ID
HvBattSoc/	HvBattSoc/;3	HvBattSoc/;	NewBattSOC/;1	
Description				
Actual state of charge of the Hv Battery.				

Port	Interface	DataElement	DataType	E2E ID
IuprDenominatorInhb/	IuprDenominatorInhb/;1	IuprDenominatorInhb/;	InhibitSts/;1	
Description				
Inhibition flag of IUPR denominator from China Six.				

Port	Interface	DataElement	DataType	E2E ID
LoadValCalcd/	LoadValCalcd/;3	LoadValCalcd/;	Perc5/;2	
Description				
The ECU shall make the signal Calculated Load Value according to SAE J1979 available on CAN.				

Port	Interface	DataElement	DataType	E2E ID
MonrCycCmpl/	MonrCycCmpl/;3	MonrCycCmpl/;	NotCmpl1/;2	
Description				
OBD Warm Up Cycle Complete. 0= Not Complete; 1= Complete.				

Port	Interface	DataElement	DataType	E2E ID
OBMPrpsnActve/	OBMPrpsnActve/;1	OBMPrpsnActve/;	Boolean/;2	
Description				
propulsion active in OBM requirement				

Port	Interface	DataElement	DataType	E2E ID
ThrPos/	ThrPos/;3	ThrPos/;	Perc2/;2	
Description				
ECM throttle position (AbsThrottlePosition) on CAN				

Port	Interface	DataElement	DataType	E2E ID
VehSpdLgt/	VehSpdLgt/;15	VehSpdLgt/;	VehSpdLgtSafe1/;3	
Description				
Vehicle speed longitudinal based on wheel speed sensors and longitudinal acceleration Note: This signal is ASIL B And for ASIL D vehicle speed signal, please use signal 'WhlFastSpdSafe'.				

6.3.3.10.3 EM2Obd2Interface: Send Ports

Port	Interface	DataElement	DataType	Generation Type	E2E ID
ElecMot2ErrIndcnReq/	ElecMot2ErrIndcnReq/;1	ElecMot2ErrIndcnReq/;	OnOff1/;2	Periodic	
Description					
MIL ilumination req from EM2					

Port	Interface	DataElement	DataType	Generation Type	E2E ID
OBMElecMot2PendingSts/	OBMElecMot2PendingSts/;1	OBMElecMot2PendingSts/;	OnOff1/;2	Periodic	

Description
OBM pending fault send to propulsion of P1 motor

6.3.3.10.4 Signal Map

252591 v2 **Signal Map [CS Released]**

Purpose: To map SystemSignal to actual signals used on CAN
Verification Method: Documentation
Legacy ID:

Signal Map

System Signal	Actual Signal	Description
SysSigDenOfDiagRateGen	DenoOfDiagcRateGen	To arbitrate the denominator flag of diagnostics rate
SysSigDrvCyc	DrvgCyc	Driving Cycle indicator
SysSigDrvCycToClrPrmntDtc	DrvgCycToClrPrmntDtc	Indicate whether the driving cycle meets the condition of clearing permenent DTCs
SysSigEngNSafe	EngNSafe	Engine Speed [RPM] and Engine Speed derivative
SysSigEngSt1WdSts	EngSt1WdSts	Engine status
SysSigEngT	EngT	Engine Temperature
SysSigEngTiOn	EngTiOn	Engine Time On
SysSigHVBattSoc	HvBattSoc	State of Charge from High Voltage Battery
SysSigClrRdyReq	DrvgCycToClrPrmntDtc	Signal to allow clearing of permanent faults
SysSigErrIndcnReq	ElecMot2ErrIndcnReq	MIL request
SysSigIUPRDenominatorInhb	IuprDenominatorInhb	Inhibition flag of IUPR denominator
SysSigLoadValCalcd	LoadValCalcd	Calculated Load Value
SysSigMonrCycCompl	MonrCycCmpl	OBD Warm Up Cycle Complete
SysSigThrPos	ThrPos	Throttle position
SysSigVehSpdLgt	VehSpdLgt	Vehicle speed longitudinal
AmbTRaw	AmbTRaw	Environment temperature

6.3.3.10.5 OBM Requirements

1051684 v1 [CS Released] **Send ElecMot2 OBM trip Pending fault code status signal to Propulsion S-512-02**

Purpose: To ElecMot2 Send OBM trip Pending fault code status signal to Propulsion
Verification Method: Test
Legacy ID:

AC1	To Send ElecMot2 OBM trip Pending fault code status signal to Propulsion	
参数标定	/	/
前提条件 Pre-Condition	/	
触发事件	HVElectricDrive子系统应接收Propulsion子系统发送的OBM行程开始结束信号 OBMPrpsnActve	
响应动作	HVElectricDrive发送本次OBM行程的“Pending fault code status”信号 OBMElecMot2PendingSts至Propulsion子系统。 “Pending fault code status”信号定义： ●0表示本次OBM行程未检测到会点亮MIL灯的Pending fault code; ●1表示本次OBM行程至少检测到1个会点亮MIL灯的Pending fault code。	

6.3.3.11 Logical component: GeneralRequirements

ID	Name	Version	Status	Last Changed
x0400000021EFE655	GeneralRequirements	EEA3.5 25R3(8 a1a6)	Released	2025-11-24

General Requirements for EDS Systems

6.3.3.11.1 Included Virtual Function Clusters(VFCs)

6.3.3.11.2 GeneralRequirements: Receiver Ports

6.3.3.11.3 GeneralRequirements: Send Ports

Port	Interface	DataElement	DataType	Generation Type	E2E ID
HVEDPrpsnDataEvtA/	HVEDPrpsnDataEvtA/ ;1	HVEDPrpsnDataEvtA/;	PrpsnDataEvt /;1	Periodic	

Description
HVED Cloud Big Data Signals.

Port	Interface	DataElement	DataType	Generation Type	E2E ID
HVEDPrpsnDataEvtB/	HVEDPrpsnDataEvtB/ ;1	HVEDPrpsnDataEvtB/;	PrpsnDataEvt /;1	Periodic	

Description
HVED Cloud Big Data Signals.

Port	Interface	DataElement	DataType	Generation Type	E2E ID
HVEDPrpsnDataEvtC/	HVEDPrpsnDataEvtC/;1	HVEDPrpsnDataEvtC/;	PrpsnDataEvt/;1	Periodic	

Description

HVED Cloud Big Data Signals.

Port	Interface	DataElement	DataType	Generation Type	E2E ID
HVEDPrpsnDataEvtD/	HVEDPrpsnDataEvtD/;1	HVEDPrpsnDataEvtD/;	PrpsnDataEvt/;1	Periodic	

Description

HVED Cloud Big Data Signals.

Port	Interface	DataElement	DataType	Generation Type	E2E ID
HVEDPrpsnDataEvtE/	HVEDPrpsnDataEvtE/;1	HVEDPrpsnDataEvtE/;	PrpsnDataEvt/;1	Periodic	

Description

HVED Cloud Big Data Signals.

Port	Interface	DataElement	DataType	Generation Type	E2E ID
HVEDPrpsnDataPrdA/	HVEDPrpsnDataPrdA/;1	HVEDPrpsnDataPrdA/;	PrpsnDataPrd/;1	Periodic	

Description

HVED Cloud Big Data Signals.

Port	Interface	DataElement	DataType	Generation Type	E2E ID
HVEDPrpsnDataPrdB/	HVEDPrpsnDataPrdB/;1	HVEDPrpsnDataPrdB/;	PrpsnDataPrd/;1	Periodic	

Description

HVED Cloud Big Data Signals.

Port	Interface	DataElement	DataType	Generation Type	E2E ID
HVEDPrpsnDataPrdC/	HVEDPrpsnDataPrdC/;1	HVEDPrpsnDataPrdC/;	PrpsnDataPrd/;1	Periodic	

Description

HVED Cloud Big Data Signals.

Port	Interface	DataElement	DataType	Generation Type	E2E ID
HVEDPrpsnDataPrdD/	HVEDPrpsnDataPrdD/;1	HVEDPrpsnDataPrdD/;	PrpsnDataPrd/;1	Periodic	

Description

HVED Cloud Big Data Signals.

Port	Interface	DataElement	DataType	Generation Type	E2E ID
HVEDPrpsnDataPrdE/	HVEDPrpsnDataPrdE/;1	HVEDPrpsnDataPrdE/;	PrpsnDataPrd/;1	Periodic	

Description

HVED Cloud Big Data Signals.

6.3.3.11.4 Carconfig info:

Parameter number	Parameter name	Parameter value	Description
4	ELECTRIC DRIVELINE	06	Electric Motor AWD
		07	Electric Motor FWD
		08	Electric Motor RWD
		80	Combustion Engine FWD and Electric Motor FWD2

Parameter number	Parameter name	Parameter value	Description
		02	Combustion Engine FWD and Electric Motor FWD
		03	Combustion Engine FWD and Electric Motor RWD
		04	Combustion Engine RWD and Electric Motor FWD
		05	Combustion Engine RWD and Electric Motor RWD
9	GEARBOX	81	8ATEAWD

6.3.3.11.5 Requirements

6.3.3.11.5.1 Measurements

242030 v2 **HVDC Current [CS Released]**

Purpose: Measure and signal the current on the HV DC link.
Verification Method: Test
Legacy ID:

The current on the HV DC link shall be measured by inverter and the value shall be sent out on ***SysSigIDc***. The current in to the inverter is defined as positive.

The accuracy of the DC current shall be as follows: If the DC current is -100A~+100A, the accuracy shall be ±5A, if the DC current is more than +100A or less than -100A, the accuracy shall be 5%.

242031 v2 **HVDC Voltage [CS Released]**

Purpose: Measure and signal the voltage on the HV DC link.
Verification Method: Test
Legacy ID:

The voltage on the HV DC link shall be measured by inverter and the value shall be sent out on ***SysSigUDc***.

The accuracy of the DC voltage shall be as follows: If the DC voltage is less than 100V, the accuracy shall be ±5V, if the DC voltage is more than 100V, the accuracy shall be 5%.

716350 v1 **Sensor raw delay [CS Released]**

Purpose: to avoid sensor initial value abnormal because of sensor supply voltage unstable
Verification Method: Analysis AND Test
Legacy ID:

To make sure Input values from sensors are accurate and the system work normally. system sensors raw value shall be used delay for a period of time after sensors power supply last stable for 500ms(tbd).

Sensor should be Include:

HVDC voltage sensor

HVDC current sensor

Phase current sensor

Resolver sensor

Temperature sensor

Note：

unstable sensor supply voltage will result in sensor raw swing which may cause torque output abnormal in the beginning of drive cycle

6.3.3.11.5.2 General ECU & SW

97966 v2 Memory area for data [CS Released]

Purpose: To reduce the amount of time in the workshop during a SW update.
Verification Method: Analysis AND Test
Legacy ID: 383931

There shall be a specific memory area in the NVRAM for unique settings/data. These unique settings are not allowed to be deleted during a SW download.

A list of all data that are stored in this area shall be provided to the OEM.

Note:

Example of settings is: Rotor position calibration, Vehicle unique keys, DRO data

97968 v3 Running reset recovery [CS Released]

Purpose: The vehicle needs the inverter to recover as fast as possible after a running reset. However, multiple running resets may be a symptom of a fault that may cause the inverter to be unsafe
Verification Method: Test
Legacy ID: 383933

The ECU shall be able to recover after a running reset and it shall

- go to Standby mode after single running reset, awaiting the mode request for Standby mode and then follow the mode request.

- go to Inverter Fault mode after repeated running resets (more than 2 resets during 60 s).

Info: The definition of a running reset is a reset that is not initiated by the “normal” reset conditions, such as power up, diagnostic reset request, startup/shut down I/O etc. 12 V dips and watch dog reset are common causes for running reset.

To improve software robustness, system reset or watch dog monitoring mechanism shall be introduced in some special situation.such as Aurix chip faults, program stuck, hardware over current. etc (these faults maybe result in ECU not be waked up or system break down). Retry monitoring mechanism can improve software perfermance and optimize vehicle performance in these conditions. Retry counter have to be limited considering actual application which can't make hardware burned or vehicle out of control. detail retry strategy should discuss with OEM necessary and when Rery occured, DTC should be reported for information.

Note: After a running reset the ECU may skip the initiation and/or running of startup tests for hardware that can be assumed to be initiated and/or cannot cause running resets. Normally RAM contents cannot be trusted after a running reset, but if the immobilizer check is OK after the reset the immobilizer does not need to be reinitialized.

97964 v2 **General availability [CS Released]**

Purpose: During running reset it can be expected that CAN signals from other ECUs are not received or have not init/default values for 500ms.
Verification Method: Test
Legacy ID: 383930

The inverter shall not set a DTC, go to failure mode or decrease torque output due to a running reset in another ECU.

See separate requirements for specific signals.

6.3.3.11.5.3 Calibrations

251536 v4 **Car configuration - Vehicle setup [CS Released]**

Purpose: To enable use of different parameter sets for the inverter when placed in different vehicle variants.
Verification Method: Test
Legacy ID: 382697

To enable use of different parameter sets for the inverter when placed in different vehicle variants. The HV Electric Drive system may map the following Car Configuration Parameters. The exact parameters which may be used in projects may different, please clarify with GEELY.
Parameters:

1.Electric Driveline (parameter number CCP#4 ELECTRIC DRIVELINE)

Use case: Recognize the different electric driveline

2.GearBox (parameter number CCP#9 GEARBOX)

Use case: Recognize the different gear box

The above two parameters can be used to calculate how many electric motors in the vehicle, used for compliance of GBT32960.3

Electric Driveline	GearBox	How many electric motors
Combustion Engine FWD and Electric Motor FWD	any value except GHS	1
Combustion Engine FWD and Electric Motor RWD	any value	1
Combustion Engine RWD and Electric Motor FWD	any value except GHS	1
Combustion Engine RWD and Electric Motor RWD	any value	1
Electric Motor AWD	any value	2
Electric Motor FWD	any value	1
Electric Motor RWD	any value	1
Combustion Engine FWD and Electric Motor FWD	any value except GHS	1
any value	GHS	2
Combustion Engine FWD and Electric Motor RWD	8ATEAWD	2

- DIAGNOSIS SYSTEM type((parameter number CCP 46# DIAGNOSIS SYSTEM)

Use case: To distinguish different OBD systems used in different markets, thereby reducing SW variants in EDS

46	01	OBD 2
	02	EOBD
	03	JOBD
	80	COBD-6

HVED shall check DIAGNOSIS SYSTEM type of this vehicle by CCP 46#
- when DIAGNOSIS SYSTEM==0x01, then HVED shall support OBD2 diagnotic protocol.
- when DIAGNOSIS SYSTEM==0x02, then HVED shall support EOBD diagnotic protocol.
- when DIAGNOSIS SYSTEM==0x03, then HVED shall support JOBD diagnotic protocol.

251537 v1 Calibration of Inverter [CS Released]

Purpose: To ensure that the inverter can be calibrated by OEM
Verification Method: Test
Legacy ID:

It shall be possible to request a calibration of the EM to match the inverter, this shall include calibrating to zero degree offset angle and any internal delay.
The request for calibration shall be possible from any of the following modes of operation Stb, TqCtrl, SpdCtrl, UDcCtrl
The inverter shall have a zero degree self learning implemented.

6.3.3.11.5.4 Vehicle Mode Management

Vehicle Mode Management
Common for all EDS

242170 v1 Evaluation of VMM signals [CS Released]

Purpose: Ensure correct behaviour by evaluating CarMode, UsageMode, ElPowerLevel, ElEnergyLevel and to make only a single decission in the same processing period.
Verification Method: Test
Legacy ID:

A subscriber to one or more VMM signals(*CarMode, UsageMode, ElPowerLevel, ElEnergyLevel)* shall have only one evaluation (taking all subscribed VMM signals into account) and take one decision leading to if the functions should be active or not.

The evaluation method shall take into account the priority order that exists for VMM signals, by evaluating them in the following order, from the signal: (***SysSigVehModMngt***)

- CarMode
- UsageMode
- ElPowerLevel and ElEnergyLevel

242175 v1 **Requirements in CarMode DYNO [CS Released]**

Purpose: To be able to get an overview of what functions that are affected by CarMode Dyno.
Verification Method: Test
Legacy ID:

All systems that have requirements in CarMode Dyno that does not exist in any VMM specification must contact the VMM team and manufacturing for approval.

CarMode and UsageMode is received in the signal ***SysSigVehModMngt***.

CarMode	UsageMode	Allowed states of subsystem
DYNO	Abandoned	Power off
	InActive	Power off, Inin, Stb
	Convenience	Power off, Inin, Stb
	Active	Power off, Inin, Stb
	Driving	Inin, Stb, Tqctrl, UDCCtrl, SpdCtrl, PreChrg

The system shall allways be allowed to enter the state Fault if one is currently present. The system shall also be able to enter the calibration state when necessary, unless there is an active fault.

242178 v1 Requirements in CarMode FACTORY [CS Released]

Purpose: To be able to get an overview of what functions that are affected by CarMode Factory
Verification Method: Test
Legacy ID:

All systems that have requirements in CarMode FACTORY that does not exist in any VM specification must contact the VMM team and manufacturing for approval.

CarMode and UsageMode is received in the signal ***SysSigVehModMngt***.

CarMode	UsageMode	Allowed states of subsystem
FACTORY	Abandoned	Power off
	InActive	Power off, Inin, Stb
	Convenience	Power off, Inin, Stb
	Active	Power off, Inin, Stb
	Driving	Driving pauses FACTORY MODE and goes temporarily to NORMAL

The system shall allways be allowed to enter the state Fault if one is currently present. The system shall also be able to enter the calibration state when necessary, unless there is an active fault.

242184 v1 Requirements in CarMode TRANSPORT [CS Released]

Purpose: To be able to get an overview of what functions that are affected by CarMode Transport
Verification Method: Test
Legacy ID:

All systems that have requirements in CarMode Transport that does not exist in any VMM specification must contact the VMM team and manufacturing for approval.

CarMode and UsageMode is received in the signal ***SysSigVehModMngt***.

CarMode	UsageMode	Allowed states of subsystem
TRANSPORT	Abandoned	Power off
	InActive	Power off
	Convenience	Power off
	Active	Power off
	Driving	Driving pauses TRANSPORT MODE and goes temporarily to NORMAL

242185 v1 Requirements in CarMode CRASH [CS Released]

Purpose: To be able to get an overview of what functions that are affected by CarMode Crash
Verification Method: Test
Legacy ID:

The inverter shall do the following actions when ***SysSigVehModMngt***, *CarModSts1* = CRASH.

- Set ***SysSigTqAvl*** = 0
- Ramp down the EM
- When ***SysSigTqAct*** = 0 the system shall do an Active Discharge
- After Active Discharge is complete, the system shall perform a 3-phase short circuit
- Set ***SysSigModSts*** = Power Down mode
- Power Down

IsgTqAvl shall be 0 at all times when *CarModSts1* = CRASH.

242186 v1 **Requirements in CarMode NORMAL [CS Released]**

Purpose: To be able to get an overview of what functions that are affected by CarMode Normal
Verification Method: Test
Legacy ID:

All systems that have requirements in CarMode Dyno that does not exist in any VMM specification must contact the VMM team and manufacturing for approval.

CarMode and UsageMode is received in the signal ***SysSigVehModMngt***.

CarMode	UsageMode	Allowed states of subsystem
NORMAL	Abandoned	Power off
	InActive	Power off, Inin, Stb
	Convenience	Power off, Inin, Stb
	Active	Power off, Inin, Stb
	Driving	Inin, Stb, Tqctrl, UDCCtrl, SpdCtrl, PreChrg

The system shall allways be allowed to enter the state Fault if one is currently present. The system shall also be able to enter the calibration state when necessary, unless there is an active fault.

6.3.3.11.5.4.1 Vehicle Mode for GHS

364314 v1 **Requirements in CarMode NORMAL / GHS2.0 [CS Released]**

Purpose: To be able to get an overview of what functions that are affected by CarMode Normal
Verification Method: Test
Legacy ID: 242186

All systems that have requirements in CarMode Dyno that does not exist in any VMM specification must contact the VMM team and manufacturing for approval.

CarMode and UsageMode is received in the signal ***SysSigVehModMngt***.

CarMode	UsageMode	Allowed states of subsystem
NORMAL	Abandoned	Power off
	InActive	Power off, Inin, Stb, Tqctrl, UDCCtrl, SpdCtrl, PreChrg
	Convenience	Power off, Inin, Stb
	Active	Power off, Inin, Stb, Tqctrl, UDCCtrl, SpdCtrl, PreChrg
	Driving	Inin, Stb, Tqctrl, UDCCtrl, SpdCtrl, PreChrg

The system shall allways be allowed to enter the state Fault if one is currently present. The system shall also be able to enter the calibration state when necessary, unless there is an active fault.

Note
in remote start scenario, UM will be the InActive mode but motor needed to use for start the engine in some cases, so the other motor mode is necessary.
在远程启动场景中，UM将会保持为InActive，但是一些情况下需要使用电机来启动发动机，因此电机的其他模式必须允许使用

364322 v1 **Requirements in CarMode FACTORY / GHS2.0 [CS Released]**

Purpose: To be able to get an overview of what functions that are affected by CarMode Factory
Verification Method: Test
Legacy ID: 242178

All systems that have requirements in CarMode FACTORY that does not exist in any VM specification must contact the VMM team and manufacturing for approval.

CarMode and UsageMode is received in the signal ***SysSigVehModMngt***.

CarMode	UsageMode	Allowed states of subsystem
FACTORY	Abandoned	Power off
	InActive	Power off, Inin, Stb, Tqctrl, UDCCtrl, SpdCtrl, PreChrg
	Convenience	Power off, Inin, Stb
	Active	Power off, Inin, Stb, Tqctrl, UDCCtrl, SpdCtrl, PreChrg
	Driving	Driving pauses FACTORY MODE and goes temporarily to NORMAL

The system shall allways be allowed to enter the state Fault if one is currently present. The system shall also be able to enter the calibration state when necessary, unless there is an active fault.

Note
in remote start scenario, UM will be the InActive mode but motor needed to use for start the engine in some cases, so the other motor mode is necessary.
在远程启动场景中，UM将会保持为InActive，但是一些情况下需要使用电机来启动发动机，因此电机的其他模式必须允许使用

6.3.3.11.5.5 Limitations

250885 v1 — HVDC Current limitation [CS Released]

Purpose: This is a feature to protect the HV battery system and the HV Electric Drive system shall not exceed the voltage limits sent from the battery system
Verification Method: Analysis AND Test
Legacy ID:

For applications where *IdcLimYesNo* equals "yes", the applied actual and estimated available torque shall be limited such that the absolute value of the inverters estimated HVDC current, ***SysSigIDc***, does not exceed the maximum HVDC current limit, *IdcMaxLim*.

The current limitation shall only act when the absolute value of the HVDC current exceeds the maximum DC current limit threshold, (*IdcMaxLim* + *IdcMaxLimOffset*).

250623 v1 — Voltage Limitation, 400V Systems [CS Released]

Purpose: This is a feature to protect the HV battery system and the HV Electric Drive system shall not exceed the voltage limits sent from the battery system
Verification Method: Analysis AND Test
Legacy ID:

The HVElectricDrive shall limit the torque to prevent the voltage on the HV-bus from exceeding the values on ***SysSigHvBattUDynMinLim*** and ***SysSigHvBattUDynMaxLim***.
If for any reason these signals are not available for use, there shall be internal fixed limit voltage levels to use as replacement values until the signals are available again.

Maximum voltage limit = 400V (this value shall be possible to calibrate)
Minimum voltage limit = 210V (this value shall be possible to calibrate)

These HVDC limits shall not be violated, there is although an exception during crank of ICE where the lower voltage level can be down to TBD V for XX s.

Note: The System Voltage shall be confirmed on a per project basis

250627 v1 — Voltage Limitation, 800V Systems [CS Released]

Purpose: This is a feature to protect the HV battery system and the HV Electric Drive system shall not exceed the voltage limits sent from the battery system
Verification Method: Analysis AND Test
Legacy ID:

The HVElectricDrive shall limit the torque to prevent the voltage on the HV-bus from exceeding the values on ***SysSigHvBattUDynMinLim*** and ***SysSigHvBattUDynMaxLim***.

If for any reason these signals are not available for use, there shall be internal fixed limit voltage levels to use as replacement values until the signals are available again.

Maximum voltage limit = 800V (this value shall be possible to calibrate)
Minimum voltage limit = 420V (this value shall be possible to calibrate)

These HVDC limits shall not be violated, there is although an exception during crank of ICE where the lower voltage level can be down to TBD V for XX s.

Note: The System Voltage shall be confirmed on a per project basis

251602 v2 **Inverter self protection at high speeds [CS Released]**

Purpose: Back EMF voltage is proportional to motor speed. Back EMF that is potentially harmful for the inverter HW shall not occur within normal operational speed
Verification Method: Test
Legacy ID:

The inverter shall protect its HW (IGBT) from high back EMF voltage in situations where this can appear.

This shall be done by applying 3--phase short circuit.

Note:

In order to high back EMF, the system should be done by applying freewheeling in Low speed and applying 3--phase short circuit in high speed.Speed threshold value depend on High Voltage Battery Voltage range and detail protect scheme should be discussed with OEM;

the system implement 3-phase short circuit strategy should consider IGBT fault status.eg.one or two high-brigde work abnormal. the system should applying 3--phase short circuit by low-brigdes. the control stategy should consider hardware secondary injury;

251603 v1 **Protection from unintended HVDC generation [CS Released]**

Purpose: Back EMF voltage is proportional to motor speed. Back EMF that is potentially harmful for the inverter HW shall not occur within normal operational speed
Verification Method: Test
Legacy ID:

The inverter shall prevent unintend rectifying to the HVDC.

Info: This may happen due to high back EMF voltage etc.

6.3.3.11.5.6 Diagnostics

General diagnostics of the Electric Drive System

97951 v2 **System specific diagnostic services - calibration [CS Released]**

Purpose: Diagnostic services is the only way for the workshop tools to initiate something in the inverter

Verification Method: Test
Legacy ID: 383021

The ECU shall have one diagnostic service for each calibration job that either needs:

- to be started manually (that cannot be performed while normal functionality is running)

- the inverter to be in a specific state that the inverter might not be in during a driving cycle

97945 v2 Detection of need of calibration [CS Released]

Purpose: The only way to signal the workshop tools that an action needs to be taken is through DTCs
Verification Method: Test
Legacy ID: 383016

The inverter shall set a DTC when any calibration is missing or not preformed.

The inverter shall set a DTC when it detects that it needs to be re-calibrated.

There shall be one DTC for each calibration job that must be started manually.

190635 v1 Shut off path test diagnostic [CS Released]

Purpose: Inverter diagnostic
Verification Method: Test
Legacy ID:

The inverter shall implement the shut-off path test during start up or power down. If test fails, transmit to Fault mode.

190638 v2 Resolver sensor diagnositc [CS Released]

Purpose: Inverter diagnostic
Verification Method: Test
Legacy ID:

The inverter shall diagnostic the resolver sensor, including:
1. Loss of tracking
2. Degradation of signal
3. Open line of sine
4. Open line of cosine

5. Open line of excitation
6. Short to battery of sine/cosine/excitation
7. Short to ground of sine/cosine/excitation
8. Raw signal out of range high of sine/cosine/excitation
9. Raw signal out of range low of sine/cosine/excitation
10. Sin*sin + cos*cos check out of range
11. Sine, cosine amplitude difference high
12. Resolver angle invalid
13. Power supply of excitation chip out of range

If HVED use Inductive Position Sensor:
The inverter shall diagnostic the Inductive Position Sensor, including:
1. Degradation of signal
2. Open line of sine
3. Open line of cosine
4. Short to battery of sine/cosine
5. Short to ground of sine/cosine
6. Sin*sin + cos*cos check out of range
7. Sine, cosine amplitude difference high
8. Resolver angle invalid

190640 v2 HVDC voltage sensor diagnostic [CS Released]

Purpose: Inverter diagnostic
Verification Method: Test
Legacy ID:

The inverter shall diagnostic the HVDC voltage sensors, including:
1. Software over voltage
2. Hardware over voltage
3. Software under voltage
4. Power supply of HVDC voltage sensor out of range
5. HVDC voltage sensor plausibility check

- Test run criteria (TRC)

1. After the corresponding PNC of diagnosed frame trigger(command to wake up) in 2 second, begin to detect whether there is any lost of communication.
2. KL30 have normal voltage from 9-16V
3. ***SysHvRlySts*** =Closed

- Detection Criteria

Software under or over voltage threshold should consider OEM High voltage strategy and confirmed with OEM

190641 v2 Phase currents sensor diagnostic [CS Released]

Purpose: Inverter diagnostic
Verification Method: Test
Legacy ID:

The inverter shall diagnostic the phase current sensors, including:
1. Phase current offset out of range A/B/C phase
2. Current sensor short to battery A/B/C phase
3. Current sensor short to ground A/B/C phase
4. Raw signal out of range high A/B/C phase
5. Raw signal out of range low A/B/C phase
6. Physical signal over current A/B/C phase
7. Sum of three phase currents rationality
8. Power supply of phase current sensors out of range
9. Missing phase current (open circuit fault)
10. Gradient of phase current high A/B/C phase

190655 v2 KL30 power supply diagnostic [CS Released]

Purpose: Inverter diagnostic
Verification Method: Test
Legacy ID:

The inverter shall diagnostic the power supply of KL30, including:
1. Battery voltage low
2. Battery voltage high
3. Battery voltage rationality check

If KL30<7.8V (TBD) & high voltage contactor open & HVDC voltage >60V, HVED subsystem should trigger Active Short Circuit to secure the high voltage safety. At the same time, a certain drag torque will implement on the wheels.

190656 v1 CPSR relay diagnostic [CS Released]

Purpose: Inverter diagnostic
Verification Method: Test
Legacy ID:

The inverter shall diagnostic the CPSR status, including:
1. CPSR voltage out of range high

190657 v1 Bus off diagnostic [CS Released]

Purpose: Inverter diagnostic
Verification Method: Test
Legacy ID:

The inverter shall diagnostic bus off failure. If a bus off event occures a DTC shall be set.

190688 v1 **Watchdog diagnostic [CS Released]**

Purpose: Inverter diagnostic
Verification Method: Test
Legacy ID:

The system shall monitor the watchdog status, including the internal watchdog and the external watchdog.

251403 v1 **Drive cycle define [CS Released]**

Purpose: To define drive cycle

Verification Method: Inspection Test
Legacy ID:

The system shall listen to port: ***SysSigDrvCyc*** which indicates whether propulsion system activation has complete or not. Used for diagnostic drive cycle counter.

251405 v1 **Alive timeout diagnostic [CS Released]**

Purpose: Inverter diagnostic
Verification Method: Test
Legacy ID:

The system shall diagnose the following received signal update bit alive timeout status, including:

SysSigCoolTSigForDtElec

SysSigCoolFlowForDtElec

EngCoolInhbFlow

SysSigEngSt1WdSts

SysSigHvBattUdc

SysSigHvBattuDynMinLim

SysSigHvBattuDynMaxLim

SysSigActvDampgModReq

SysSigEngSpdOfActvDmpg

SysSigModReq

SysSigPwrLimMin

SysSigPwrLimMax

SysSigSpdReq

SysSigTqReq

SysSigFbLimnIndcn

SysSigVehModMngt

SysSigDrvCyc

660145 v4 **Lost of communication diagnostic [CS Released]**

Purpose: From vehicle level to unify all the component diagnostic for lost communication. The purpose is help to judge the who is the ECU has bad CAN connect or CAN chip fault happen in vehicle for after market etc.When one of the ECU has CAN problem , all other signal receive ECUs will record lost communication DTC to point out the culprit.

Verification Method: Test

Legacy ID:

This is general requirement for lost of communication in CAN or CANFD network.

The CAN or CANFD frame defined to be diagnosed for lost of communication shall obey below requirement to define DTC criteria.

- Test run criteria (TRC)

1. After the corresponding PNC of diagnosed frame trigger(command to wake up) in 2 second, begin to detect whether there is any lost of communication.
2. KL30 have normal voltage from 9-16V
3. No bus off failure refer to Req ID 190657 Bus off diagnostic

- Detection Criteria

Lost of frame or timeout for 250ms (depend on Supplier design) or update ub always zero for five frame consecutively.

Note:If anyone ECU have special communication, EDS system should support to relax threshold or debounce time to improve software robustness after discussing with OEM

- Fault effect

If **CrashStsSafeSts** frame lost have be checked, It shouldn't triggle active discharge function in order to avoid High Voltage Energy system pre-charging failed.

If **SysSigTqReq** frame lost have be checked,torque output shall derating to Zero;

If **SysSigModReq** frame lost have checked, the post-processing should be discuss with OEM;

1043692 v1 **DTC Black Box [CS Released]**

Purpose: DTC Black Box

Verification Method: Test

Legacy ID:

为方便调查 开发过程中 或 售后 的问题原因，

当出现电驱故障时，应可以记录 内部关键数据信息 到存储设备中，此数据成为 “黑匣子”数据。

根据不同的电驱故障，可记录不同的内部关键数据信息。

根据硬件存储能力，配置存储时长。

1043693 v1 Phase currents HW Over Current diagnostic [CS Released]

Purpose: Phase currents HW Over Current diagnostic
Verification Method: Test
Legacy ID:

DTC Code	tbd
DTC描述	硬件过流
DTC使能条件 Test run criteria (TRC)	❖ 以下条件全部满足： 1.ECU Supply > 9V 2.Mode not equal to PwrDwn
DTC报出逻辑	开管（Tq/Spd/Udc/ASC模式），检测到任意一相硬件过流引脚电平由高变为低电平。
DTC诊断进入故障保护	记录DTC，LOS2（ASC/FW）。
DTC诊断退出故障保护	故障不锁存，可恢复。 同时满足以下条件，退出故障保护： ● 车速<3kph(滞环1km/h)； ● 电机转速<50rpm(滞环10rpm)； ● 恢复次数<=3次； ● P档； ● 电机模式不在Torque 或 Boost； ● 关断路径请求为FW； ● 无休眠请求； ● 硬件故障触发保护时间>200ms； ● 硬件过流故障恢复正常。
DTC清除逻辑	故障恢复后，第40（TBD）个驾驶循环
DTC存储黑匣子	记录DTC时，应至少记录前20ms（TBD）内部关键数据。 应包含：电流传感器采样电压，相电流，三相开管状态等关键信息。

DTC Code	tbd
DTC描述	硬件过流
DTC使能条件 Test run criteria (TRC)	❖ 以下条件全部满足： 1.ECU Supply > 9V 2.Mode not equal to PwrDwn
DTC报出逻辑	不开管（Stb模式），检测到任意一相硬件过流引脚电平由高变为低电平。
DTC诊断进入故障保护	仅记录DTC，无故障保护。

DTC诊断退出故障保护	仅记录DTC，无故障保护。
DTC清除逻辑	故障恢复后，第40（TBD）个驾驶循环
DTC存储黑匣子	记录DTC时，应至少记录前20ms（TBD）内部关键数据。 应包含：电流传感器采样电压，相电流，三相开管状态等关键信息。

6.3.3.11.6 Signal Map

251550 v1 Reference to Signals [CS Released]

Purpose: To show which interfaces that contains the signals
Verification Method: Documentation
Legacy ID:

Signal Map is found within

LC: EM[Number]GeneralInterface

EM1 (Main Front EM system)
EM2 (Auxillary Front EM system)
EM3 (Main Rear EM system)

Depending on System Voltage Signals are also mapped in

400V
LC: EM[Number]400VInterface

800V
LC: EM[Number]800VInterface

713484 v2 Signal Map [CS Released]

Purpose: To map signal
Verification Method: Test
Legacy ID:

Signal Map

System Signal	Actual Signal	Description
	HVEDPrpsnDataEvtA	
	HVEDPrpsnDataPrdA	
	HVEDPrpsnDataEvtB	
	HVEDPrpsnDataPrdB	
	HVEDPrpsnDataEvtC	
	HVEDPrpsnDataPrdC	
	HVEDPrpsnDataEvtD	
	HVEDPrpsnDataPrdD	
	HVEDPrpsnDataEvtE	
	HVEDPrpsnDataPrdE	

6.3.3.11.7 DTC requirements for GHS

Only for GHS projects
Note：PHEV/HEV/EREV

3DHT100
3DHT160
DHT120
DHT160
DHT160B
DHT260

SO：lilei11

6.3.3.11.7.1 DTC For Process

843542 v1 **DFC_EV_MOT1_OVER_SPEED/GHS [Released]**

Purpose: Define the DTCs of Function Motor Over Speed
Verification Method: Analysis AND Test
Legacy ID:

DTC List:

Function ID (关联故障ID)	DTC (来源与GNDP编码)	Description (对应编码的描述)
M-22	0A4492	Drive Motor "A" Position Sensor Circuit Overspeed Component Failures Performance or incorrect operation
Warning Indicator (是否有故障指示)	NotificationId (故障灯&故障文言警示ID)	**Fault indicator signal (触发CSD/HMI故障灯/文言指示的信号)**
Yes	tell-tale-19	PrpsnErrIndcnReq=2
Test Run Criteria (故障使能条件)	1. 车辆模式转换5±2s内应该限制故障检测到。 2. 在车辆使用模式转换的5±2s内应该限制故障检测到。 3. 车辆模式处于工厂、运输、碰撞模式时应禁止故障监控。 4. 当电源可用性受到限制时，所有DTC测试应被禁止。 5. 车辆电压处于9 ~16V正常电压阈值内才可检测到。 6. 车辆处于驾驶使用模式。	
Detection Criteria (故障上报条件)	1.通过电涡流传感器计算发电机P1的转速超速（监测其转数大于12500rpm）。 Note：具体上报条件见SSDS	
Failed Criteria (故障失败条件)	电涡流传感器计算发电机P1的转速超速（监测其转数Nspd>=12500rpm）。	
Passed Criteria (故障成功条件)	电涡流传感器计算发电机P1的转速超速（监测其转数Nspd<10000rpm）。	

说明:
① Funtion ID: 故障发生时影响的整车功能。
② DTC: 故障码。

DTC High Byte								DTC Middle Byte								DTC Low Byte							
0x9				0x2				0x3				0x4				0x1				0x1			
1	0	0	1	0	0	1	0	0	0	1	1	0	1	0	0	0	0	0	1	0	0	0	1
B		1		2				3				4				1				1			
前两个字节由GMRDB数据库定义，GMRDB数据包含SAE标准实施的排放相关故障信息，及Geely自定义诊断故障信息。其主要包含**故障组件、功能**两大类。**具体故障类型由DTC Low Byte定义实施**。																故障类别，此故障类型描述**大体**故障方向。				故障子类别，此故障类型**细化**了故障方向。			

③ Description: 故障码对应的故障描述，每一个码对应一个故障描述，由电子电器诊断开发团队定义。
④ Test Run Criteria: 故障使能条件，当满足该标准时执行故障检测：
下述为基础技术团队强制要求的故障使用条件，如有偏差需联系电子电器团队认可:
1、车辆模式转换5±2s内应该限制故障检测。
2、在车辆使用模式转换的5±2s内应该限制故障检测。
3、当电源可用性受到限制时，所有DTC测试应被禁止。
4、车辆使用模式处于"运输"、"工厂"、"碰撞"时应禁止监控。
5、当ECU供电电压低于内部组件可正常执行的最小限值时。

包含但不仅限于上述条件限制，设计方应考虑一些独立场景限制，以防正常情况下的DTC上报。
eg.
车辆配置信息，如未装配相关执行器，则不应该执行该部件故障监控。
车辆模式信息，如便利模式下相关功能不可用，则不应该执行该功能故障监控。

⑤ Detection Criteria：故障码上报条件，当故障维持一段时间后（经过故障消抖处理，以防误报），如相关故障参数无法在需求中明确，应提醒实施方结合实际条件制订合理的标准，保障故障上报精确。

⑥ Failed Criteria：故障失败条件，当处于何种条件下应判定成故障场景。

⑦ Passed Criteria：故障恢复条件，当处于何种条件下应判定成正常场景。

⑧ Warning Indicator：是否设计故障指示。(故障灯&故障文言)

⑨ Fault indicator signal：故障指示触发信号，如设计故障灯&故障文言的故障，应将触发HMI/CSD的信号与值定义清晰。

⑩ NotificationId：故障指示ID，对应CSD/HMI灯或文言的标准ID。

843546 v1 **DFC_EV_MOT1_CRIT_OT/GHS [Released]**

Purpose: Define the DTCs of Function Motor Over Temperature
Verification Method: Analysis AND Test
Legacy ID:

DTC List:

Function ID (关联故障ID)	**DTC (来源与GNDP编码)**	**Description (对应编码的描述)**
M-14	0A2F98	Drive Motor "A" Over Temperature Component Failures Component or system over temperature
Warning Indicator (是否有故障指示)	**NotificationId (故障灯&故障文言警示ID)**	**Fault indicator signal (触发CSD/HMI故障灯/文言指示的信号)**
No	tell-tale-24	HybErrIndcnReq-TelltlSysHybFailr=1
Test Run Criteria (故障使能条件)	1. 车辆模式转换5±2s内应该限制故障检测到。 2. 在车辆使用模式转换的5±2s内应该限制故障检测到。 3. 车辆模式处于工厂、运输、碰撞模式时应禁止故障监控。 4. 当电源可用性受到限制时，所有DTC测试应被禁止。 5. 车辆电压处于9 ~16V正常电压阈值内才可检测到。 6. 车辆处于驾驶使用模式。	

Detection Criteria (故障上报条件)	1. 监控到发电机P1温度二级过温警告（监测其温度值高于120℃）。 Note：具体上报条件见SSDS
Failed Criteria (故障失败条件)	发电机P1温度二级过温警告（监测其温度值Thi>120℃）。
Passed Criteria (故障成功条件)	发电机P1温度二级过温警告（监测其温度值Thi<110）。

说明:
① Funtion ID: 故障发生时影响的整车功能。
② DTC: 故障码。

DTC High Byte				DTC Middle Byte				DTC Low Byte							
0x9				0x2				0x3				0x4				0x1				0x1			
1	0	0	1	0	0	1	0	0	0	1	1	0	1	0	0	0	0	0	1	0	0	0	1
B		1		2				3				4				1				1			
前两个字节由GMRDB数据库定义，GMRDB数据包含SAE标准实施的排放相关故障信息，及Geely自定义诊断故障信息。其主要包含**故障组件、功能**两大类。**具体故障类型由DTC Low Byte定义实施**。																故障类别，此故障类型描述**大体**故障方向。				故障子类别，此故障类型**细化**了故障方向。			

③ Description: 故障码对应的故障描述，每一个码对应一个故障描述，由电子电器诊断开发团队定义。
④ Test Run Criteria: 故障使能条件，当满足该标准时执行故障检测:
下述为基础技术团队强制要求的故障使用条件，如有偏差需联系电子电器团队认可:
1、车辆模式转换5±2s内应该限制故障检测。
2、在车辆使用模式转换的5±2s内应该限制故障检测。
3、当电源可用性受到限制时，所有DTC测试应被禁止。
4、车辆使用模式处于"运输"、"工厂"、"碰撞"时应禁止监控。
5、当ECU供电电压低于内部组件可正常执行的最小限值时。

包含但不仅限于上述条件限制，设计方应考虑一些独立场景限制，以防正常情况下的DTC上报。
eg.
车辆配置信息，如未装配相关执行器，则不应该执行该部件故障监控。
车辆模式信息，如便利模式下相关功能不可用，则不应该执行该功能故障监控。

⑤ Detection Criteria：故障码上报条件，当故障维持一段时间后（经过故障消抖处理，以防误报），如相关故障参数无法在需求中明确，应提醒实施方结合实际条件制订合理的标准，保障故障上报精确。

⑥ Failed Criteria：故障失败条件，当处于何种条件下应判定成故障场景。

⑦ Passed Criteria：故障恢复条件，当处于何种条件下应判定成正常场景。

⑧ Warning Indicator：是否设计故障指示。(故障灯&故障文言)

⑨ Fault indicator signal：故障指示触发信号，如设计故障灯&故障文言的故障，应将触发HMI/CSD的信号与值定义清晰。

⑩ NotificationId：故障指示ID，对应CSD/HMI灯或文言的标准ID。

843568 v1 **DFC_EV_MOT1_INVERTER_CRIT_OT/GHS [Released]**

Purpose: Define the DTCs of Function Inverter Over Temperature
Verification Method: Analysis AND Test
Legacy ID:

DTC List:

Function ID (关联故障ID)	DTC (来源与GNDP编码)	Description (对应编码的描述)

M-14	0A3C98	Drive Motor "A" Inverter Over Temperature Component Failures Component or system over temperature
Warning Indicator (是否有故障指示)	NotificationId **(故障灯&故障文言警示ID)**	**Fault indicator signal (触发CSD/HMI故障灯/文言指示的信号)**
No	tell-tale-24	HybErrIndcnReq-TelltlSysHybFailr=1
Test Run Criteria (故障使能条件)	1. 车辆模式转换5±2s内应该限制故障检测到。 2. 在车辆使用模式转换的5±2s内应该限制故障检测到。 3. 车辆模式处于工厂、运输、碰撞模式时应禁止故障监控。 4. 当电源可用性受到限制时，所有DTC测试应被禁止。 5. 车辆电压处于9 ~16V正常电压阈值内才可检测到。 6. 车辆处于驾驶使用模式。	
Detection Criteria (故障上报条件)	1. 监控到发电机P1 IGBT逆变器温度二级过温警告（监测其温度大于120℃）。 Note：具体上报条件见SSDS	
Failed Criteria (故障失败条件)	发电机P1 IGBT逆变器温度二级过温警告（监测其温度TivtrOT>=120℃）。	
Passed Criteria (故障成功条件)	发电机P1 IGBT逆变器温度二级过温警告（监测其温度TivtrOT<110℃？）。	

说明:
① Funtion ID: 故障发生时影响的整车功能。
② DTC: 故障码。

DTC High Byte				**DTC Middle Byte**				**DTC Low Byte**							
0x9				0x2				0x3				0x4				0x1				0x1			
1	0	0	1	0	0	1	0	0	0	1	1	0	1	0	0	0	0	0	1	0	0	0	1
B		1		2				3				4				1				1			
前两个字节由GMRDB数据库定义，GMRDB数据包含SAE标准实施的排放相关故障信息，及Geely自定义诊断故障信息。其主要包含**故障组件、功能**两大类。**具体故障类型由DTC Low Byte定义实施**。																故障类别，此故障类型描述**大体**故障方向。				故障子类别，此故障类型**细化**了故障方向。			

③ Description: 故障码对应的故障描述，每一个码对应一个故障描述，由电子电器诊断开发团队定义。
④ Test Run Criteria: 故障使能条件，当满足该标准时执行故障检测:
下述为基础技术团队强制要求的故障使用条件，如有偏差需联系电子电器团队认可:
1、车辆模式转换5±2s内应该限制故障检测。
2、在车辆使用模式转换的5±2s内应该限制故障检测。
3、当电源可用性受到限制时，所有DTC测试应被禁止。
4、车辆使用模式处于"运输"、"工厂"、"碰撞"时应禁止监控。
5、当ECU供电电压低于内部组件可正常执行的最小限值时。

包含但不仅限于上述条件限制，设计方应考虑一些独立场景限制，以防正常情况下的DTC上报。
eg.
车辆配置信息，如未装配相关执行器，则不应该执行该部件故障监控。
车辆模式信息，如便利模式下相关功能不可用，则不应该执行该功能故障监控。

⑤ Detection Criteria：故障码上报条件，当故障维持一段时间后（经过故障消抖处理，以防误报），如相关故障参数无法在需求中明确，应提醒实施方结合实际条件制订合理的标准，保障故障上报精确。

⑥ Failed Criteria：故障失败条件，当处于何种条件下应判定成故障场景。

⑦ Passed Criteria：故障恢复条件，当处于何种条件下应判定成正常场景。

⑧ Warning Indicator：是否设计故障指示。(故障灯&故障文言)

⑨ Fault indicator signal：故障指示触发信号，如设计故障灯&故障文言的故障，应将触发HMI/CSD的信号与值定义清晰。

⑩ NotificationId：故障指示ID，对应CSD/HMI灯或文言的标准ID。

6.3.3.12 Logical component: ThermalManagementRequirements

ID	Name	Version	Status	Last Changed
x04000000231892D3	ThermalManagementRequirements	(8a4a3a3)	CS Released	2026-01-27

The thermal system in the HVElectric drive is interfacing a thermal system in the vehicle which provides a coolant media to the Electric Drive system. The control of the coolant (flow and temperature) media lies in the external system.
The basic idea is that the HVElectricDrive request the coolant performance it needs to keep system operational with full performance. The request shall then cover that the near time future torque request are covered in the margins. To create a well functioning thermal system on the outside the coolant performance request need to be well synchronized so the temperature regulation can be optimized for performance.

6.3.3.12.1 Included Virtual Function Clusters(VFCs)

6.3.3.12.2 ThermalManagementRequirements: Receiver Ports

6.3.3.12.3 ThermalManagementRequirements: Send Ports

6.3.3.12.4 Requirements

6.3.3.12.4.1 Measurements

242041 v2 **Temperature of the EM [CS Released]**

Purpose: To signal the EM temperature
Verification Method: Test
Legacy ID:

The most relevant (describing performance and life time aspects) temperature of the EM shall be signalled on To signal the EM ***SysSigMotT***. Several measurement points might be needed to find the most relevant temperature.
The temperature shall reflect the therrmal status of the EM and in case derating of performance due to temperature it shall be visible in this temperature.

242047 v1 **Temperature of the Inverter [CS Released]**

Purpose: To signal the component temperature
Verification Method: Test
Legacy ID:

The most relevant (describing performance and life time aspects) temperature of the inverter shall be signalled on ***SysSigInvrT***. Several measurement points might be needed to find the most relevant temperature.
The temperature shall reflect the thermal status of the inverter and in case derating of performance due to temperature it shall be visible in this temperature.

242050 v2 **Temperature of the Coolant [CS Released]**

Purpose: To signal the temperature of the coolant flow through the component
Verification Method: Test
Legacy ID:

The coolant temperature at the inlet to the component shall be signalled on ***SysSigCoolt***
The measured and reported value shall have an accuracy better than +-InvTempError in the range InvTempRange (calibratable), the accuracy is compared to the actual coolant temperature at the inlet to the component.

242079 v2 **Coolant Flow Actual [CS Released]**

Purpose: To be able to predict a thermal limitation
Verification Method: Test
Legacy ID:

If the Electric drive system coolant circuit is in series,it shall receive the predicted coolant flow on port ***SysSigCooltFlowElecDt*** .

If the Electric drive system coolant circuit is in parallel,it shall receive the predicted coolant flow on port ***SysSigCooltFlowForFrntDtElec*** for front electric drive system and ***SysSigCooltFlowForReDtElec*** for rear electric drive system.

6.3.3.12.4.2 Controls

242076 v5 **Cooling Requests [CS Released]**

Purpose: To signal the requested coolant flow for the components
Verification Method: Test
Legacy ID:

The requested cooling for the component shall be reported via port ***SysSigCoolgReq***. The cooling request shall state No-, Low-, Mid-, High-coolant flow need of the component.

No increased flow rate requested , ***SysSigCoolgReq*** = 0
Below situation component shall request no increased flow rate, ***SysSigCoolgReq*** = 0:

- If no high voltage and motor speed ＜200rpm(TBD)；
-
- For EDS combines Boost : When the DC Charger have been connected without Active heating request and Boost charging request, ***SysSigDCChrgnHndlSts*** ="1/3" & ***SysSigHeatModEnad*** ≠ "Heatg" & ***SysSigBoostMdReq*** ≠ “0x01 BoostMdReq_BoosterOrHvBattDirectCharging” or “0x02 BoostMdReq_BoosterCharging”.
- For EDS combines Boost : If the DC Charger have been connected without Active heating request and Boost discharging request , ***SysSigDCChrgnHndlSts*** ="4/5" & ***SysSigHeatModEnad*** ≠"0x1 Enable"& ***SysSigBoostMdReq*** ≠ “0x03 BoostMdReq_Discharging”
- For EDS doesn't combine Boost : If the DC Charger have been connected without Active heating request ,

SysSigDCChrgnHndlSts ="1/3/4/5" & ***SysSigHeatModEnad*** ≠"0x1 Enable"

- When the AC Charger have been connected without Active heating request , ***SysSigOnBdChrgrHndlSts1*** ="1/2/3/4/5/6/7/9" & ***SysSigHeatModEnad*** ≠ "Heatg" , the component shall request no increased flow rate.

Low increase rate requested, ***SysSigCoolgReq*** = 1, allowed for ***SysSigCooltFlowMinReq*** >=TBD1 L/min
High increase rate requested, ***SysSigCoolgReq*** = 2, allowed for ***SysSigCooltFlowMinReq*** >=TBD2 L/min
Immediate full flow requested, ***SysSigCoolgReq*** = 3, allowed for ***SysSigCooltFlowMinReq*** >=TBD3 L/min

TBD3>TBD2>TBD1
Hysteresis shall be considered.

The component shall be able to request cooling in all modes of operation, above sleep.

242070 v1 **Request cooling Electric Motor [CS Released]**

Purpose: Manage coolant flow in the high temperature cooling circuit.
Verification Method: Test
Legacy ID:

The requested cooling for the electrical motor shall be reported via port ***SysSigMotCooltFlowReq***.
This is requested as cooling flow needed to remain in a good thermal state.
Flow is requested as l/min.

Note: This requirement is only for platforms where the EM & Inverter are on different cooling circuits.

242073 v1 **Request lower Coolant temperature Electric Motor [CS Released]**

Purpose: Manage coolant flow in the high temperature cooling circuit.
Verification Method: Test
Legacy ID:

If a high thermal event occurs the system shall be able to flag an extra need for cooling by the signal ***SysSigCooltTFlg***

When active the need for extra cooling is requested to the thermal system.

Signal coding:
0 = EngTLoFlg_ActvLoTForEng
1 = EngTLoFlg_DeactvLoTForEng

Note: This requirement is only for platforms where the EM & Inverter are on different cooling circuits.

242075 v1 **Inhibit Engine Coolant circuit flow [CS Released]**

Purpose: Inhibit Engine Coolant circuit flow
Verification Method: Test
Legacy ID:

Inhibitation of non-crucial coolant flow requests. ***SysSigEngCooltInhbFlow*** sent out to inform inverter that the engine is in heat up phase and that all additional coolant flow will postphone the time when the engine has reached normal operating temperature.

This requirement applies where where the Electric Drive System is placed in the vicinity of the combustion engine.

312946 v1 External Cooling Requests [CS Released]

Purpose: To receive signals that requests coolant flow for other components integrated in the same closed loop cooling system as HVED component.
Verification Method: Test
Legacy ID:

The requested cooling from the Transmission Control component shall be reported via port ***ExtSysSigCoolgReq***.
The cooling request shall state No-, Low-, Mid-, High-coolant flow need of the component.

No increased flow rate requested,***ExtSysSigCoolgReq*** = 0
Low increase rate requested, ***ExtSysSigCoolgReq*** = 1, allowed for ***ExtSysSigCoolgReq*** >=TBD1 C Deg. (TBC)
High increase rate requested, ***ExtSysSigCoolgReq*** = 2, allowed for ***ExtSysSigCoolgReq***>=TBD2 C Deg. (TBC)
Immidiate full flow requested, ***ExtSysSigCoolgReq*** = 3, allowed for ***ExtSysSigCoolgReq***>=TBD3 C Deg. (TBC)

TBD3>TBD2>TBD1
Hystersis shall be implemented with an 5 C Degrees limit

The component shall be able to request cooling in all modes of operation, above sleep

321432 v1 Oil Cooling Request [CS Released]

Purpose: Provide the rear motor oil cooling request to transmission control module
Verification Method: Analysis AND Test
Legacy ID:

Inverter should calculate the motor cooling oil flow and send it to the transmission control moudle by ***SysSigOilCoolgReq.***

1040558 v1 Required Zero or Low coolant flow [CS Released]

Purpose: Set Zero/Lower coolant flow request to reduce the power consumption in no cooling needed but with high voltage condition.
Verification Method: Analysis AND Test
Legacy ID:

Signal	Description
SysSigDrvrSeatSts	Describes if an occupant is detected in the driver seat. Detect if the seat switch sensor closed or open.
SysSigVehModMngtGlbSafe1::UsgModSts1	usage mode state

为了减少功耗，当存在高压电源时，HVED 系统应考虑驾驶员状态和功耗模式，以减少最小冷却液流量请求。

❖ **场景 1：驾驶员不在座椅上**

如果满足以下所有条件，所需的最小冷却液流量 ***SysSigCoolt FlowMinReq*** 应为 0 L/min：

- ***SysSigVehModMngtGlbSafe1::UsgModSts1*** = "Inactive" or "Abandoned";
- ***SysSigDrvrSeatSts*** = "OccptNotPrsnt"；
- 非Torque模式
- 未在升压充电状态（如果组件具有此功能）；
- 无主动加热请求（如果组件具有此功能）；

❖ **场景 2：驾驶员在座椅上且 HVED 冷却**

如果满足以下所有条件，所需的最小冷却液流量 ***SysSigCoolt FlowMinReq*** 应为 2 L/min：

- 存在高压电；
- ***SysSigDrvrSeatSts*** = "OccptPrsnt" 或 "Undefd1" 或 "Undefd2" 或 与***SysSigDrvrSeatSts*** 通信丢失；
- 非Torque模式
- 未在升压充电状态（如果组件具有此功能）；
- 无主动加热请求（如果组件具有此功能）；
- 无 AC/DC 充电器连接；

❖ **场景 3：车辆在转毂上**

满足以下所有条件，所需的最小冷却液流量***SysSigCoolt FlowMinReq*** 应为 2 L/min（待定）：

- ***SysSigVehModMngtGlbSafe1::Carmode*** = CarModDyno；
- 未在升压充电状态（如果组件具有此功能）；
- 无主动加热请求（如果组件具有此功能）；
- 非Torque模式
  或者
  Torque模式，但逆变器/电机/水温等温度较低，未触发242052/706412流量提升。

To reduce the power consumption, HVED system shall consider state of Driver and Usage mode to reduce minimum coolant flow request, when HV power exist.

Scene 1: No driver on seat

The minimum coolant flow required ***SysSigCoolt FlowMinReq*** shall be **0 L/min**, if all below condition fulfilled:

- ***SysSigVehModMngtGlbSafe1::UsgModSts1*** = "Inactive" or "Abandoned";
- ***SysSigDrvrSeatSts*** = "OccptNotPrsnt";
- not Torque mode
- No Boost Charging(If component has this function);
- No Active Heating request(If component has this function);

Scene 2: Driver on seat And HVED is Cool

The minimum coolant flow required ***SysSigCoolt FlowMinReq*** shall be **2 L/min**(TBD), if all below condition fulfilled:

- High voltage exist;
- ***SysSigDrvrSeatSts*** = "OccptPrsnt"or"Undefd1"or"Undefd2" or lose communication with ***SysSigDrvrSeatSts***
- not Torque mode
- Not in Boost Charging State(If component has this function);
- Not in Active Heating request(If component has this function);
- No AC/DC Charger connect (Detail to see Req ID 242052 706412);

Scene 3: Vehicle on dyno

The minimum coolant flow required ***SysSigCoolt FlowMinReq*** shall be **2 L/min**(TBD), if all below condition fulfilled:

- ***SysSigVehModMngtGlbSafe1::Carmode1*** = CarModDyno;
- Not in Boost Charging State(If component has this function);
- Not in Active Heating request(If component has this function);
- Not Torque mode
  or
  in Torque mode,but the inverter/motor/water temperature is low, and flow rate increase 242052/706412 has not been triggered.

6.3.3.12.4.3 Limitations

250808 v1	EM temperature limitation [CS Released]

Purpose: To prevent the EM overheating but still have maximum available torque as far as possible. To ensure that maximum torque is available as far as possible, when the temperature is rising, the temperature rise shall not be stopped until the EM temp has reached the max EM temp (ElecMotTMax-ElecMotTCrkMarigin), given that the enough torque is requested.

Verification Method: Test

Legacy ID:

If the temperature of the electric machine exceeds *EmTempLimThreshold* the available and applied torque shall be limited such that the electric machine temperature does not exceed *EmTempLimMax*.
The EM temperature limitation shall be designed to halt the increase of temperature, i.e. the available torque shall stabilize at a non-zero level with the temperature stabilizing somewhere in the EM temperature limitation range (between *EmTempLimThreshold* and *EmTempLimMax*), depending on the actual cooling condition.

The limitation shall ensure that the full temperature range of the EM is used. The *EM temperature* is not allowed to overshoot the max value.

The *EM temperature* is available in port ***SysSigMotT***.

Note:

EmTempLimThreshold shall be set by calibration based on exposure to high temperatures in relevant applications.

The EM temperature limitation functionality is not intended to intentionally reduce the temperature to below *EmTempLimThreshold* when active.

However, the level of variation around the stabilization point depends on calibration of the EM temperature limitation which is set by balancing between drivability and availability of torque while ensuring that the maximum temperature limit is not exceeded.

250845 v1 **Inverter temperature limitation [CS Released]**

Purpose: Temperature protection of the inverter with a smooth degradation scheme.
Verification Method: Analysis AND Test
Legacy ID:

When the inverter temperature exceeds *InvTempLimThreshold* the available and applied torque shall be limited such that the inverter temperature estimate does not exceed *InvTempLimMax*. The limitation shall ensure that the full temperature range of the EM is used. The inverter temperature is not allowed to overshoot the max value.

Torque levels which does not cause the inverter temperature to exceed *InvTempLimThreshold* shall not be affected by this limitation if the coolant flow exceeds *the minimum required coolant flow*.

The coolant flow and temperature information over CAN, ***SysSigCooltFlowForDtElec*** and ***SysSigTSigForDtElec***, must not be used in a way such that a low or unavailable signal value can lead to a long term significant impact on available torque.

The *Inv temperature* is in signal ***SysSigInvrT***.

Note: *InvTempLimMax* and *InvTempLimThreshold* shall be set by the hardware design and be correlated with the assumptions used for analysing the fulfilment of the performance requirements (drive cycles etc.).

This functionality shall in a controlled way avoid scenarios leading to higher temperatures of the power electronics than can be accepted without risk for unforeseen permanent degradation of remaining life time.

The coolant part of this requirement is minimum capability requirement and does **NOT** imply that a different logical behaviour shall be applied at lower coolant flow levels (e.g. turning of torque complete when exceeding the threshold whenever the coolant flow level is below the base flow).

6.3.3.12.4.4 Diagnostics

190636 v1 **Motor temperature sensor diagnositc [CS Released]**

Purpose: Inverter diagnostic
Verification Method: Test
Legacy ID:

The inverter shall diagnostic the electric motor temperature sensors, including:
1. Short to battery
2. Short to ground
3. Raw signal out of range high
4. Raw signal out of range low

5. Physical signal over temperature
6. Motor temperature signal plausibility check

190637 v1 **Inverter temperature sensor diagnostic [CS Released]**

Purpose: Inverter diagnostic
Verification Method: Test
Legacy ID:

The inverter shall diagnostic the inverter temperature sensors, including:
1. Short to battery
2. Short to ground
3. Raw signal out of range high
4. Raw signal out of range low
5. Physical signal over temperature
6. Inverter temperature signal plausibility check
7. Two or more inverter temperature sensors fail

190643 v1 **PCB temperature sensor diagnostic [CS Released]**

Purpose: Inverter diagnostic
Verification Method: Test
Legacy ID:

The inverter shall diagnostic the PCB temperature sensors, including:
1. Short to battery
2. Short to ground
3. Raw signal out of range high
4. Raw signal out of range low
5. Physical signal over temperature
6. Inverter temperature signal plausibility check

190652 v1 **Coolant temperature sensor diagnostic [CS Released]**

Purpose: Inverter diagnostic
Verification Method: Test
Legacy ID:

The inverter shall diagnostic the coolant temperature sensor, including:
1. Short to battery
2. Short to ground
3. Raw signal out of range high
4. Raw signal out of range low
5. Physical signal over temperature

6.3.3.12.4.5 ActiveHeatingControl

659800 v2 **Active heating Scenario [CS Released]**

Purpose: Scenario to show when active heating will be acting.
Verification Method: Test
Legacy ID:

整车存在动态加热和静态加热场景，HVED子系统应接收整车加热场景信号***VehSceneForEDSHeatg***，通过该场景信号识别主动加热场景：

- ***VehSceneForEDSHeatg*** = **0x1 VehstatOrDyn_statsceneOK**，表示允许进入静态加热场景；
- ***VehSceneForEDSHeatg*** = **0x2 VehstatOrDyn_DynsceneOK**，表示允许进入动态加热场景；

317350 v4 **Available Heating Power [CS Released]**

Purpose: To provide the current available active heating power
Verification Method: Test
Legacy ID:

如果 ***SysSigUDc*** > 特定电压(400V 平台：200V(TBD)，800V 平台：380V(TBD))，逆变器应监控驱动系统的主动加热能力。

- 如果驱动系统能够产生主动加热，并且 ***SysSigHeatFdBck*** = 0x0 开启或 0x1 关闭，逆变器应提供 HVDC 侧当前可用的主动加热功率，并通过信号***SysSigHeatPwrMax*** 发送，在此状态下 ***SysSigHeatPwrMax*** ≠ 0。
- 如果驱动系统不能产生主动加热（**Inhibit**状态），提供的可用加热功率 ***SysSigHeatPwrMax*** = 0。
- **SysSigHeatPwrMax**的计算不能变化过快。此外也允许其随着系统温度保护进行derating；
- 不同的电驱产品有不同的能力，应根据实际能力发送至***SysSigHeatPwrMax***。
  （比如单电机发热能力4kW，***SysSigHeatPwrMax***应发送4kW。）
  （比如双电机单边发热能力2kW，***SysSigHeatPwrMax***应发送2kW。）

注意：这里的加热功率并不等于实际传递给冷却液的热量。加热功率仅指驱动系统的 HVDC 功耗。

If the "***SysSigHvSysRlySts = 0x1 Clsd***" **and** ***SysSigUDc*** **>** 特定电压（400V平台：200V(tbd)，800V平台：380V(tbd)）, inverter shall monitor the ability of active heating of Drive System.

- If the Drive System could produce active heat, and***"SysSigHeatFdBck = 0x0 On or 0x1 off"***, inverter shall provide the current available active heating power of HVDC side and send out by signal **SysSigHeatPwrMax**, in this state **SysSigHeatPwrMax≠0 .**
- If the Drive System can NOT produce active heat (**Inhibit state**), the provided available heat power **SysSigHeatPwrMax = 0**.
- The calculation of Signal **SysSigHeatPwrMax** can't change in large amplitude according with Torque power response and execute. But it can derating according to temperature protection.
- Different electric drive products have different capabilities, and ***SysSigHeatPwrMax*** should be sent based on the actual capabilities.

(For example, if a single motor has a heating capability of 4kW, ***SysSigHeatPwrMax*** should send 4kW.)
(For example, if a dual motor has a heating capability of 2kW on one side, ***SysSigHeatPwrMax*** should send 2kW.)

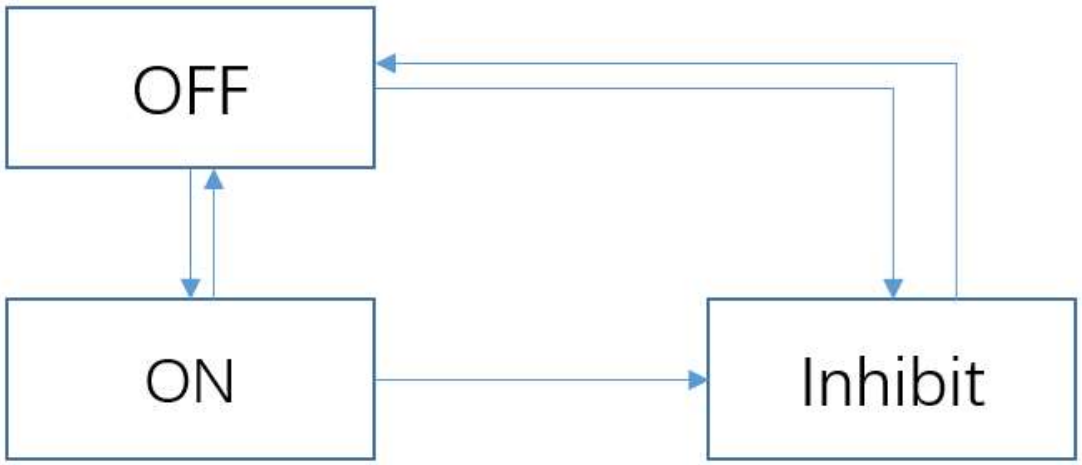
Note: Heating power in here isn't equal to the actual heat transfer to coolant. Heating power only means HVDC power consumption of Drive System.

317353 v2a1 Active heating state feedback [CS Released]

Purpose: To feedback the actual heating status of EM
Verification Method: Test
Legacy ID:

Inverter shall feedback the active heating state by signal **SysSigHeatFdBck**:

1) When inverter's active heating function can be used, ***"SysSigHeatFdBck = 0x1 Off or 0x0 On"*** , but receive the enabling signal ***"SysSigHeatModEnadReq*** = "***0x0 NotEnable"*** or ***"SysSigModReq ≠ 0x2 TqCtrl"*** mode request , active heating turn off state ***"SysSigHeatFdBck = 0x1 Off"***;
2) When inverter's active heating function can be used, ***"SysSigHeatFdBck = 0x1 Off or 0x0 On"***, and receives the enabling signal ***"SysSigHeatModEnadReq*** = "***0x1 Enable"*** turn on request and ***"SysSigModReq = 0x2 TqCtrl"*** Torque control mode request, If active heating is turned on correctly, feedback the active heating state turn on ***"SysSigHeatFdBck = 0x0 On"***;
3) When inverter detects the active heating function cannot be used because of internal faults or protection **(LOS Signal Bit1/2/3 trigger in SysSigLimnIndcn)**, feedback the active heating state inhibit ***"SysSigHeatFdBck = 0x2 Inhibit"*** .
4) When inverter's internal faults or protection recover **(LOS Signal Bit1/2/3 clear in SysSigLimnIndcn)**, feedback the active heating state Off ***"SysSigHeatFdBck = 0x1 Off"*** .
5) State switch process:
   State Off or State On to Off: according the condition 1).
   State On or State Off to On: according the condition 2).
   State On/Off to inhibit: according the condition 3).
   State Inhibit to Off: according the condition 4).



645846 v2 Active heating mode inhibition for Clutch [CS Released]

Purpose: inhibit HVED entering into active heating mode.
Verification Method: Test
Legacy ID:

For EDS with E-clutch, the E-clutch status SysSigCluSts shall be added as one of the conditions for the EDS entering into the active heating.

1. If the E-clutch status ***SysSigCluSts*** equals 'Closed' or 'StuckClosed':
   In this situation, Signal ***SysSigHeatFdBck*** sending method according to *Requirement ID 317353.*
2. If the E-clutch status ***SysSigCluSts*** equals 'Open' or 'StuckOpen' or 'Undefined' or 'Ongoing'
   Active heating function is not allowed to be turn on. In this situation ,if also without inverter's internal faults or protection,Inverter shall feedback the active heating state"***SysSigHeatFdBck = 0x1 Off"***.
3. Other situation, Signal ***SysSigHeatFdBck*** sending method according to *Requirement ID 317353.*

655513 v2 Active Heating Mode Inhibition for Booster [CS Released]

Purpose: To let boost charging have higher priority if EDS have boost function but can't boost and active heating in the same time.
Verification Method: Analysis AND Test
Legacy ID:

For EDS integrated with booster, the boost request signal ***SysSigBoostMdReq*** shall be added as one of the conditions for the EDS entering into the active heating. Boost charging has higher priority than EDS active heating.

1) If EDS has received the boost charge mode request from HVES on port: ***SysSigBoostMdReq != "0x00 BoostMdReq_Default",*** EDS shall not allow the active heating, send ***"SysSigHeatFdBck = 0x2 Inhibit"*** to Thermal Management.
   While active heating function is on working, ***"SysSigHeatFdBck = 0x0 On"*** and received the boost charge mode request ***SysSigBoostMdReq != "0x00 BoostMdReq_Default"***. EDS should turn off active heating power by the rate of descent = **4kW/S（TBD）to Zero watt**.
2) If EDS has received the boost charge mode request from HVES on port: ***SysSigBoostMdReq = "0x00 BoostMdReq_Default"***. Signal ***SysSigHeatFdBck*** sending method according to *Requirement ID 317353.*

655511 v2 Active heating power execution [CS Released]

Purpose: To define the priority of active heating execution.
Verification Method: Test
Legacy ID:

- 当 HVED 处于无扭矩输出状态时，***SysSigTqReq*** = 0，每个 HVED 应根据热管理分配的主动加热功率 ***SysSigFrntHeatPwrAllwd***执行主动加热功能。
- 当 HVED 处于扭矩输出状态时，每个 HVED 应优先执行来自推进系统的扭矩请求 ***SysSigTqReq***。如果有剩余电能，这些剩余电能可以按照热管理分配的主动加热功率 ***SysSigFrntHeatPwrAllwd*** 用于主动加热功能。

- 当 HVED 处于动态加热场景时，考虑主动加热影响扭矩精度，
  当处于小扭矩输出时（电机端扭矩绝对值小于6Nm），应线性限制加热功率
  （电机端扭矩绝对值0~5Nm对应加热功率0kW (TBD)，电机端扭矩绝对值6Nm对应最大加热功率Max kW）。

- 为避免突加电机加热功率造成的整车抖动，在内部加热交流侧电流指令卸载斜率为400A/s。

- When EDS is in no torque output scenario, "***SysSigTqReq = 0"***, each EDS should execute active heating function according to the active heating power distributed by thermal management, ***SysSigHeatPwrAllwd.***

- When EDS is in torque output scenario, each EDS should execute the torque request ***SysSigTqReq*** from Propulsion in priority. If have some remaining electric energy, this remaining electric energy can be used into active heating function according to the active heating power distributed by thermal management, ***SysSigHeatPwrAllwd.***

- When HVED is in the dynamic heating scenario, considering the impact of active heating on torque accuracy, when the motor is in a low torque output state (absolute value of motor torque less than 6Nm TBD), the heating power should be linearly limited (absolute value of motor torque 0~5Nm corresponds to heating power 0kW, absolute value of motor torque 6Nm corresponds to maximum heating power Max kW).

- To avoid vehicle shaking caused by sudden increases in motor heating power, the internal heating AC current command unload rate is set to 400 A/s.

317354 v4 **Actual heating power feedback [CS Released]**

Purpose: To provide the actual active heating power
Verification Method: Test
Legacy ID:

Inverter shall receive the allow active heating power on port **SysSigHeatPwrAllwd**.
The Thermal manager system use infinitely variable control, so the power request signal **SysSigHeatPwrAllwd** will change constantly, inverter shall respond the change in time.
If the Drive System active heating is turned on:

- When **SysSigHeatPwrMax** < **SysSigHeatPwrAllwd**: Drive System shall warm up the coolant using its available maximum heating ability.
- When **SysSigHeatPwrMax** ≥ **SysSigHeatPwrAllwd**: Drive System shall warm up the coolant according to the distributed electrical energy from ECM.

Inverter shall send the actual active heating power of HVDC side on port **SysSigHeatPwrAct**.
Actual active heating power shall be calculate by internal

- When in Vehicle Static state, **SysSigHeatPwrAct = U*I** ,
- When in Vehicle dynamic state, **SysSigHeatPwrAct** shall be calculate out the additional power consume only produce by active heating.(Not contain the passive heating produce by driving)

The accuracy of the active heating power shall be less than 10%(TBD).

Note: Heating power in here isn't equal to the actual heat transfer to coolant. Heating power only means HVDC power consumption of Drive System.

655512 v1 Sum Heating Power Feedback [CS Released]

Purpose: To calculate the sum heating power (power loss) during driving. Vehicle Thermal management will consider active and passive heat which affect or may be reused in vehicle.

Verification Method: Analysis AND Test

Legacy ID:

No matter whether the active heating is turned on or not, after each EDS is awakened, the total power loss of the EDS shall feedback in real time. The power loss refers to the high-voltage power that is not converted into mechanical energy, but transform to heat.

The EDS shall send the power loss as sum heating power on signal port **SysSigHeatPwrSumAct**.

- When in Vehicle Static state, **SysSigHeatPwrSumAct = SysSigHeatPwrAct** ,
- When in Vehicle dynamic state, **SysSigHeatPwrSumAct = U*I –Tq*n/9550**

  (Total heat power = HVDC power – Actual Torque power , value of **" U I Tq n "**shall be calculate with a sign)

Note: Heating power in here isn't equal to the actual heat transfer to coolant. Heating power only means HVDC power consumption of Drive System.

655515 v1a1 Active Heating Bypass immobiliser to Torque Mode for Special Case [CS Released]

Purpose: In Special Case , For example request active heating during On board Charging. Propulsion will request torque mode to execute active heating. But in this situation , even if EDS send immobiliser challenge to CEM ,but CEM won't response.EDS shall bypass immobiliser to active heating.

Verification Method: Analysis AND Test

Legacy ID:

For Mode state:

The condition switch from standby mode to torque mode :

- If ModReq="TqCtrl" & HvSysRlySts="Closed" & ImobEngSts3 = ImobMtn,

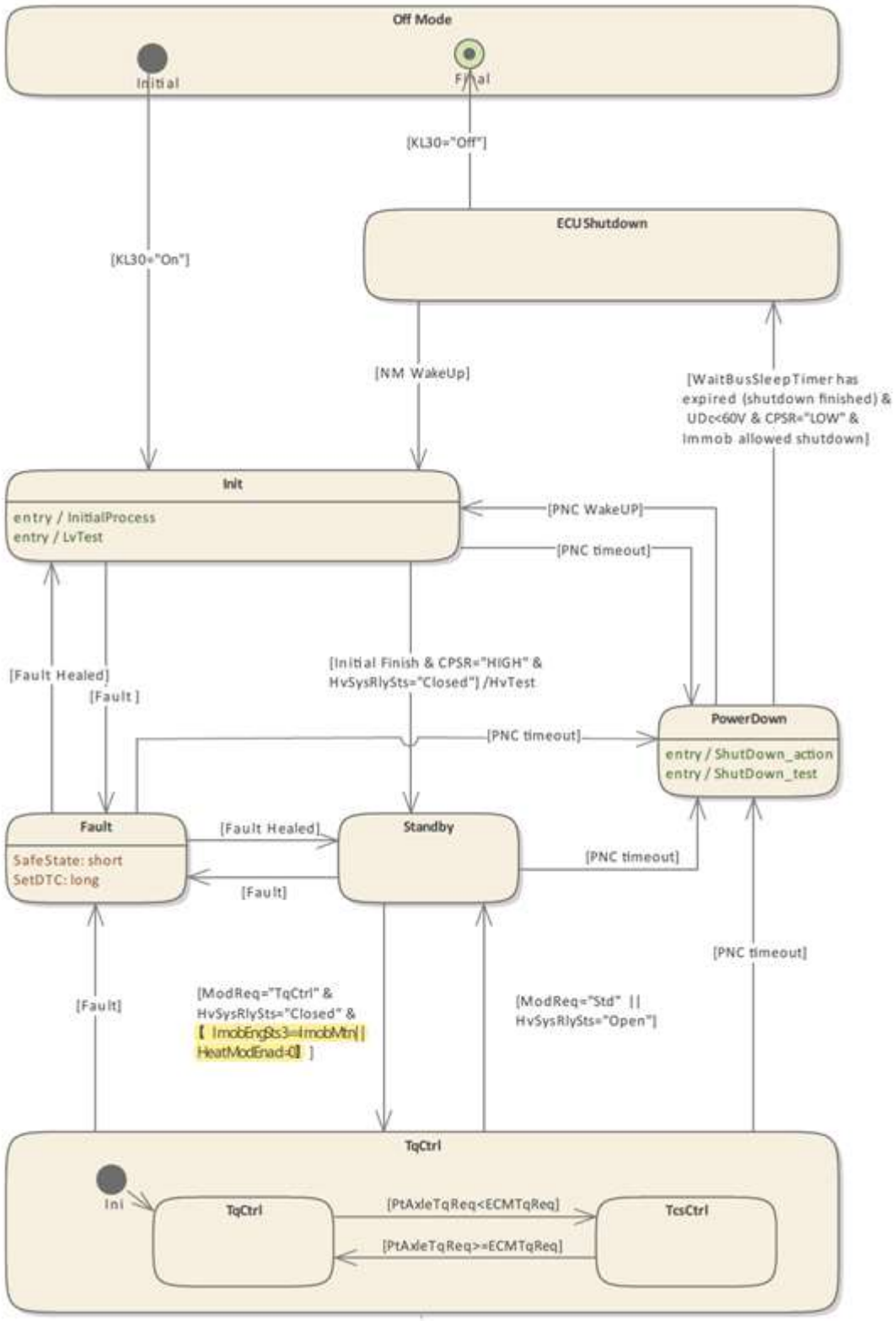
  ModSts="TqCtrl".
- If ModReq="TqCtrl" & HvSysRlySts="Closed" & ***SysSigHeatModEnadReq*** = "***0x1 Enable*** & ImobEngSts3 ≠

ImobMtn,

Bypass immobiliser and ModSts="TqCtrl".

- If ModReq="TqCtrl" & HvSysRlySts="Closed" & ***SysSigHeatModEnadReq*** = "***0x0 NotEnable*** & ImobEngSts3 ≠ ImobMtn,

  ModSts="Standby".



1041514 v1 **Active Heating Mode Inhibition for Oil pump [CS Released]**

Purpose: To protect the motor and bearing which oil pump have serious fault.

Verification Method: Test
Legacy ID:

For EDS integrated with Oil pump,

- If Oil pump have serious fault, EDS shall not allow active heating, send ***"SysSigHeatFdBck = 0x2 Inhibit"*** to Thermal Management.
- If active heating function is on working and oil pump have serious fault, EDS should turn off active heating power by the rate of descent = **4kW/S（TBD）to Zero watt**. Send ***"SysSigHeatFdBck = 0x2 Inhibit"*** to Thermal Management.
- If Oil pump fault clear, EDS shall be able to allow active heating again. Signal ***SysSigHeatFdBck*** sending method according to *Requirement ID 317353.*

1041515 v1 Active Heating Mode Inhibition for Dual-Motors [CS Released]

Purpose: To protect the dual motors have driving coordinate issue ,if one work and other not working.
Verification Method: Test
Legacy ID:

For the EDS with Single shaft dual-motors condition,

1) when one of motors inhibit active heating, ***"SysSigHeatFdBck = 0x2 Inhibit"***. The other motor shall also inhibit active heating, ***"SysSigHeatFdBck = 0x2 Inhibit".***
2) If both motor recover and be able to active heating, their active heat feedback signal shall switch from inhibit to Off, ***"SysSigHeatFdBck = 0x1 Off"***. After that, ***SysSigHeatFdBck*** sending method according to *Requirement ID 317353.*

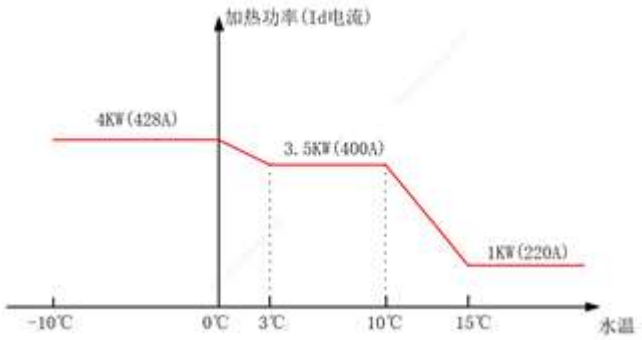
1041521 v1 Available Heating Power Derating for Coolant Temperature [CS Released]

Purpose: Available Heating Power Derating for Coolant Temperature
Verification Method: Test
Legacy ID:

当HVED系统长时间主动加热，且水温较高时，存在铜排过温风险。

因此HVED系统对水温进行识别，当水温达到一定温度时，限制主动加热功率。
此需求适用于单个电机主动加热能力最大为4kW的产品。

❖ 静态加热：主动加热功率根据水温做限制。
- 当***SysSigCoolt*** <= 0℃(TBD)，主动加热功率最大4kW
- 当0℃(TBD)<***SysSigCoolt*** <= 3℃(TBD)，主动加热功率 4kW 线性 降额至3.5kW(TBD)
- 当3℃(TBD)<***SysSigCoolt*** <= 10℃(TBD)，主动加热功率 最大3.5kW(TBD)
- 当10℃(TBD)<***SysSigCoolt*** <= 15℃(TBD)，主动加热功率 3.5kW(TBD) 线性 降额至1kW(TBD)



❖ 动态加热（TBD）：主动加热功率不做限制。

1043694 v1 Active Heating Mode Inhibition for Oscillatory Heating [CS Released]

Purpose: Active Heating Mode Inhibition for Oscillatory Heating
Verification Method: Test
Legacy ID:

对于具备振荡加热的ECU：

1. 当收到***RearPlsHeatgCmd*** = 0x1 On， ***WhlMotSysHeatFdBck*** = 0x2 禁止。
2. 当收到***RearPlsHeatgCmd*** ≠ 0x1 On， ***WhlMotSysHeatFdBck*** 不再强制发禁止。

6.3.3.12.4.6 Oil Pump

1040542 v1 Oil Pump Information [CS Released]

Purpose: Oil pump information
Verification Method: Test
Legacy ID:

If HVED have oil pump to cool motor ,HVED shall send Oil Pump information on below port in form.

Real signal please check SDB and signal map.

System Signal (Send port)	Description
SysSigSpdTar	Oil pump target speed
SysSigRstrtCmd	Oil pump reset command
SysSigSleepCmd	Oil pump sleep command
SysSigFbSpdAct	Oil pump actual speed
SysSigFbIAct	Oil pump actual current
SysSigFbUAct	Oil pump actual voltage
SysSigFbTAct	Oil pump MCU temperature
SysSigFbActModSts	Oil pump mode state
SysSigSleepAck	Oil pump sleep confirm
SysSigSupplier	Oil pump supplier
SysSigLimnIndcn	Oil pump Los
SysSigVersion	Oil pump version
SysSigLinTxSts	Oil pump Transimit state

SysSigLinRxSts	Oil pump Receive state

1040543 v1 **Oil Pump Diagnostic [CS Released]**

Purpose:
Verification Method:
Legacy ID:

If EDS hardware part combine oil pump to cool motor and lubricate bearing, HVED shall diagnose the oil pump, including:
1. Self-check;
2. Lose LIN communication;
3. Over voltage fault;
4. Under voltage fault;
5. Over temperature fault;
6. Stuck fault;
7. Over current fault;
8. Over load fault;
9. Senor fault;
10. Power limitation.

If fault of oil pump won't affect motor working speed, Bit 0 shall be trigger in Port ***SysSigLimnIndcn***.

If fault of oil pump affect motor working speed, Bit 5 shall be trigger in Port ***SysSigLimnIndcn***.

Detail to be handshake in SSDS.

1040544 v1 **Oil Pump SW OTA [CS Released]**

Purpose: Oil pump SW OTA
Verification Method: Test
Legacy ID:

The software of oil pump shall support OTA software update.
To solve the oil pump software bug for aftermarket by OTA.

Oil pump software shall be finish updated during inverter software update process.

Each oil pump software shall be finish updated in 1minute.(TBD, defined according to the size of oil pump SW)

油泵软件应支持OTA软件更新。
通过OTA解决售后油泵软件漏洞。

在逆变器软件更新过程中，应更新油泵软件。

每个油泵更新应在1分钟内完成。（待定,具体根据油泵供应商的软件大小定义）

1040554 v1 **Oil pump protection [CS Released]**

Purpose: Oil pump protection
Verification Method: Test
Legacy ID:

为避免整车上电时，油泵预润滑失败，增加识别DcDcActvd状态和电压充足时，确保有电源供给的情况下，再开始油泵启动（预润滑）。

Information：整车上电时，DCDC进入运行所需时间较长，因而可能导致油泵预润滑时报出欠压故障。

1043079 v1 Oil Pump Speed request from EEM [CS Released]

Purpose: Oil Pump Speed request from thermal management
Verification Method: Test
Legacy ID:

代替信号	信号定义
SysSigReqOilSpdSts	来着EEM的油泵转速请求使能
以下为各个ECU的正式信号 XCM2/MGM: ***EEMReqIsgOilSpd.EEMReqOilSpdSts*** IFDS2: ***EEMReqIsgRiOilSpd.EEMReqOilSpdSts*** IEM2/IEM3: ***EEMReqWhlMotSysOilSpd.EEMReqOilSpdSts*** IEM2: ***EEMReqWhlMotSysRiOilSpd.EEMReqOilSpdSts*** XCM2: ***EEMReqElecMot2OilSpd.EEMReqOilSpdSts***	

代替信号	信号定义
SysSigReqOilSpdVal	来着EEM的油泵转速请求
以下为各个ECU的正式信号 XCM2/MGM: ***EEMReqIsgOilSpd.EEMReqOilSpdVal*** : ***EEMReqIsgRiOilSpd.EEMReqOilSpdVal*** IEM2/IEM3: ***EEMReqWhlMotSysOilSpd.EEMReqOilSpdVal*** IEM2: ***EEMReqWhlMotSysRiOilSpd.EEMReqOilSpdVal*** XCM2: ***EEMReqElecMot2OilSpd.EEMReqOilSpdVal***	

- 当HVED接收到***SysSigReqOilSpdSts*** = 0x1:On， 应控制油泵转速为：
  = 允许的最大油泵转速；
- 当HVED接收到***SysSigReqOilSpdSts*** != 0x1:On，应控制油泵转速为：
  = 自身散热需求计算的油泵转速请求。

若箱体内同时存在机械油泵，电子油泵的最大工作转速应不使机械油泵空吸。

为适应不同供应商的油泵，EEM发送的***SysSigReqOilSpdVal***仅作为参考，最大工作转速取决于油泵自身能力。

6.3.3.12.4.7 Active Store Oil Heat

1043124 v1 Active Store Oil Heat [CS Released]

Purpose: To store oil heat when low temperature
Verification Method: Test
Legacy ID:

SysSigMotHeatStorgReq Signal Value Description	Description
0: No request	Heat storage control request

1: Request HVED control valve to bypass coolant flow	主动蓄热请求
Front motor signal:FrntMotHeatStorgReq Rear motor signal:ReaMotHeatStorgReq	

SysSigHeatStorgAllwd Signal Value Description	Description
0: Default 1: Allow 2: NotAllow	Heat storage allow 主动蓄热允许
Front motor signal:IsgHeatStorgAllwd Rear motor signal:WhlMotSysHeatStorgAllwd	

SysSigOilT Signal Value Description	Description
-50~200℃	Motor Oil temperature 电机油温
Front motor signal:IsgOilT Rear motor signal:WhlMotSysOilT Rear right motor signal: WhlMotSysRiOilT	

- HVED shall send motor oil temperature to energy Electrical Energy Management on port ***SysSigOilT***.
- 电驱系统应通过***SysSigOilT***信号将电机油温发送给电力能源管理系统。

If HVED have motor oil temperature sensor, shall report actual oil temperature.
如果HVED有电机油温传感器，应报告实际油温。

If HVED don't have motor oil temperature sensor but supplier have ability to estimate oil temperature, shall report estimated oil temperature.
如果HVED没有电机油温传感器，但供应商有能力估算油温，则应报告估算的油温。

If none of above is feasible, HVED shall send motor temperature as oil temperature.
如果以上均不可用，HVED应将电机温度作为油温发送。

- HVED shall send the heat storage allow to Thermal Management on port ***SysSigHeatStorgAllwd***
- 电驱系统应通过***SysSigHeatStorgAllwd***信号将主动蓄热允许发送给热管理系统。

- If during initial, ***SysSigHeatStorgAllwd*** = "0: Default";
- If motor temperature is hot, ***SysSigHeatStorgAllwd*** = "2: NotAllow";
- If HVED can detect coolant valve have fault, ***SysSigHeatStorgAllwd*** = "2: NotAllow";
- If HVED enter to fail safety state and need to execute Active Short Circuit, ***SysSigHeatStorgAllwd*** = "2: NotAllow";
- If HVED receive boost charging or buck discharging request, ***SysSigHeatStorgAllwd*** = "2: NotAllow";
- If HVED receive active heat request, ***SysSigHeatStorgAllwd*** = "2: NotAllow";
- If HVED receive pulse heat request, ***SysSigHeatStorgAllwd*** = "2: NotAllow";
- If none of above , ***SysSigHeatStorgAllwd*** = "1: Allow".

- 如果在初始化期间，***SysSigHeatStorgAllwd*** =“0:Default”；
- 如果电机温度过高，***SysSigHeatStorgAllwd*** =“2:NotAllow”；
- 如果电驱系统可以检测到冷却液阀有故障，***SysSigHeatStorgAllwd*** =“2:NotAllow”；

- 如果电驱系统进入故障安全状态并需要执行主动短路，***SysSigHeatStorgAllwd*** =“2:NotAllow”；
- 如果电驱系统收到升压充电或降压放电请求，***SysSigBoostMdReq*** = “0x01 BoostMdReq_BoosterOrHvBattDirectCharging” or “0x02 BoostMdReq_BoosterCharging” or “0x03 BoostMdReq_Discharging”，***SysSigHeatStorgAllwd*** =“2:NotAllow”；
- 如果电驱系统收到主动加热请求***SysSigHeatModEnad*** = "0x0 Heatg"，***SysSigHeatStorgAllwd*** =“2:NotAllow”；
- 如果电驱系统收到振荡加热请求***RearPlsHeatgCmd***= 0x1 On，***SysSigHeatStorgAllwd*** =“2:NotAllow”；
- 如果以上都没有，***SysSigHeatStorgAllwd*** =“1: Allow”。

❖ HVED shall receive the heat storage control request from Thermal Management on port ***SysSigMotHeatStorgReq***, and control coolant valve on and off.(Only applicable for IEM7)
电驱系统应接收***SysSigMotHeatStorgReq***信号来自热管理，并控制冷却液阀开关。(仅使用于IEM7后驱)

If ***SysSigMotHeatStorgReq*** = "0: No request", HVED control coolant valve to cool motor.
If ***SysSigMotHeatStorgReq*** = "1: Request", HVED control coolant valve not to cool motor(heat storage).

如果***SysSigMotHeatStorgReq*** ="0: No request"，电驱系统控制冷却液阀给电机散热。
如果***SysSigMotHeatStorgReq*** ="1: Request"，电驱系统控制冷却液阀不给电机散热，即蓄热。



6.3.3.12.5 Signal Map

251600 v1 **Reference to Signals [CS Released]**

Purpose: To show which interfaces that contains the signals
Verification Method: Documentation
Legacy ID:

Signal Map is found within LC: EM[Number]ThermalInterface

EM1 (Main Front EM system)
EM2 (Auxillary Front EM system)
EM3 (Main Rear EM system)

Depending on System Voltage Signals are also mapped in

400V
LC: EM[Number]400VInterface

800V
LC: EM[Number]800VInterface

6.3.3.13 Logical component: MotorControlRequirements

ID	Name	Version	Status	Last Changed
x0400000025F1CA27	MotorControlRequirements	(10a4a11)	Released	2026-05-21

Motor control requirements for EDS.

6.3.3.13.1 Included Virtual Function Clusters(VFCs)

6.3.3.13.2 MotorControlRequirements: Receiver Ports

6.3.3.13.3 MotorControlRequirements: Send Ports

6.3.3.13.4 Carconfig info:

Parameter number	Parameter name	Parameter value	Description
9	GEARBOX	A4	FSSR+R2SR
		A5	LFWD+R2SR

6.3.3.13.5 Requirements

6.3.3.13.5.1 Mode Manager

Note: Review if Sync Control shall remain. If it is removed specify the non-requestable modes Pre-charge & Powerdown states

242345 v2 Inverter Mode request [CS Released]

Purpose: To define the inverter operating modes that can be requested by ECM.
Verification Method: Test
Legacy ID:

The inverter shall accept the mode requests received on port ***SysSigModeReq***

The valid mode requests are as follows: Stb, TqCtrl, SpdCtrl, UDcCtrl

242349 v2 Inverter Mode status [CS Released]

Purpose: To signal the current status of the node.
Verification Method: Test
Legacy ID:

The inverter mode status shall be reported on port ***SysSigModSts***.

The following mode states are allowed: Inin, PreChrg, Stb, TqCtrl, SpdCtrl, UDcCtrl, Flt, PwrDwn, TcsCtrl.

242350 v2 Startup event [CS Released]

Purpose: Define startup event for the node to wake up and start communicating
Verification Method: Test
Legacy ID: 396452
Network Activation Type: VFC Wake Up

The Wake-up and shut-down of the node shall be based on Partial Network Clusters (PNC) concept. The startup event is defined as when:

- A PNC CAN frame has been received with at least one of the bits the node shall listen to set.

Note: The Virtual Function Clusters (VFC's) that the ECU is part of will be mapped to Partial Network Clusters (PNC's).

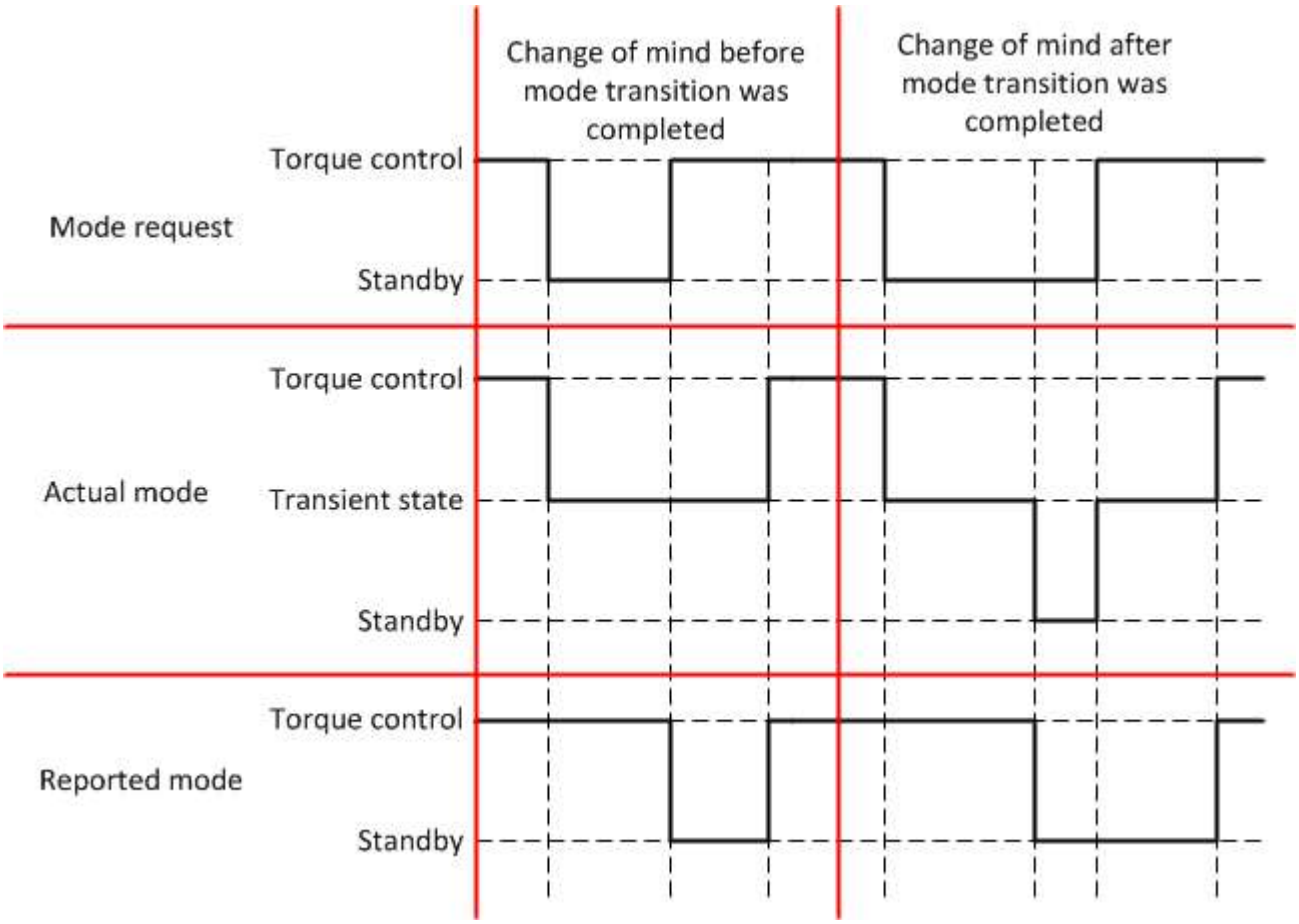
242351 v2 Mode transition and request [CS Released]

Purpose: The ECM (mode requester) consider the Electric Drive System(EDS) being in mode transition while the mode request is not the same as the reported mode. As soon as the reported mode is the same as the requested the ECM sees that as a confirmation and start acting as the inverter really has reached that mode. In case ECM need to change the request during transistion the EDS need to react in a certain way to make it work.

Verification Method: Analysis
Legacy ID:

If a request to transistion to the previous mode is recieved on ***SysSigModReq*** before the new state transition is completed it shall complete the mode transition before going back to the previous mode. The reported mode on ***SysSigModSts*** shall be changed to the mode the inverter is going to, insted of the mode it is going from.

Example:



242352 v1 **Mode reported during transition [CS Released]**

Purpose: To make sure that all components using the operational modes of the HVElectric Drive system are coordinated when a new state is valid
Verification Method: Test
Legacy ID:

The value sent on ***SysSigModSts*** shall change only when all prerequisistes for the new state have been fullfilled.

Note: That is, if the inverter is in a transient state the last valid state will then be signalled until another valid state has been reached. "Valid" in this case is a state which there is a state in the signal definition of ***SysSigModSts***)

242353 v2 **Complete an ongoing transition [CS Released]**

Purpose: To ensure compatibility with the requesting systems view on request and transition
Verification Method: Test
Legacy ID:

An initiated or ongoing transition shall be completed even if conditions that would initiate a different transition are fulfilled during the transition.

242354 v1 **Mode based estimations [CS Released]**

Purpose: The receiver of some signals assumes that the reported value is free to use as is and therefore a filter of the active mode is needed, specifically this i the available torque
Verification Method: Test
Legacy ID:

The available torque estimation shall report 0 on both min and max in ***SysSigTqAvl*** when the active mode is Init, Fault or Powerdown regardless what the estimated torque is.

Note:
This makes the signal reflect the available torque that the propulsive manager can utilize.

242355 v2 **HV System power up sequence [CS Released]**

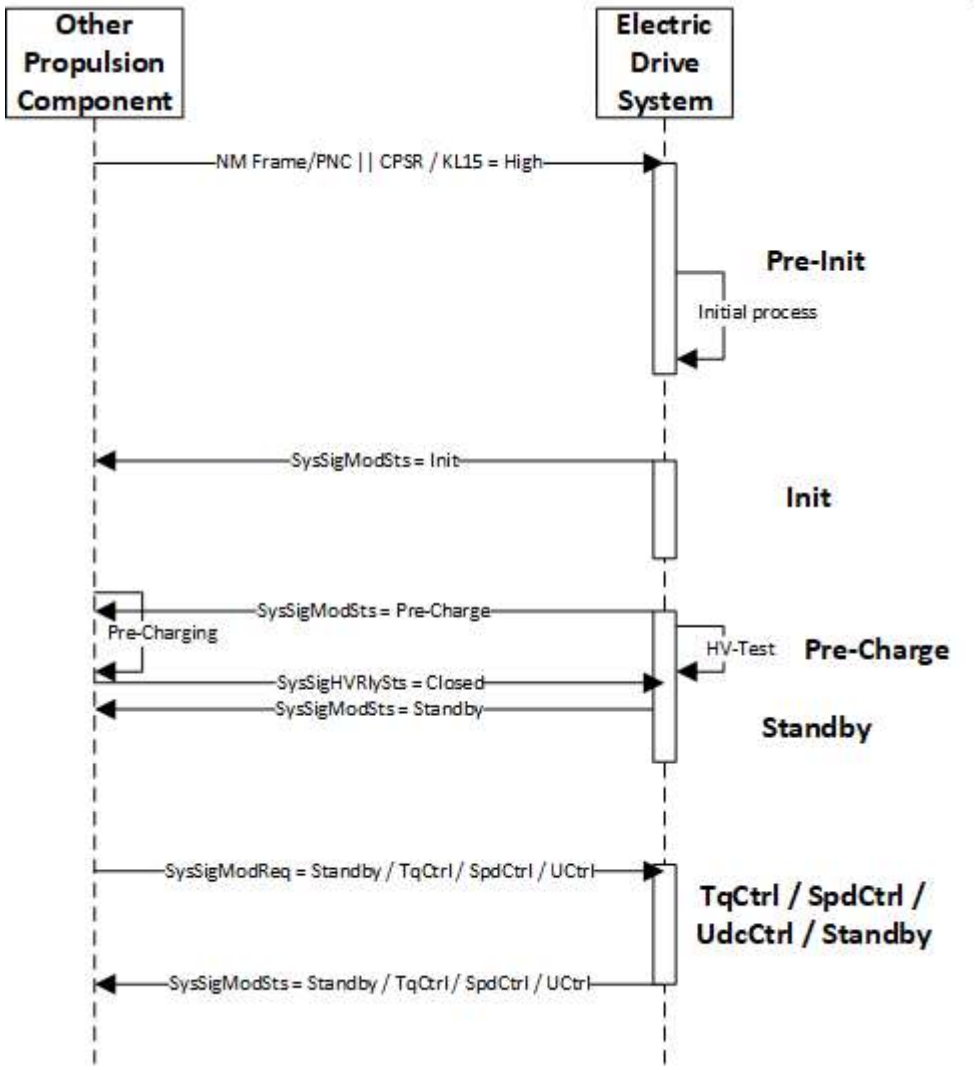
Purpose: To prepare the vehicle for driving
Verification Method: Test
Legacy ID:

The wake up shall be enabled by the PNC concept.
When the inverter receives wake up request then the following procedure shall be followed;

- PNC active or CPSR go to high
- Init all necessary components of the SW
- Check internal HW/SW to ensure proper functionality such as memory check etc. all checks that not need the DC-link up and running (LV test)
- Perform 3-phase Isolation test
- HV test to ensure proper functionality (HV test), BECM test of HV-system.
- Wait for HV relay to close
- HV relay closed

If all tests are OK, then the inverter is allowed to enter Standby Mode otherwise enter Fault Mode

information: It is not allowed that any component reports Status==Standby before power up checks and/or calibrations are complete. Standby shall be reached within 200ms from receiving ContactorStatus==Closed from BECM. Status Standby means that the unit is operational and ready to receive/carry out commands.



242373 v2 **Quick restart [CS Released]**

Purpose: To handle quick restart requests during power down
Verification Method: Test
Legacy ID:

The Electric Drive System(EDS) shall be able to handle quick restart (driver change of mind) requests during power down (less than 3 seconds) or opposite situation when shutting down quickly precisely when the system is initiating.
EDS shall respond to 12-volt power applied to CPSR Power input when CPSR power is cycled from on to off to on and send the mode status of the motor message on propulsion CAN within 500ms of receiving the power/wake-up signal (PNC Start and PNC Driving).

The message the mode status of the motor shall be sent to the ECM 800ms after power/wake-up signal (PNC Start and PNC Driving).

6.3.3.13.5.1.1 Initialization

242379 v2 — Time to signal init [CS Released]

Purpose: To ensure the vehicle start time.
Verification Method: Test
Legacy ID:
Network Activation Type: VFC Wake Up

The inverter shall send the first mode status set to "Init mode" by port ***SysSigModSts*** within 180 ms from receiving power/wake-up signal (PNC-Start).

242383 v2 — Wake up timing [CS Released]

Purpose: Define the wake up timing upon receiving startup event in order to meet vehicle start time requirements.
Verification Method: Test
Legacy ID: 393812
Network Activation Type: VFC Wake Up

The node shall initialize and begin communication (Propulsion CAN) within 150ms of receiving the power/wake-up signal (PNC-Start).

242457 v1 — Self diagnosis [CS Released]

Purpose: To make sure that the system/component is properly initialized and diagnosed for errors before leaving initialization state
Verification Method: Test
Legacy ID:

The component shall initialize and run self tests on all ingoing SW and HW components in the inverter and EM to ensure that they are working properly, this shall be performed while in Initialization state, until this is finalized it shall remain there and communicate Init state on signal ***SysSigModSts***.

Note: Some tests that require HV voltage will need to be coordinated with the signal ***SysSigHvRlySts*** when it is set to closed.

242459 v1 — Quick restart during initialization [CS Released]

Purpose: If the user performs a key on/off/on procedure the time of initialization shall be as short as possible
Verification Method: Analysis AND Test
Legacy ID:

If a quick restart, power down followed by a power up before the component has completed the power down procedure, occurs then the initialization procedure shall not restart all initialization tests, instead it shall reuse the applicable test results from the last test.

242461 v6 — 3-phase short circuit for isolation tests [CS Released]

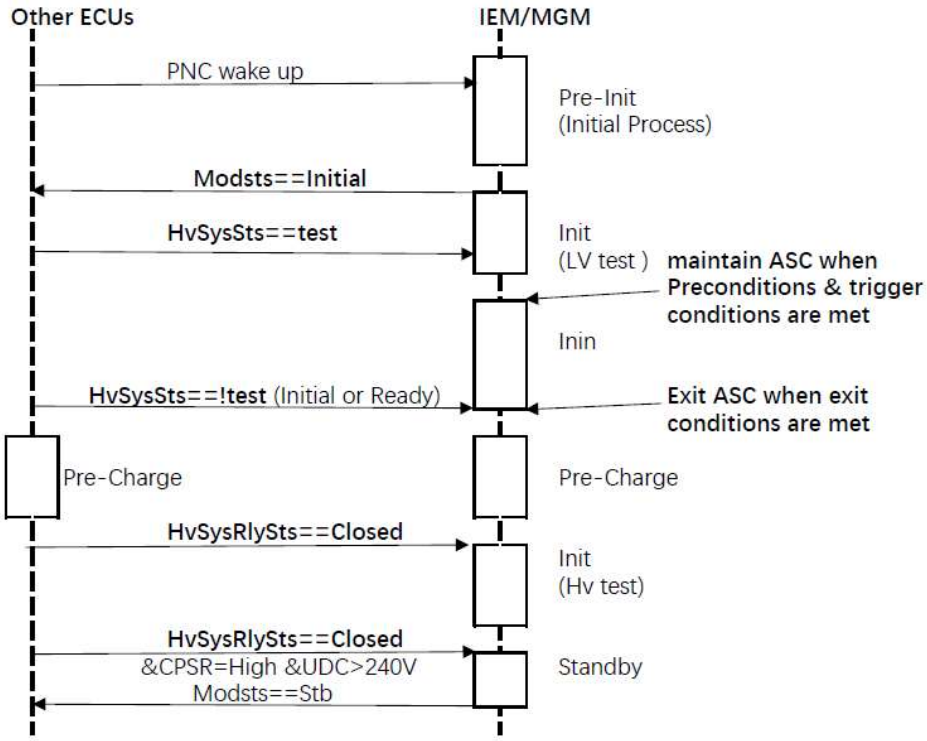
Purpose: The system isolation is monitored during vehicle startup and this test includes the HVAC part of the system and the EM. Doing a three-phase short circuit allows enables that part to be diagnosed as well.

Verification Method: Test

Legacy ID:

For application marked as applicable, according to *HvSysAscRequestApplicable*
The Electric Drive System(EDS) shall maintain a 3-phase short circuit when ***HvSysSts*** is "Test". This shall be done by closing the upper or lower 3 IGBTs.（Upper IGBTs is more suggested to be priority.）

"3-phase short circuit for isolation tests" Activated sequence



1) Trigger condition: HvSysSts from "Rdy" or "Inin" to "test"
EDS will response "test" request after LV test finished.
The timer shall re-timing.(The timer in here is to calculate how long HvSysSts="test" is sending)

2) Exit condition: any one of the following conditions
HvSysSts="Rdy" or "Inin" .
When EDS detecting HV(＞60v).
When EM speed is higher than 500 rpm.
When "test" time expired/timeout.(＞80s)
When lost communication with BECM.(＞250ms)

3) Precondition of this "test"
EDS's own LV test must passed
EM speed is valid (no resolver fault or EM speed can be measured)
EDS Without HV(≤60v).

EDS shall feedback the three phase active short circuit state on part ***SysSig3PhaShoCircSt***

Signal Value Description	Description
0x0:Initial	EDS's own LV test has not not finish, shall report Initial

0x1:NoShoCirc	no three phase short circuit
0x2:UpprShoCirc	upper bridge three phase short circuit
0x3:UndrShoCirc	under bridge three phase short circuit
0x4:Other	3 phase power electronic switches are controlled to be working other specific state
0x5:Unknown	SW lose detection of 3 phase power electronic switches
0x6:Reserve	Reserve
0x7:Reserve	Reserve

- If EDS's own LV test has not finish, ***"SysSig3PhaShoCircSt = 0x0 Initial "***
- If 3 phase power electronic switches don't have upper or under bridge three phase short circuit, ***"SysSig3PhaShoCircSt = 0x1 NoShoCirc"***
- If 3 phase power electronic switches have upper bridge three phase short circuit,***"SysSig3PhaShoCircSt = 0x2 UpprShoCirc"***
- If 3 phase power electronic switches have under bridge three phase short circuit,***"SysSig3PhaShoCircSt = 0x3 UndrShoCirc"***
- If 3 phase power electronic switches are controlled to be working other specific state but neither no three phase short circuit, nor upper/under bridge three phase short circuit,***"SysSig3PhaShoCircSt = 0x4 Other"***
- If SW lose detection of 3 phase power electronic switches,***"SysSig3PhaShoCircSt = 0x5 Unknown"***

Diagnosis:
- If lost signal of HvSysSts from BECM more than 250ms, EDS shall record DTC :lost of Communication with BECM.
- If HvSysSts = Test more than 80s,EDS shall record DTC :The command of 3-phase short circuit for isolation tests timeout.

Note:

Diagnostics for timeout monitoring of relevant HV-system signaling shall be considered whenever not doing so leads to a risk for being stuck in the short circuit state

The isolation test will be executed after crash event or isolation failure in last driving cycle. Three-phase short circuit is defined by activating 3 power electronic switches in either the upper or lower row. This will cause the electric machine to be electrically visible by the HV isolation resistance monitoring system.

6.3.3.13.5.1.2 Standby

This is a ready to operate state, no actual torque is delivered in this state, the system waits for a request to go to a torque producing state. Transitioning into this state also requires that torque is ramped down before entering.

242474 v2 **Standby mode prerequisites [CS Released]**

Purpose: To clarify when the node is allowed to enter Standby mode and be ready to transition to a mode where it deliver torque.
Verification Method: Test
Legacy ID:

The component shall transition to Standby mode when the following is fulfilled

- Initialization is finalized and no faults are present AND ***SysSigHvRlySts*** *status transitions to closed*
- HV Test is finalized;

- ***SysSigHvRlySts*** *status transitions to closed;*

The component shall transition to Standby mode from Non Initialization mode or remain in Standby mode when either of the following is fulfilled

- Standby is requested on ***SysSigModReq*** *and* ***SysSigHvRlySts*** *status is closed and* no faults are present.
- ***SysSigHvRlySts*** *status transitions to open and request active discharge when in any of the Speed, Torque or Voltage control modes*

242478 v1 **Time to standby [CS Released]**

Purpose: To ensure the vehicle's ready for driving timing constraints
Verification Method: Test
Legacy ID:

The mode standby shall be entered and reported in ***SysSigModSts*** no later than 550ms after a startup event is received.

242480 v2 **Standby [CS Released]**

Purpose: Standby shall behave in a stabile way regardless of control input and act as the default "safe" mode.
Verification Method: Test
Legacy ID:

When transitioning to Standby any residual torque shall be ramped out in a smooth* way before entering Standby.

When the inverter is in standby and residual torque has been ramped out the torque control shall be disabled but the inverter remain operational (fully powered up and initialized) and ready to transition to a torque producing control mode if conditions for this are fulfilled.

Note:

*Smoothness is not defined in this specification but set by the calibration after balancing with drivability, where drivability has priority over the second priority, speed.

Disabled torque control implies that the normal PWM-control scheme is inhibited, either by short circuiting the 3 phases or by disabling switch pulses for the power switches of the inverter.

6.3.3.13.5.1.2.1 PMA/SEA

676986 v1 — Asynchronous Motor Standby control during Driving [CS Released]

Purpose: To reduce power consumption for asynchronous motor.
Verification Method: Analysis AND Test
Legacy ID:

For the vehicle use asynchronous motor:
During driving, when asynchronous motor don't need to output torque,the Propulsion System will require ***SysSigModReq*** = "Standby" to reduce power consumption.
So the asynchronous motor controller will work with Power module in (IGBT or Mosfet etc.) shut off state.

6.3.3.13.5.1.3 Torque Control

250332 v3 — Torque Control mode prerequisites [CS Released]

Purpose: Describe the conditions for the HV Electric Drive to enter Torque control mode, the requests will come from other subsystems.
Verification Method: Test
Legacy ID:

The node shall enter Torque Control mode if and only if all of the following conditions are fulfilled (at the same time):

- (***SysSigModSts*** = Stb OR SpdCtrl OR UDcCtrl) AND (***SysSigModReq*** = TqCtrl)
- ***SysSigHvRlySts*** = Clsd
- ***SysSigTqAllwd*** = (Propulsive torque allowed) OR (Regenerative torque allowed) OR (Propulsive and regenerative torque allowed)
- E2E protection of ***SysSigTqAllwd*** is fulfilled
- ***SysSigCluSts***=Closed If the Electric Drive System is with clutch.

250333 v1 — Torque request [CS Released]

Purpose: Allows the propulsive management to control the torque produced by the electrical machine
Verification Method: Test
Legacy ID:

When the inverter's mode of operation is torque control the actual produced torque shall be controlled to match the requested torque, defined by ***SysSigTqReq***.

250334 v1 — Dynamic and steady state definitions [CS Released]

Purpose: Definition needed for the torque accuracy requirements
Verification Method: Documentation
Legacy ID:

Steady state requirements apply when abs(d/dt torque request) <= 20 Nm/s for > 500 ms.

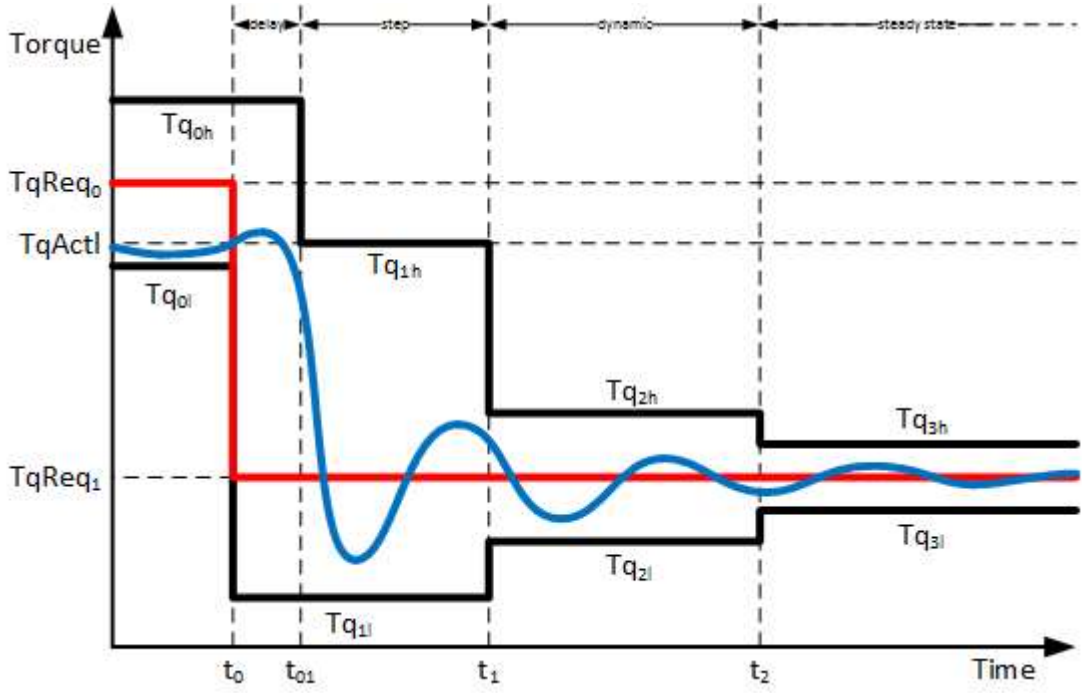
Dynamic requirements apply when abs(d/dt torque request) <= 20 Nm/s for > 50 ms and <= 500 ms.

In other cases the step response requirements apply.

130923 v1 **Negative step response accuracy [CS Released]**

Purpose: Generic Torque Accuracy Requirement
Verification Method: Analysis AND Test
Legacy ID:

When there is a negative step on the torque request the torque on the motor shaft shall be within the boundaries in the graph below (TqxI < TqActI < Tqxh).



$TqReq_0$ = the torque request before the step
$TqReq_1$ = the torque request after the step
TqActI = the actual torque in the domain applicable for the application, as measured with a torque sensor

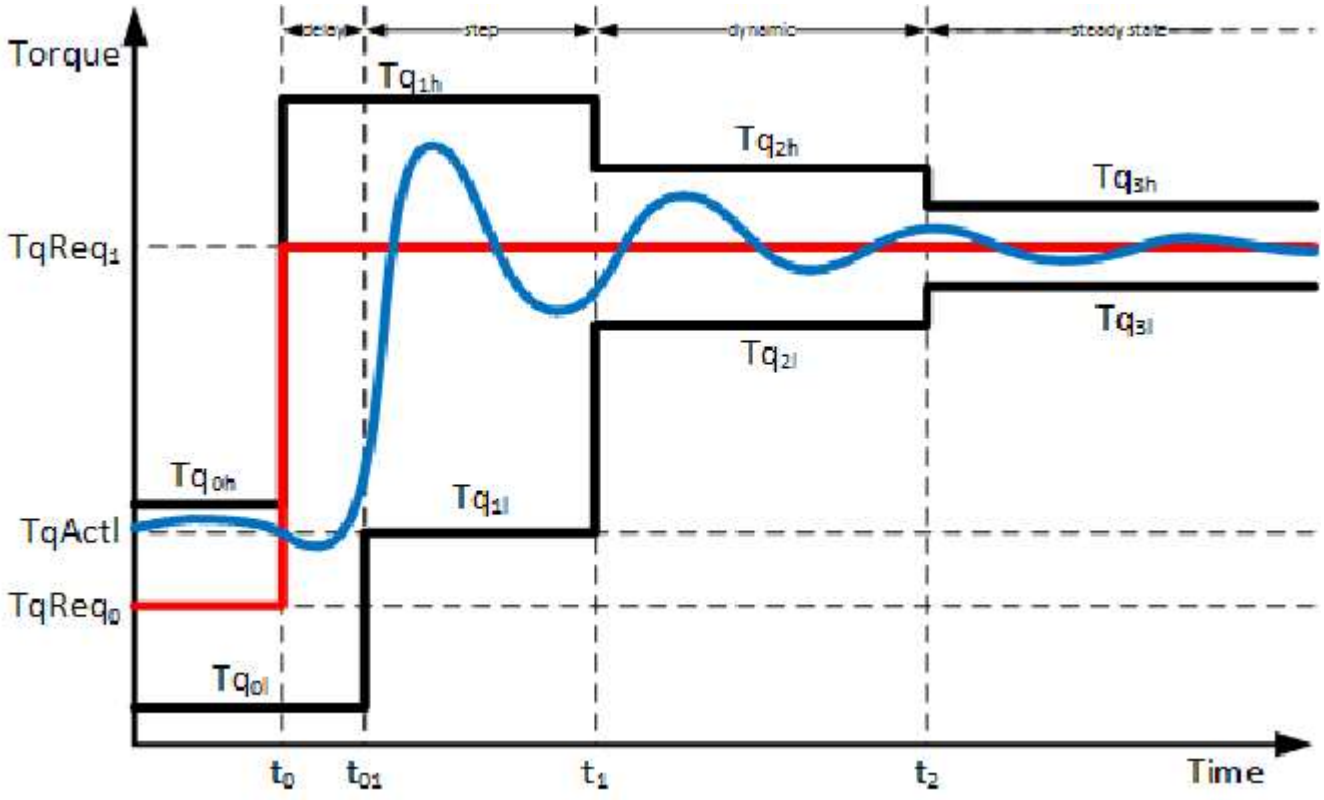
Parameter	Comment
t_0 = the time of the step on CAN Tq_{0I} = defined by conditions before the step Tq_{0h} = defined by conditions before the step	For testing, put system in steady state before the step.
$t_{01} = t_0 + 15$ ms Tq1I = min(TqReq1+0.2*(TqReq1-TqReq0), Tq3I) Tq_{1h} = max(TqActI @ t_0, Tq_{3h})	Delay from torque request signal to actuating part of the software (including sampling time on the signal). Limits can not be harder than the steady state limits.

$t_1 = t_0 + 50$ ms Tq_{2l} = min(dynamic accuracy, Tq_{1l}) Tq_{2h} = max(dynamic accuracy, Tq_{1h})	The time for the step to settle. Between t_1 and t_2 dynamic accuracy limits apply. Limits can not be more generous than the step limits.
$t_2 = t_0 + 500$ ms Tq_{3l} = steady state accuracy Tq_{3h} = steady state accuracy	After t_2 steady state limits apply.

130930 v1 **Positive step response accuracy [CS Released]**

Purpose: Generic Torque Accuracy Requirement
Verification Method: Analysis AND Test
Legacy ID:

When there is a positive step on the requested torque the actual torque on the motor shaft shall be within the boundaries in the graph below (Tq_{xl} < TqActl < Tq_{xh}).



$TqReq_0$ = the torque request before the step

$TqReq_1$ = the torque request after the step

TqActl = the actual torque in the domain applicable for the application, as measured with a torque sensor

Parameter	Comment
t_0 = the time of the step on CAN Tq_{0l} = defined by conditions before the step Tq_{0h} = defined by conditions before the step	For testing, put system in steady state before the step.
$t_{01} = t_0 + 15$ ms Tq_{1l} = min(TqActl @ t_0, Tq_{3l}) Tq_{1h} = max($TqReq_1$+0.2*($TqReq_1$-$TqReq_0$), Tq_{3h})	Delay from torque request signal to actuating part of the software (including sampling time on the signal). Limits can not be harder than the steady state limits.
$t_1 = t_0 + 50$ ms Tq_{2l} = max(dynamic accuracy, Tq_{1l}) Tq_{2h} = min(dynamic accuracy, Tq_{1h})	The time for the step to settle. Between t_1 and t_2 dynamic accuracy limits apply. Limits can not be more generous than the step limits.
$t_2 = t_0 + 500$ ms Tq_{3l} = steady state accuracy Tq_{3h} = steady state accuracy	After t_2 steady state limits apply.

6.3.3.13.5.1.3.1 PMA/SEA

676988 v1 **Torque response slope - PMA/SEA 400V [CS Released]**

Purpose: To define the torque slope for different situation.
Verification Method: Test
Legacy ID:

HVED system shall have different torque slope to handle different situation.

❖ Be able to provide quickest motor shaft **"Torque Drop"** response slope.For vehicle drive stable ,
When TCS active,

- When ***SysSigTqAct*** ＞ ***SysSigPtTqAtAxleMaxReq*** ≥ 0Nm, -10200Nm/s(TBD,according HW ultimate ability) .

When propulsion system request,

- When ***SysSigTqAct*** ＞ ***SysSigTqReq*** ≥ 0Nm,-10200Nm/s(TBD,according HW ultimate ability) ;
- When ***SysSigTqAct*** ＜ ***SysSigTqReq*** ≤ 0Nm,+10200Nm/s(TBD,according HW ultimate ability) .

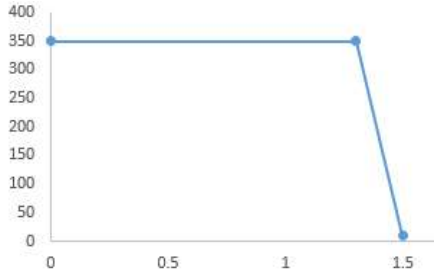
When DTCS or EDTC or Anti-Spin active,

IntTqTarATS : Internal torque target to avoid tire slip,

- When ***SysSigTqAct*** ＞ *IntTqTarATS* ≥ 0Nm,-10200Nm/s(TBD,according HW ultimate ability) ;
- When ***SysSigTqAct*** ＜ *IntTqTarATS* ≤ 0Nm,+10200Nm/s(TBD,according HW ultimate ability) .

Set as the maximum limitation of motor shaft "Torque Drop" response slope.

Example and explanation for **"Torque Drop"**: From +350Nm to +10Nm, From -350Nm to -10Nm.



From +350Nm to +10Nm torque drop response



From -350Nm to -10Nm torque drop response

❖ Be able to provide quickest motor shaft **"Torque Pull-up"** response slope .For vehicle driveability , When propulsion system request,

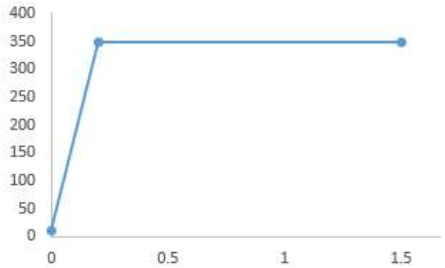
- When ***SysSigTqReq*** ＞ ***SysSigTqAct*** ≥ 0Nm,+2125Nm/s(TBD,according HW ultimate ability) ;
- When ***SysSigTqReq*** ＜ ***SysSigTqAct*** ≤ 0Nm,-2125Nm/s(TBD,according HW ultimate ability) .

When DTCS or EDTC or Anti-Spin active,
IntTqTarATS : Internal torque target,

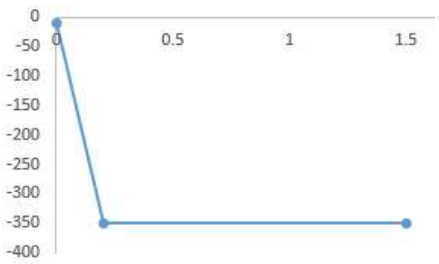
- When *IntTqTarATS* ＞ ***SysSigTqAct*** ≥ 0Nm,+2125Nm/s(TBD,according HW ultimate ability) ;
- When *IntTqTarATS* ＜ ***SysSigTqAct*** ≤ 0Nm,-2125Nm/s(TBD,according HW ultimate ability) .

Set as the maximum limitation of motor shaft "Torque Pull-up" response slope.

Example and explanation for **"Torque Pull-up"**: From +10Nm to +350Nm, From -10Nm to -350Nm.



From +10Nm to +350Nm torque pull-up response



From -10Nm to -350Nm torque pull-up response

❖ Be able to provide quickest motor shaft **"Torque Zero-Cross"** response slope .For vehicle driveability ,

Zero-Cross buffer function maybe omitted during low motor speed. This need ECU owner to check with VCU/ECM(Propulsion system master) whether omit Zero-Cross buffer.
Zero-Cross buffer description: When motor torque in (-20Nm,20Nm) and motor speed in (-1600rpm,1600rpm), motor response slope is low in (50Nm/s, 200Nm/s).

When propulsion system request,

- When ***SysSigTqReq*** ＞ 0Nm ＞ ***SysSigTqAct*** ,
  - From negative torque to 0Nm, +10200Nm/s(TBD,according HW ultimate ability);
  - From 0Nm to positive torque,+2125Nm/s(TBD,according HW ultimate ability) .

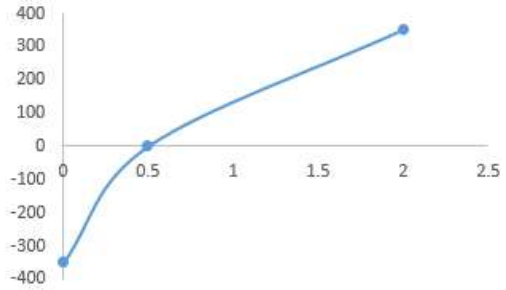
- When ***SysSigTqReq*** < 0Nm < ***SysSigTqAct*** ,
  - From positive torque to 0Nm, -10200Nm/s(TBD,according HW ultimate ability);
  - From 0Nm to negative torque,-2125Nm/s(TBD,according HW ultimate ability) .

When DTCS or EDTC or Anti-Spin active,
IntTqTarATS : Internal torque target,

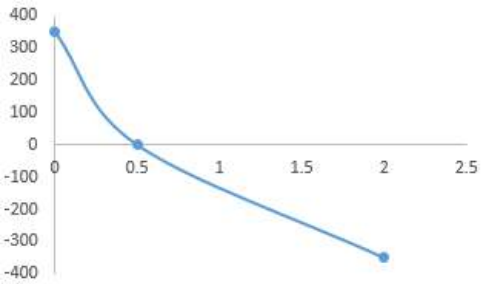
- When *IntTqTarATS* > 0Nm > ***SysSigTqAct*** ,
  - From negative torque to 0Nm, +10200Nm/s(TBD,according HW ultimate ability);
  - From 0Nm to positive torque,+2125Nm/s(TBD,according HW ultimate ability) .
- When *IntTqTarATS* < 0Nm < ***SysSigTqAct*** ,
  - From positive torque to 0Nm, -10200Nm/s(TBD,according HW ultimate ability);
  - From 0Nm to negative torque,-2125Nm/s(TBD,according HW ultimate ability) .

Set as the maximum limitation of motor shaft "Torque Zero-Cross" response slope.

Example and explanation for **"Torque Zero-Cross"**: From -350Nm to +350Nm, From +350Nm to -350Nm.



From -350Nm to +350Nm



From +350Nm to -350Nm

676989 v1 **Torque response slope - PMA/SEA 800V [CS Released]**

Purpose: To define the torque slope for different situation.
Verification Method: Test
Legacy ID:

HVED system shall have different torque slope to handle different situation.

❖ Be able to provide quickest motor shaft **"Torque Drop"** response slope.For vehicle drive stable ,
When TCS active,

- When ***SysSigTqAct*** > ***SysSigPtTqAtAxleMaxReq*** ≥ 0Nm, -10200Nm/s(TBD,according HW ultimate ability) .

When propulsion system request,

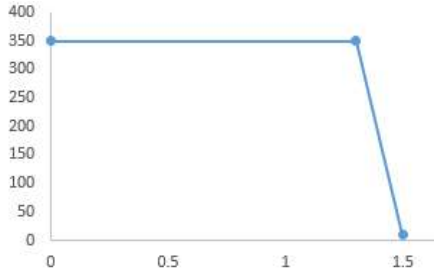
- When ***SysSigTqAct*** > ***SysSigTqReq*** ≥ 0Nm,-10200Nm/s(TBD,according HW ultimate ability) ;
- When ***SysSigTqAct*** < ***SysSigTqReq*** ≤ 0Nm,+10200Nm/s(TBD,according HW ultimate ability) .

When DTCS or EDTC or Anti-Spin active,
IntTqTarATS : Internal torque target to avoid tire slip,

- When ***SysSigTqAct*** > *IntTqTarATS* ≥ 0Nm,-10200Nm/s(TBD,according HW ultimate ability) ;
- When ***SysSigTqAct*** < *IntTqTarATS* ≤ 0Nm,+10200Nm/s(TBD,according HW ultimate ability) .

Set as the maximum limitation of motor shaft "Torque Drop" response slope.

Example and explanation for **"Torque Drop"**: From +350Nm to +10Nm, From -350Nm to -10Nm.



From +350Nm to +10Nm torque drop response



From -350Nm to -10Nm torque drop response

❖ Be able to provide quickest motor shaft **"Torque Pull-up"** response slope .For vehicle driveability , When propulsion system request,

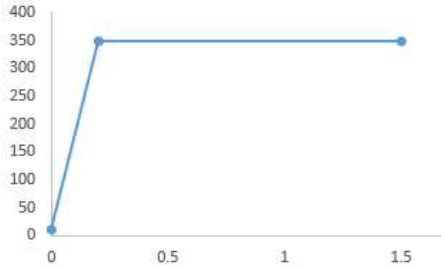
- When ***SysSigTqReq*** ＞ ***SysSigTqAct*** ≥ 0Nm,+4000Nm/s(TBD,according HW ultimate ability) ;
- When ***SysSigTqReq*** ＜ ***SysSigTqAct*** ≤ 0Nm,-4000Nm/s(TBD,according HW ultimate ability) .

When DTCS or EDTC or Anti-Spin active,
IntTqTarATS : Internal torque target,

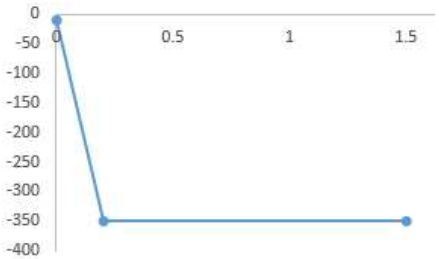
- When *IntTqTarATS* ＞ ***SysSigTqAct*** ≥ 0Nm,+4000Nm/s(TBD,according HW ultimate ability) ;
- When *IntTqTarATS* ＜ ***SysSigTqAct*** ≤ 0Nm,-4000Nm/s(TBD,according HW ultimate ability) .

Set as the maximum limitation of motor shaft "Torque Pull-up" response slope.

Example and explanation for **"Torque Pull-up"**: From +10Nm to +350Nm, From -10Nm to -350Nm.



From +10Nm to +350Nm torque pull-up response



From -10Nm to -350Nm torque pull-up response

❖ Be able to provide quickest motor shaft **"Torque Zero-Cross"** response slope .For vehicle driveability ,

Zero-Cross buffer function maybe omitted during low motor speed. This need ECU owner to check with PCMU(Propulsion system master) whether omit Zero-Cross buffer.
Zero-Cross buffer description: When motor torque in (-20Nm,20Nm) and motor speed in (-1600rpm,1600rpm), motor response slope is low in (50Nm/s, 200Nm/s).

When propulsion system request,

- When ***SysSigTqReq*** ＞ 0Nm ＞ ***SysSigTqAct*** ,
  - From negative torque to 0Nm, +10200Nm/s(TBD,according HW ultimate ability);
  - From 0Nm to positive torque,+4000Nm/s(TBD,according HW ultimate ability) .
- When ***SysSigTqReq*** ＜ 0Nm ＜ ***SysSigTqAct*** ,
  - From positive torque to 0Nm, -10200Nm/s(TBD,according HW ultimate ability);
  - From 0Nm to negative torque,-4000Nm/s(TBD,according HW ultimate ability) .

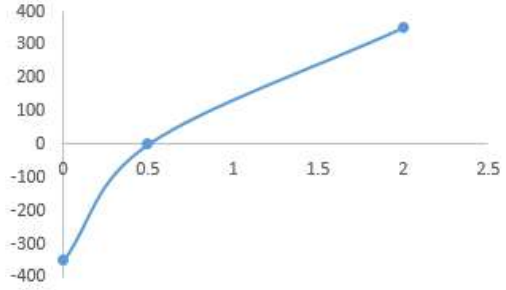
When DTCS or EDTC or Anti-Spin active,

IntTqTarATS : Internal torque target,

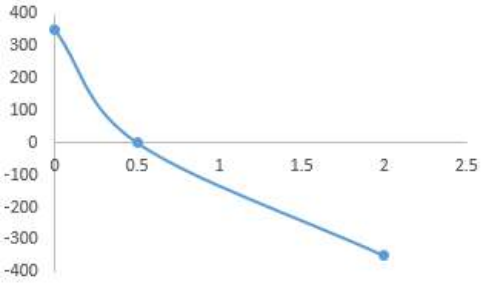
- When *IntTqTarATS* ＞ 0Nm ＞ ***SysSigTqAct*** ,
  - From negative torque to 0Nm, +10200Nm/s(TBD,according HW ultimate ability);
  - From 0Nm to positive torque,+4000Nm/s(TBD,according HW ultimate ability) .
- When *IntTqTarATS* ＜ 0Nm ＜ ***SysSigTqAct*** ,
  - From positive torque to 0Nm, -10200Nm/s(TBD,according HW ultimate ability);
  - From 0Nm to negative torque,-4000Nm/s(TBD,according HW ultimate ability) .

Set as the maximum limitation of motor shaft "Torque Zero-Cross" response slope.

Example and explanation for **"Torque Zero-Cross"**: From -350Nm to +350Nm, From +350Nm to -350Nm.



From -350Nm to +350Nm



From +350Nm to -350Nm

685557 v2 **Torque request 800V Advanced [CS Released]**

Purpose: Allows the propulsive management to control the torque produced by the electrical machine.
Advanced requirement aim to get more accurate data and enough torque during accelerating.

Verification Method: Test

Legacy ID:

When the inverter's mode of operation is torque control the actual produced torque shall be controlled to match the requested torque, defined by ***SysSigTqReq***.

SysSigTqReq from propulsion is the actual wanted torque request on the reference point (Wheel Axle for PMA) .

HVED system shall consider all losses to calculate the motor torque request.

PMA/SEA 800V :

- **When request output propulsive torque:**

Motor Axle torque request
= Wheel Axle torque request / Gear ratio /(Torque accuracy Low limit * Transmission efficiency)

- **When request output regenerate torque:**

Motor Axle torque request
= Wheel Axle torque request / Gear ratio * (Torque accuracy Low limit * Transmission efficiency)

For example:

If request output propulsive torque:
Wheel Axle torque request = 3000Nm , Gear ratio = 10.33 , Torque accuracy Low limit = 100% - 3% = 97% , Transmission efficiency = 97%.
So
Motor Axle torque request = 3000/10.33/(97%*97%) Nm = 309 Nm.

If request output regenerate torque:
Wheel Axle torque request = 3000Nm , Gear ratio = 10.33 , Torque accuracy Low limit = 100% - 3% = 97% , Transmission efficiency = 97%.
So
Motor Axle torque request = 3000/10.33*(97%*97%) Nm = 273Nm.

685558 v1 Torque request E2E protection 800V Advanced [CS Released]

Purpose: E2E protect
Verification Method: Test
Legacy ID:

When the inverter's mode of operation is torque control the actual produced torque shall be controlled to match the requested torque, defined by ***SysSigTqReq***.

SysSigTqReq from propulsion have E2E protection .

HVED system shall confirm if E2E of ***SysSigTqReq*** is ok.

If E2E have fault less than max(250ms,5 frame), keep previous value;
If E2E have fault more than or equal to max(250ms,5 frame), enter to fault handling state according to SSDS.

685559 v2 Torque request Over range [CS Released]

Purpose: For drive safety, if propulsion requested torque value over range, HVED shall keep in safe Zero torque state.

Verification Method: Test
Legacy ID:

When the inverter's mode of operation is torque control the actual produced torque shall be controlled to match the requested torque, defined by ***SysSigTqReq***.

SysSigTqReq from propulsion is the actual wanted torque request on the reference point (Axle for PMA) .

HVED system shall consider if ***SysSigTqReq*** from propulsion is in valid range.

If ***SysSigTqReq*** ＞ (100%+10%) *Maximum value of ***SysSigTqAvl::MaxAvailTorq*** ,enter to Zero torque state.
If ***SysSigTqReq*** ＜ (100%+10%) *Minimum value of ***SysSigTqAvl::MinAvailTorq*** ,enter to Zero torque state.

Note:
The Maximum value of "SysSigTqAvl" refers to the maximum torque capacity of HVED without faults, derating, and operating in the constant torque zone of external characteristics.
The Minimum value of "SysSigTqAvl" refers to the minimum torque capacity of HVED without faults, derating, and operating in the constant torque zone of external characteristics.

6.3.3.13.5.1.4 Speed Control

250344 v2 **Speed request [CS Released]**

Purpose: Allows the propulsive management to control the speed produced by the electrical machine
Verification Method: Test
Legacy ID:

When the inverter is in Speed Control mode the inverter shall control the torque so that the EM speed matches value given on port ***SysSigSpdReq***.

particular case

In the project of the rear motor + rear gearbox, the rear motor needs to receive transmission subsystem sended mode request SysSigTrsmWhlMotSysModReq , and in 200 ms (TBD) within the motor speed adjustment to the target speed SysSigTrsmWhlMotSysSpdReq, When the rear motor speed reaches the target speed (Max deviation is ±50rpm, TBD), the speed control mode will automatically exit and respond to the mode request(SysSigModReq) of the Propulsion subsystem
在后电机+后变速箱项目中，后电机需要接收子TransmissionControl子系统发送的转速模式请求 SysSigTrsmWhlMotSysModReq=SpdCtrl，并在200ms (TBD)内将电机转速调整到目标转速 SysSigTrsmWhlMotSysSpdReq，当后电机转速达到目标转速(最大偏差为±50rpm, TBD)时，转速控制模式将自动退出，并响应Propulsion子系统的模式请求(SysSigModReq)

250345 v1 **Speed Control mode prerequisites [CS Released]**

Purpose: Describe the conditions for the HV Electric Drive to enter Speed control mode, the requests will come from other subsystems.
Verification Method: Test
Legacy ID:

The node shall enter Speed control mode if and only if all of the following conditions are fulfilled (at the same time):

- (***SysSigModSts*** = Stb OR TqCtrl) AND (***SysSigModReq*** = SpdCtrl)
- ***SysSigHvRlySts*** = Clsd
- ***SysSigTqAllwd*** = (Propulsive torque allowed) OR (Regenerative torque allowed) OR (Propulsive and regenerative torque allowed)
- E2E protection of ***SysSigTqAllwd*** is fulfilled

250374 v1 **Speed control - control delay [CS Released]**

Purpose: Maintaining speed synchronization towards a dynamically changing target.
Verification Method: Test
Legacy ID:

If the speed control set point is updated with a derivative with an absolute value less than *SpdCtrlSetPointDynamicChangeRate* when the speed control is in steady state, the speed control shall return to steady state again within *SpdCtrlSetPointDelay*.

The speed control is considered to be in steady state when:

SysSigModSts = *SpdCtrl* and the absolute value of the speed difference between the estimated EM speed and the speed control set-point is less than *SpdCtrlSteadyStateErr*

Note:

*The offset speed value is a drivability and gear shift performance parameter which is set together with transmission control through calibration.

Calibration target

SpdCtrlSetPointDynaminChangeRate corresponds to the worst realistic rate of change in speed-set point that the controller shall be able to handle. The value comes from worst case acceleration/decerelleration scenarios of the vehicle when considering total speed ratio from wheels to electric machine reference point.

SpdCtrlSetPointDelay corresponds to a short enough time for the overall control sequence to be able to handle intermediate lost synchronization and must be aligned with the relevant signal latency for the speed-set point.

134137 v2 Speed control - speed range limitation [CS Released]

Purpose: To avoid unnecessary calibration dependencies between speed limitation and speed control
Verification Method: Test
Legacy ID:

The speed range of the EM shall be limited by EM speed limitation to a speed less than ± *EMSpeedLimitMax*.

Speed control must behave as if the speed set point was limited to this speed range without introducing unexpected behaviour if the external set point is set outside the speed range.

Note:

Unexpected behaviour could include insufficient anti-windup to handle integrated correction components from trying to realize a set point outside the range.

6.3.3.13.5.1.5 Voltage Control

250564 v2 Voltage control request [CS Released]

Purpose: Define by which subsystem the HVElectricDrive is requested to enter Voltage control mode
Verification Method: Test
Legacy ID:

The propulsion system will request for Voltage control via setting ***SysSigModReq*** to UDcCtrl and the

HVElectricDrive shall enter voltage control mode.

250575 v1 **Voltage control setpoint [CS Released]**

Purpose: Defines how to get a correct setpoint to control the voltage vs in the transition phase
Verification Method: Analysis AND Test
Legacy ID:

The HVElectricDriveSystem shall set the control setpoint for the voltage according to the following requirements after receiving a request to go to Voltage control mode (or when in Voltage control mode):

As long as the the HV-battery contactor status, ***SysSigHvRlySts***, is "closed" the local HVDC voltage, at the inverter shall be used as set point voltage for voltage control (i.e. a constant, low, torque shall be applied).

When the HV-battery contactor status, ***SysSigHvRlySts*** "open" and the inverter operates in voltage control, the set point voltage for voltage control shall be set to *UdcCtrlSetPointFix (calibratable) and* the inverter shall apply torque to control the HVDC voltage to the set point.

250576 v1 **Voltage control status report [CS Released]**

Purpose: Define how to communicate the active operational mode
Verification Method: Test
Legacy ID:

When the HVElectricDrive system is in Voltage control mode it shall report UDcCtrl in signal **SysSigModSts**.

6.3.3.13.5.1.6 Fault/Error

250577 v3 **Fault Mode [CS Released]**

Purpose: In case of failure in inverter a failure mode shall be entered.

Verification Method: Test
Legacy ID:

In failure mode the inverter shall keep active short circuit or freewheeling.
The inverter shall remain in fault state as long as at least one of the fault on Fault Mode Triggers is fulfilled. If all fault are healed according to fault matrix it is OK to leave Fault Mode.
Signal ***SysSigModSts*** = fault state

118495 v3 **Fault Mode Trigger [CS Released]**

Purpose: To define which fault will trigger fault mode
Verification Method: Analysis AND Test
Legacy ID:

The supplier shall provide a fault matrix showing all faults that shall lead to fault mode.

123249 v1 **Fault Control [CS Released]**

Purpose: Protect the HV Electric Drive System itself.

Verification Method: Test
Legacy ID:

The HV Electric Drive System shall protect itself from damage by having fault control functionality in the SW.

The HV Electric Drive System shall enter the fault control (Failure Mode) when the HV Electric Drive System and HV System occurs fault.

The Fault Matrix will be agreed on between the supplier and GEEA Project Group.

6.3.3.13.5.1.7 Powerdown

118477 v2 **Communication after CPSR goes low [CS Released]**

Purpose: Allow diagnostic functions in other ECUs to shutdown before the inverter stops communication.
Verification Method: Test
Legacy ID:

The inverter shall communicate on CAN at least *CanNmTimeoutTime* after the shutdown event (info: prepare bus-sleep mode in AUTOSAR).
Communication shall be stopped at latest *CanNmTimeoutTime + CanNmWaitBusSleepTime* after the shutdown event (info: bus-sleep mode in AUTOSAR).

97494 v2 **Shutdown event [CS Released]**

Purpose: Describe how the node shall receive shutdown event in PNC concept.
Verification Method: Test
Legacy ID: 396453

The shut down event is defined as when:

- The node is going to "Prepare Bus-Sleep Mode" after getting no PNC CAN frame with any of the bits the node shall listen to set for CanNmTimeoutTime

Note: Details on Prepare Bus-SleepMode and CanNmTimeoutTime can be found in Autosar Network Management

97508 v3 **Shutdown timing [CS Released]**

Purpose: To shut down EM in a safe way
Verification Method: Test
Legacy ID: 401392

All modules on the Propulsion CAN that perform linked (via CAN) functions after key-off during vehicle Power-down [BECM, IEM, IGM/DCDC and OBC], must continue to broadcast CAN messages at least until 200ms after CPSR power is off.

123940 v1 Abort shutdown [CS Released]

Purpose: In case the driver changes the request from key off to key on the HV electric drive shall abort the shutdown process to speed up the startup sequence
Verification Method: Test
Legacy ID:

If a startup event occurs during powerdown but before the powerdown is completed it shall abort and restart as a quick restart during initalization but the test results of initialization must consider if there has been any errors detected during operation since last initialization.

250580 v2 Shutdown actions [CS Released]

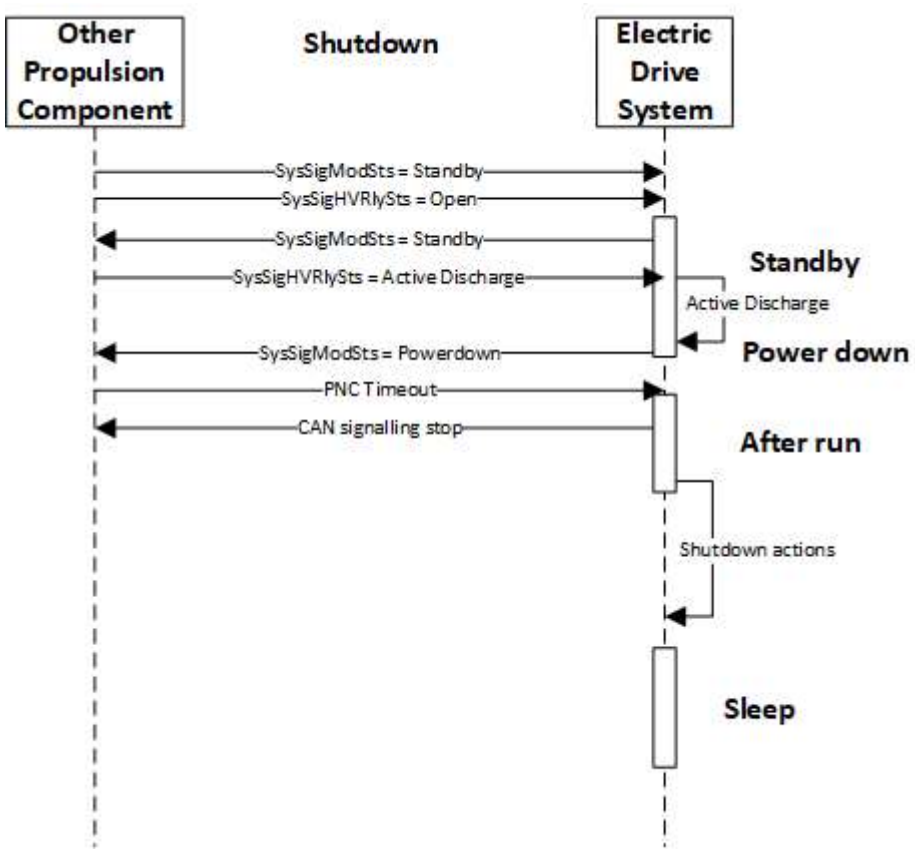
Purpose: Inverter needs to peform actions before it powers off itself during an shutdown event both to fulfill the system requirements as well as being able to startup from a well known status again.
Verification Method: Test
Legacy ID:

The following stated actions shall be started after the communication of the inverter has stopped and finish before the inverter enters sleep:

- Activities needed for Standby fulfillment (torque production disabled and residual, actively produced, torque removed)
- If active discharge is requested or ongoing then the active discharge shall be finalized
- Update non-volatile memory (if needed)
- Enter Immobilized state

Note:

Immobilizer is depending on subsystem Usage Authorization.



6.3.3.13.5.1.8 Sync Control

Sync Control requirements are only appliacble for EDS together with 7DCTH

252628 v2 **Sync control mode prerequisites [CS Released]**

Purpose: Describe the conditions for the HV Electric Drive to enter Sync control mode, the requests will come from other subsystems.
Verification Method: Test
Legacy ID:

The node shall enter SyncCtrl mode if and only if all of the following conditions are fulfilled (at the same time):

- (***SysSigModSts*** = Stb OR SpdCtrl OR UDcCtrl) AND (***SysSigModReq***= PreChrg)
- ***SysSigHvRlySts*** = Clsd
- ***SysSigTqAllwd*** = (Propulsive torque allowed) OR (Regenerative torque allowed) OR (Propulsive and regenerative torque allowed)
- E2E protection of ***SysSigTqAllwd*** is fulfilled

Note: This requirement is only valid for 7DCT gearbox.

252629 v1 **Sync control [CS Released]**

Purpose: Sync control is used to sync speed during gear shifts.
Verification Method: Test
Legacy ID:

When receiving an mode request for *sync control mode from ECM*, the Electric Drive System shall:
Control the torque so that *Estimated EM speed* matches *requested EM speed from EC*M.
The torque request shall be modified with limitations as specified in separate requirement Torque prioritizations.

Time from fulfilling the conditions to follow the request until the difference between estimated EM speed and requested EM speed is less than *SyncSetCtrlMrgn* and the software sends out speed control mode shall be less than the time specified by *SyncSetCtrlTiMax*.

Note: This requirement is only valid for 7DCT gearbox.

6.3.3.13.5.1.9 Traction Control

250583 v2 — Traction Control request [CS Released]

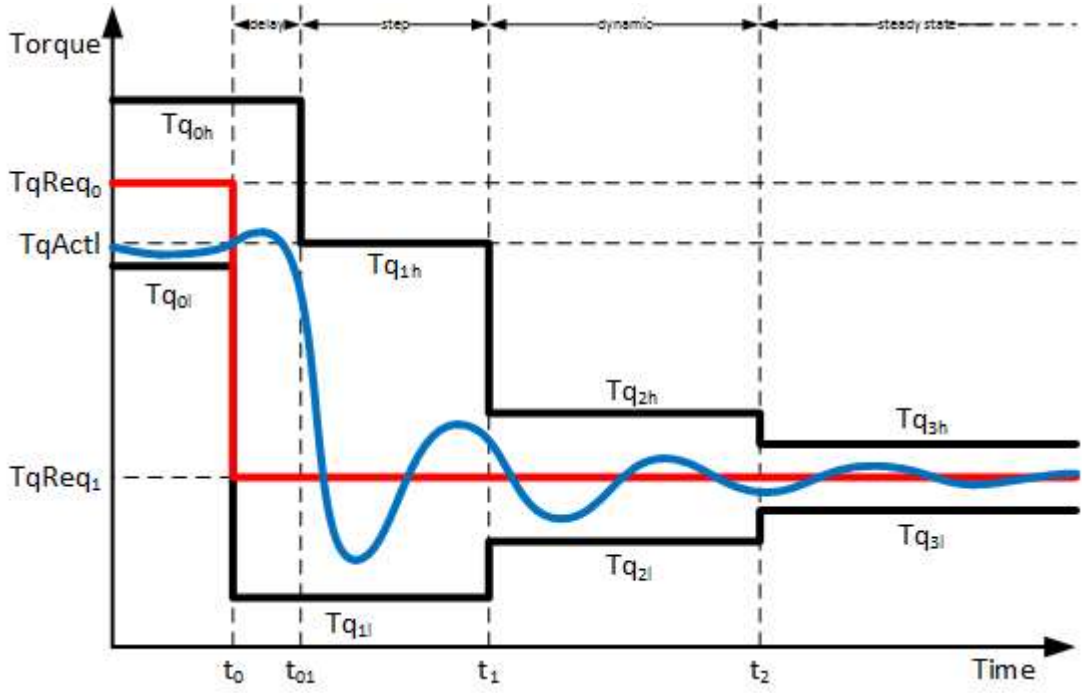
Purpose: Allows the traction control function to limit the torque produced by the electrical machine during slip
Verification Method: Test
Legacy ID:

When the inverter's mode of operation is TcsCtrl the actual produced torque shall be controlled to not surpass the torque limit on ***SysSigPtTqAtAxleMaxReq*** (PtTqAtAxleFrntReq)

130923 v1 — Negative step response accuracy [CS Released]

Purpose: Generic Torque Accuracy Requirement
Verification Method: Analysis AND Test
Legacy ID:

When there is a negative step on the torque request the torque on the motor shaft shall be within the boundaries in the graph below (Tqxl < TqActl < Tqxh).



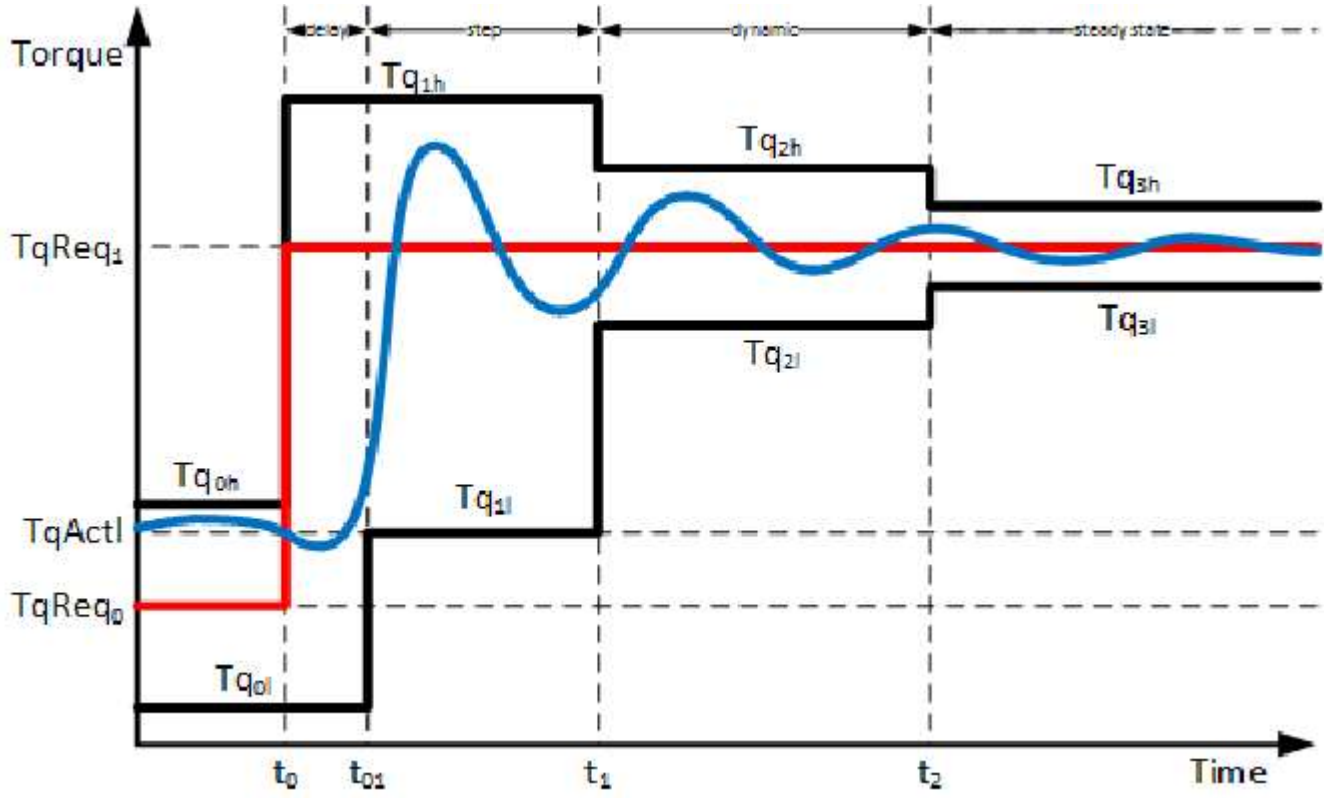
$TqReq_0$ = the torque request before the step
$TqReq_1$ = the torque request after the step
TqActl = the actual torque in the domain applicable for the application, as measured with a torque sensor

Parameter	Comment
t_0 = the time of the step on CAN Tq_{0l} = defined by conditions before the step Tq_{0h} = defined by conditions before the step	For testing, put system in steady state before the step.
$t_{01} = t_0 + 15$ ms Tq1l = min(TqReq1+0.2*(TqReq1-TqReq0), Tq3l) Tq_{1h} = max(TqActl @ t_0, Tq_{3h})	Delay from torque request signal to actuating part of the software (including sampling time on the signal). Limits can not be harder than the steady state limits.
$t_1 = t_0 + 50$ ms Tq_{2l} = min(dynamic accuracy, Tq_{1l}) Tq_{2h} = max(dynamic accuracy, Tq_{1h})	The time for the step to settle. Between t_1 and t_2 dynamic accuracy limits apply. Limits can not be more generous than the step limits.
$t_2 = t_0 + 500$ ms Tq_{3l} = steady state accuracy Tq_{3h} = steady state accuracy	After t_2 steady state limits apply.

130930 v1 **Positive step response accuracy [CS Released]**

Purpose: Generic Torque Accuracy Requirement
Verification Method: Analysis AND Test
Legacy ID:

When there is a positive step on the requested torque the actual torque on the motor shaft shall be within the boundaries in the graph below (Tq_{xl} < TqActl < Tq_{xh}).



$TqReq_0$ = the torque request before the step

$TqReq_1$ = the torque request after the step

TqActl = the actual torque in the domain applicable for the application, as measured with a torque sensor

Parameter	Comment
t_0 = the time of the step on CAN Tq_{0l} = defined by conditions before the step Tq_{0h} = defined by conditions before the step	For testing, put system in steady state before the step.
$t_{01} = t_0 + 15$ ms	Delay from torque request signal to actuating part of the software (including sampling time on the signal).

Tq_{1l} = min(TqActl @ t_0, Tq_{3l}) Tq_{1h} = max($TqReq_1$+0.2*($TqReq_1$-$TqReq_0$), Tq_{3h})	Limits can not be harder than the steady state limits.
$t_1 = t_0$ + 50 ms Tq_{2l} = max(dynamic accuracy, Tq_{1l}) Tq_{2h} = min(dynamic accuracy, Tq_{1h})	The time for the step to settle. Between t_1 and t_2 dynamic accuracy limits apply. Limits can not be more generous than the step limits.
$t_2 = t_0$ + 500 ms Tq_{3l} = steady state accuracy Tq_{3h} = steady state accuracy	After t_2 steady state limits apply.

250584 v1 **Traction Control Prerequisites [CS Released]**

Purpose: Describe the conditions for the HV Electric Drive to enter Traction control mode, the requests will come from other subsystems.
Verification Method: Test
Legacy ID:

The node shall enter Traction Control mode if and only if all of the following conditions are fulfilled (at the same time):

- ***SysSigPtTqAtAxleMaxReq*** = 0 (PtTqAtAxleFrntReq if installed in front, PtTqAtAxleRearReq if installed in rear)
- ***SysSigModReq*** = TqCtrl, SpdCtrl OR UDcCtrl
- ***SysSigModSts =*** TqCtrl, SpdCtrl OR UDcCtrl
- ***SysSigHvRlySts*** = Clsd
- ***SysSigTqAllwd*** = (Propulsive torque allowed) OR (Regenerative torque allowed) OR (Propulsive and regenerative torque allowed)
- E2E protection of ***SysSigTqAllwd*** is fulfilled

250590 v1 **Leaving Traction Control [CS Released]**

Purpose: To clarify which mode to go to after a Traction Control event
Verification Method: Test
Legacy ID:

The Electric Drive System shall go the the mode requested on ***SysSigModReq*** when the conditions for TcsCtrl no longer are fullfilled.

1045438 v1 Traction Control Prerequisites Max and Min [CS Released]

Purpose: Describe the conditions for the HV Electric Drive to enter Traction control mode, the requests will come from other subsystems.

Verification Method: Test

Legacy ID:

TCS	Traction Control System 牵引力控制系统
EDC	E-Motor Drag Control_EDC 电机拖曳控制 （极氪/博世 叫法）
EDTR	E-Motor Drag Torque Restrain 电机拖曳扭矩抑制 （采埃孚 叫法）

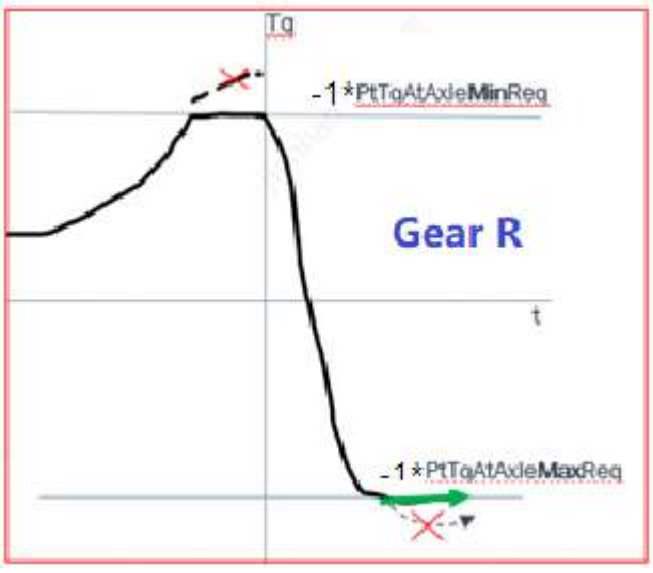
Represented Signal in Requirement	CAN Signal	Description
SysSigPtTqAtAxleMinReq	PtTqAtAxleMinReq ***::PtTqAtAxleFrntReq***	-6000~0 Nm (-6001~-19999 define invalid value,-20000 define as default value) For Front Motor **(MGM/XCM2)** Minimum regenerative wheel torque allowed by Brake Control.
	PtTqAtAxleMinReq ***::PtTqAtAxleReReq***	-6000~0 Nm(-6001~-19999 define invalid value,-20000 define as default value) For Rear Motor **(IEM3/XCUP)** Minimum regenerative wheel torque allowed by Brake Control.
	PtTqAtWhlMinReq ***::FrntLeMinReq***	-6000~0 Nm (-6001~-19999 define invalid value,-20000 define as default value) For Front left motor Minimum regenerative wheel torque allowed by Brake Control..
	PtTqAtWhlMinReq ***::FrntRiMinReq***	-6000~0 Nm(-6001~-19999 define invalid value,-20000 define as default value) For Front right Motor Minimum regenerative wheel torque allowed by Brake Control.
	PtTqAtWhlMinReq ***::ReLeMinReq***	-6000~0 Nm (-6001~-19999 define invalid value,-20000 define as default value) For Rear left motor **(IEM2XCUP2)** Minimum regenerative wheel torque allowed by Brake Control.
	PtTqAtWhlMinReq ***::ReRiMinReq***	-6000~0 Nm(-6001~-19999 define invalid value,-20000 define as default value) For Rear right Motor **(IEM2/XCUP2)** Minimum regenerative wheel torque allowed by Brake Control.
SysSigPtTqAtAxleMaxReq	PtTqAtAxleMaxReq ***::PtTqAtAxleFrntReq***	0~6000 Nm(6001~19999 define invalid value,20000 define as default value) For Front Motor **(MGM/XCM2)** Maximum Propulsive wheel torque allowed by Brake Control.
	PtTqAtAxleMaxReq ***::PtTqAtAxleReReq***	0~6000 Nm(6001~19999 define invalid value,20000 define as default value) For Rear Motor **(IEM3/XCUP)** Maximum Propulsive wheel torque allowed by Brake Control.
	PtTqAtWhlMaxReq ***::FrntLeMaxReq***	0~6000 Nm(6001~19999 define invalid value,20000 define as default value) For Front left Motor Maximum Propulsive wheel torque allowed by Brake

		Control.
	PtTqAtWhlMaxReq ***::FrntRiMaxReq***	0~6000 Nm(6001~19999 define invalid value,20000 define as default value) For Front right Motor Maximum Propulsive wheel torque allowed by Brake Control.
	PtTqAtWhlMaxReq ***::ReLeMaxReq***	0~6000 Nm(6001~19999 define invalid value,20000 define as default value) For Rear left Motor **(IEM2/XCUP2)** Maximum Propulsive wheel torque allowed by Brake Control.
	PtTqAtWhlMaxReq ***::ReRiMaxReq***	0~6000 Nm(6001~19999 define invalid value,20000 define as default value) For Rear right Motor **(IEM2/XCUP2)** Maximum Propulsive wheel torque allowed by Brake Control.

Four Situation:

GearLvrIndcn	Function	Signal change from Brake Control	E-motor shall
Gear D	TCS	***SysSigPtTqAtAxleMax Req*** reduce from 20000 to *DTcsTar*	Limit ***SysSigTqAct*** less than *DTcsTar*
Gear R	TCS	***SysSigPtTqAtAxleMax Req*** reduce from 20000 to *RTcsTar*	Limit ***SysSigTqAct*** more than (-1)**RTcsTar*
Gear D	EDTR	***SysSigPtTqAtAxleMin Req*** reduce from -20000 to *DEdtrTar*	Limit ***SysSigTqAct*** more than *DEdtrTar*
Gear R	EDTR	***SysSigPtTqAtAxleMin Req*** reduce from -20000 to *REdtrTar*	Limit ***SysSigTqAct*** less than (-1)* *REdtrTar*





❖ HVED shall enter Traction Control (TCS) mode(Send ***SysSigMotSts*** = TcsCtrl) ,if all of the following conditions are fulfilled (at the same time):

- ***GearLvrIndcn*** = D;
- For Each motor: 0≤***SysSigPtTqAtAxleMaxReq***≤6000
- For Each motor: ***SysSigPtTqAtAxleMaxReq*** < ***SysSigTqReq***;
- E2E of ***SysSigPtTqAtAxleMaxReq*** no fault;
- ***SysSigModSts*** = TqCtrl;
- ***SysSigHvRlySts*** = Clsd;
- ***SysSigTqAllwd*** = (Propulsive torque allowed) OR (Regenerative torque allowed) OR (Propulsive and regenerative torque allowed);
- E2E protection of ***SysSigTqAllwd*** is fulfilled.

When enter Traction Control mode in Gear D, HVED target torque output shall follow ***SysSigPtTqAtAxleMaxReq*** .

❖ HVED shall enter Traction Control (TCS) mode(Send ***SysSigMotSts*** = TcsCtrl) ,if all of the following conditions are fulfilled (at the same time):

- ***GearLvrIndcn*** = R;
- For Each motor: 0≤***SysSigPtTqAtAxleMaxReq***≤6000
- For Each motor: (-1)****SysSigPtTqAtAxleMaxReq*** > ***SysSigTqReq***;
- E2E of ***SysSigPtTqAtAxleMaxReq*** no fault;
- ***SysSigModSts =*** TqCtrl;
- ***SysSigHvRlySts*** = Clsd;
- ***SysSigTqAllwd*** = (Propulsive torque allowed) OR (Regenerative torque allowed) OR (Propulsive and regenerative torque allowed);
- E2E protection of ***SysSigTqAllwd*** is fulfilled.

When enter Traction Control mode in Gear R, HVED target torque output shall follow (-1)****SysSigPtTqAtAxleMaxReq*** .

❖ HVED shall enter Drag Control (EDTR) mode(Send ***SysSigMotSts*** = TcsCtrl) and if all of the following conditions are fulfilled (at the same time):

- ***GearLvrIndcn*** = D;
- For Each motor: -6000≤***SysSigPtTqAtAxleMin***≤0
- For Each motor: ***SysSigPtTqAtAxleMin*** > ***SysSigTqReq***;
- E2E of ***SysSigPtTqAtAxleMinReq*** no fault;
- ***SysSigModSts*** = TqCtrl;
- ***SysSigHvRlySts*** = Clsd;
- ***SysSigTqAllwd*** = (Propulsive torque allowed) OR (Regenerative torque allowed) OR (Propulsive and regenerative torque allowed);
- E2E protection of ***SysSigTqAllwd*** is fulfilled.

When enter Drag Control mode in Gear D, HVED target torque output shall follow ***SysSigPtTqAtAxleMinReq*** .

❖ HVED shall enter Drag Control (EDTR) mode(Send ***SysSigMotSts*** = TcsCtrl) and if all of the following conditions are fulfilled (at the same time):

- ***GearLvrIndcn*** = R;
- For Each motor: -6000≤***SysSigPtTqAtAxleMinReq***≤0
- For Each motor: (-1)****SysSigPtTqAtAxleMinReq*** < ***SysSigTqReq***;
- E2E of ***SysSigPtTqAtAxleMinReq*** no fault;
- ***SysSigModSts*** = TqCtrl;
- ***SysSigHvRlySts*** = Clsd;
- ***SysSigTqAllwd*** = (Propulsive torque allowed) OR (Regenerative torque allowed) OR (Propulsive and regenerative torque allowed);

- E2E protection of ***SysSigTqAllwd*** is fulfilled.

When enter Drag Control mode in Gear R, HVED target torque output shall follow (-1)*<u>***SysSigPtTqAtAxleMinReq***</u> .

❖ HVED shall handle over range fault of TCS or EDTR request,:

- If <u>***SysSigPtTqAtAxleMaxReq***</u>＜0,
  - set :<u>***SysSigPtTqAtAxleMaxReq***</u> = 20000Nm ;
  - 或者 认为<u>***SysSigPtTqAtAxleMaxReq***</u> 为异常值，不响应。
- If <u>***SysSigPtTqAtAxleMaxReq***</u>＜0,
  - set :<u>***SysSigPtTqAtAxleMaxReq***</u> = 20000Nm;
  - 或者 认为<u>***SysSigPtTqAtAxleMaxReq***</u> 为异常值，不响应。
- If <u>***SysSigPtTqAtAxleMinReq***</u>＞0,
  - set :<u>***SysSigPtTqAtAxleMinReq***</u> = -20000Nm;
  - 或者 认为<u>***SysSigPtTqAtAxleMinReq***</u> 为异常值，不响应。
- If<u>***SysSigPtTqAtAxleMinReq***</u>＞0,
  - set :<u>***SysSigPtTqAtAxleMinReq***</u> = -20000Nm.
  - 或者 认为<u>***SysSigPtTqAtAxleMinReq***</u> 为异常值，不响应。

❖ HVED shall handle E2E fault or Lose communication fault of TCS or EDTR request,:

- If E2E fault or Lose communication fault of <u>***SysSigPtTqAtAxleMaxReq***</u>,
  - set :<u>***SysSigPtTqAtAxleMaxReq***</u> = 20000Nm;
  - 或者 认为<u>***SysSigPtTqAtAxleMaxReq***</u> 为异常值，不响应。
- If E2E fault or Lose communication fault of <u>***SysSigPtTqAtAxleMinReq***</u>,
  - set:<u>***SysSigPtTqAtAxleMinReq***</u> = -20000Nm;
  - 或者 认为<u>***SysSigPtTqAtAxleMinReq***</u> 为异常值，不响应。

1045439 v1 **Traction Control and DTCS/EDTC [CS Released]**

Purpose: Traction Control and DTCS/EDTC
Verification Method: Test
Legacy ID:

Represented Signal in Requirement	CAN Signal	Description
<u>***SysSigPtTqAtAxleMinReq***</u>	PtTqAtAxleMinReq <u>***::PtTqAtAxleFrntReq***</u>	-6000~0 Nm (-6001~-19999 define invalid value,-20000 define as default value) For Front Motor **(MGM/XCM2)** Minimum regenerative wheel torque allowed by Brake Control.
	PtTqAtAxleMinReq <u>***::PtTqAtAxleReReq***</u>	-6000~0 Nm(-6001~-19999 define invalid value,-20000 define as default value) For Rear Motor **(IEM3/XCUP)** Minimum regenerative wheel torque allowed by Brake Control.
	PtTqAtWhlMinReq <u>***::FrntLeMinReq***</u>	-6000~0 Nm (-6001~-19999 define invalid value,-20000 define as default value) For Front left motor Minimum regenerative wheel torque allowed by Brake Control..
	PtTqAtWhlMinReq <u>***::FrntRiMinReq***</u>	-6000~0 Nm(-6001~-19999 define invalid value,-20000 define as default value) For Front right Motor Minimum regenerative wheel torque allowed by

		Brake Control.
	PtTqAtWhlMinReq ***::ReLeMinReq***	-6000~0 Nm (-6001~-19999 define invalid value,-20000 define as default value) For Rear left motor **(IEM2XCUP2)** Minimum regenerative wheel torque allowed by Brake Control.
	PtTqAtWhlMinReq ***::ReRiMinReq***	-6000~0 Nm(-6001~-19999 define invalid value,-20000 define as default value) For Rear right Motor **(IEM2/XCUP2)** Minimum regenerative wheel torque allowed by Brake Control.
SysSigPtTqAtAxleMaxReq	PtTqAtAxleMaxReq ***::PtTqAtAxleFrntReq***	0~6000 Nm(6001~19999 define invalid value,20000 define as default value) For Front Motor **(MGM/XCM2)** Maximum Propulsive wheel torque allowed by Brake Control.
	PtTqAtAxleMaxReq ***::PtTqAtAxleReReq***	0~6000 Nm(6001~19999 define invalid value,20000 define as default value) For Rear Motor **(IEM3/XCUP)** Maximum Propulsive wheel torque allowed by Brake Control.
	PtTqAtWhlMaxReq ***::FrntLeMaxReq***	0~6000 Nm(6001~19999 define invalid value,20000 define as default value) For Front left Motor Maximum Propulsive wheel torque allowed by Brake Control.
	PtTqAtWhlMaxReq ***::FrntRiMaxReq***	0~6000 Nm(6001~19999 define invalid value,20000 define as default value) For Front right Motor Maximum Propulsive wheel torque allowed by Brake Control.
	PtTqAtWhlMaxReq ***::ReLeMaxReq***	0~6000 Nm(6001~19999 define invalid value,20000 define as default value) For Rear left Motor **(IEM2/XCUP2)** Maximum Propulsive wheel torque allowed by Brake Control.
	PtTqAtWhlMaxReq ***::ReRiMaxReq***	0~6000 Nm(6001~19999 define invalid value,20000 define as default value) For Rear right Motor **(IEM2/XCUP2)** Maximum Propulsive wheel torque allowed by Brake Control.

Represented Signal in Requirement	CAN Signal	Description
SysSigDMCTarTq	DMCTarTqFrnt	For Front Motor **(MGM/XCM2-P3)** DMC target torque
	DMCTarTqRear	For Rear Motor **(IEM3/XCUP)** DMC target torque
		For Front left motor DMC target torque
		For Front right Motor DMC target torque
	TqTarFromMotDynCtrlReLe	For Rear left motor **(IEM2/XCUP2)** DMC target torque
	TqTarFromMotDynCtrlReRi	For Rear right Motor **(IEM2/XCUP2)** DMC target torque
SysSigDmcSts	DmcStsFrntToEsc	For Front Motor **(MGM/XCM2-P3)** Dmc Status

	DmcStsRearToEsc	For Rear Motor **(IEM3/XCUP)** Dmc Status
		For Front left Motor Dmc Status
		For Front right Motor Dmc Status
	MotDynCtrlStsReLe	For Rear left Motor **(IEM2/XCUP2)** Dmc Status
	MotDynCtrlStsReRi	For Rear right Motor **(IEM2/XCUP2)** Dmc Status

- When driving in Gear D, if both Traction control(TCS) and DTCS active(***SysSigDmcSts*** = OnDec), to compare ***SysSigPtTqAtAxleMaxReq*** and wheel torque request of DMC calculated by ***SysSigDMCTarTq*** HVED target torque output shall follow the smaller one.
- When driving in Gear R, if both Traction control(TCS) and DTCS active(***SysSigDmcSts*** = OnDec), to compare (-1)****SysSigPtTqAtAxleMaxReq*** and wheel torque request of DMC calculated by ***SysSigDMCTarTq*** HVED target torque output shall follow the bigger one.
- When driving in Gear D, if both Drag Control (DTC/EDTR) and EDTC active(***SysSigDmcSts*** = OnInc), to compare ***SysSigPtTqAtAxleMinReq*** and wheel torque request of DMC calculated by ***SysSigDMCTarTq*** HVED target torque output shall follow the bigger one.
- When driving in Gear R, if both Drag Control (DTC/EDTR) and EDTC active(***SysSigDmcSts*** = OnInc), to compare (-1)****SysSigPtTqAtAxleMinReq*** and wheel torque request of DMC calculated by ***SysSigDMCTarTq*** HVED target torque output shall follow the smaller one.

Note: Traction control and Drag Control active condition please check:

1045438 Traction Control Prerequisites Max and Min
1045440 Traction Control Prerequisites Max and Min - XCM

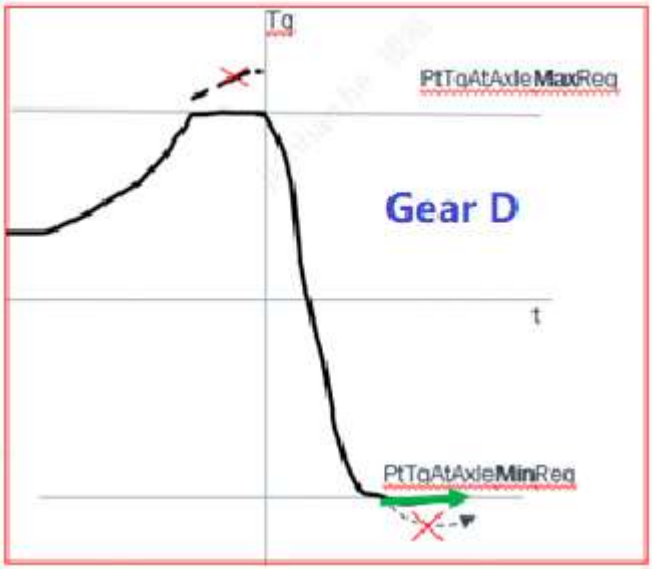
1045440 v1 Traction Control Prerequisites Max and Min - XCM [CS Released]

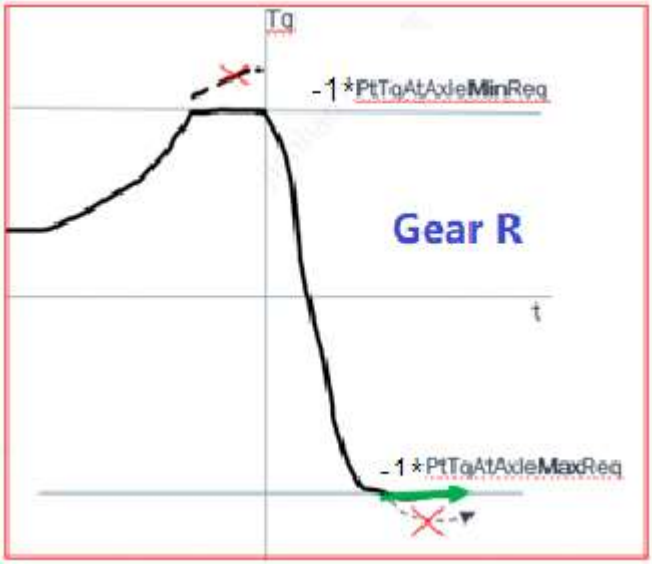
Purpose: Describe the conditions for the HV Electric Drive to enter Traction control mode, the requests will come from other subsystems.
Verification Method: Test
Legacy ID:

Signal	Description
PtTqAtAxleMinReq ***::PtTqAtAxleFrntReq***	-6000~0 Nm (-6001~-19999 define invalid value,-20000 define as default value) For Front Motor Minimum regenerative wheel torque allowed by Brake Control..
PtTqAtAxleMaxReq ***::PtTqAtAxleFrntReq***	0~6000 Nm(6001~19999 define invalid value,20000 define as default value) For Front Motor Maximum Propulsive wheel torque allowed by Brake Control.

Four Situation:			
GearLvrIndcn	Function	Signal change from Brake Control	E-motor shall

Gear D	TCS	满足（***TrsmClu0Sts*** = Disengaged）&（***PtClu0TqReq***＜1）& debounce 500ms（TBD）时： ***PtTqAtAxleMaxReq*** reduce from 20000 to *DTcsTar*	Limit ***SysSigTqAct*** less than *DTcsTar*
Gear R	TCS	满足（***TrsmClu0Sts*** = Disengaged）&（***PtClu0TqReq***＜1） & debounce 500ms（TBD）时： ***PtTqAtAxleMaxReq*** reduce from 20000 to *RTcsTar*	Limit ***SysSigTqAct*** more than (-1)**RTcsTar*
Gear D	DTC	***PtTqAtAxleMinReq*** reduce from -20000 to *DEdtrTar*	Limit ***SysSigTqAct*** more than *DEdtrTar*
Gear R	DTC	***PtTqAtAxleMinReq*** reduce from -20000 to *REdtrTar*	Limit ***SysSigTqAct*** less than (-1)* *REdtrTar*





❖ HVED shall enter Traction Control (TCS) mode(Send ***SysSigMotSts*** = TcsCtrl) ,if all of the following conditions are fulfilled (at the same time):

- ***GearLvrIndcn*** = D;
- 满足（***TrsmClu0Sts*** = Disengaged）&（***PtClu0TqReq***＜1） & debounce 500ms（TBD）时：
- For Front motor: 0≤***PtTqAtAxleMaxReqPtTqAtAxleFrntReq***≤6000
- For Front motor: ***PtTqAtAxleMaxReqPtTqAtAxleFrntReq***＜***SysSigTqReq***;
- E2E of ***PtTqAtAxleMaxReq*** no fault;
- ***SysSigModSts*** = TqCtrl;
- ***SysSigHvRlySts*** = Clsd;
- ***SysSigTqAllwd*** = (Propulsive torque allowed) OR (Regenerative torque allowed) OR (Propulsive and regenerative torque allowed);
- E2E protection of ***SysSigTqAllwd*** is fulfilled.

When enter Traction Control mode in Gear D, HVED target torque output shall follow ***PtTqAtAxleMaxReq*** .

❖ HVED shall enter Traction Control (TCS) mode(Send ***SysSigMotSts*** = TcsCtrl) ,if all of the following conditions

are fulfilled (at the same time):

- ***GearLvrIndcn*** = R;
- 满足（***TrsmClu0Sts*** = Disengaged）&（***PtClu0TqReq***＜1） & debounce 500ms（TBD）时：
- For Front motor: 0≤***PtTqAtAxleMaxReqPtTqAtAxleFrntReq***≤6000
- For Front motor: (-1)*PtTqAtAxleMaxReqPtTqAtAxleFrntReq＞***SysSigTqReq***;
- E2E of ***PtTqAtAxleMaxReq*** no fault;
- ***SysSigModSts =*** TqCtrl;
- ***SysSigHvRlySts*** = Clsd;
- ***SysSigTqAllwd*** = (Propulsive torque allowed) OR (Regenerative torque allowed) OR (Propulsive and regenerative torque allowed);
- E2E protection of ***SysSigTqAllwd*** is fulfilled.

When enter Traction Control mode in Gear R, HVED target torque output shall follow (-1)****PtTqAtAxleMaxReq*** .

❖ HVED shall enter Drag Control (EDTR) mode(Send ***SysSigMotSts*** = TcsCtrl) and if all of the following conditions are fulfilled (at the same time):

- ***GearLvrIndcn*** = D;
- For Front motor: -6000≤***PtTqAtAxleMinReqPtTqAtAxleFrntReq***≤0
- For Front motor: ***PtTqAtAxleMinReqPtTqAtAxleFrntReq***＞***SysSigTqReq***;
- E2E of ***PtTqAtAxleMinReq*** no fault;
- ***SysSigModSts =*** TqCtrl;
- ***SysSigHvRlySts*** = Clsd;
- ***SysSigTqAllwd*** = (Propulsive torque allowed) OR (Regenerative torque allowed) OR (Propulsive and regenerative torque allowed);
- E2E protection of ***SysSigTqAllwd*** is fulfilled.

When enter Drag Control mode in Gear D, HVED target torque output shall follow ***PtTqAtAxleMinReq*** .

❖ HVED shall enter Drag Control (EDTR) mode(Send ***SysSigMotSts*** = TcsCtrl) and if all of the following conditions are fulfilled (at the same time):

- ***GearLvrIndcn*** = R;
- For Front motor: -6000≤***PtTqAtAxleMinReqPtTqAtAxleFrntReq***≤0
- For Front motor: (-1)****PtTqAtAxleMinReqPtTqAtAxleFrntReq***＜***SysSigTqReq***;
- E2E of ***PtTqAtAxleMinReq*** no fault;
- ***SysSigModSts =*** TqCtrl;
- ***SysSigHvRlySts*** = Clsd;
- ***SysSigTqAllwd*** = (Propulsive torque allowed) OR (Regenerative torque allowed) OR (Propulsive and regenerative torque allowed);
- E2E protection of ***SysSigTqAllwd*** is fulfilled.

When enter Drag Control mode in Gear R, HVED target torque output shall follow (-1)****PtTqAtAxleMinReq*** .

❖ HVED shall handle over range fault of TCS or EDTR request,:

- If ***PtTqAtAxleMaxReq::PtTqAtAxleFrntReq***＜0, :
  - set :***PtTqAtAxleMaxReq::PtTqAtAxleFrntReq*** = 20000Nm ;
  - 或者 认为***PtTqAtAxleMaxReq::PtTqAtAxleFrntReq*** 为异常值，不响应。
- If ***PtTqAtAxleMinReq::PtTqAtAxleFrntReq***＞0, set :
  - set :***PtTqAtAxleMinReq::PtTqAtAxleFrntReq*** = -20000Nm;
  - 或者 认为***PtTqAtAxleMinReq::PtTqAtAxleFrntReq*** 为异常值，不响应。

❖ HVED shall handle E2E fault or Lose communication fault of TCS or EDTR request,:

- If E2E fault or Lose communication fault of ***PtTqAtAxleMaxReq***,
  - set :***PtTqAtAxleMaxReq::PtTqAtAxleFrntReq*** = 20000Nm ;
  - 或者 认为***PtTqAtAxleMaxReq::PtTqAtAxleFrntReq*** 为异常值，不响应。

- If E2E fault or Lose communication fault of ***PtTqAtAxleMinReq***,
  - set :***PtTqAtAxleMinReq::PtTqAtAxleFrntReq*** = -20000Nm;
  - 或者 认为***PtTqAtAxleMinReq::PtTqAtAxleFrntReq*** 为异常值，不响应。

6.3.3.13.5.1.10 Clutch Operation

This is for EDS mode manage which with clutch

342958 v1 Clutch Operation mode [CS Released]

Purpose: To define the conditions clutch shall transit to clutch operation mode
Verification Method: Analysis AND Test
Legacy ID:

The requirement is applicable for EDS which is with clutch.

The state clutch operation indicate that the clutch is operating.

If clutch is open and the following conditions are met,the inverter state transit to Clutch Operation from Standby:

- HV main relay status ***HvSysRlySts***="Closed"
- Clutch state request ***SysSigCluOperTypReq***="Engage" or ***SysSigModeReq***="TqCtrl"

If clutch is closed and the following conditions are met,the inverter state transit to Clutch Operation from Standby:

- HV main relay status ***HvSysRlySts***="Closed"
- ***SysSigModeReq***="TqCtrl"

If the following conditions are met,the inverter state transit to Clutch Operation from Torque Control mode:

- ***SysSigCluOperTypReq***="DisEngage"

342960 v1 Standby mode control with clutch [CS Released]

Purpose: To define the conditions the inverter shall transit to standby mode when EDS is with clutch.
Verification Method: Analysis AND Test
Legacy ID:

The requirement is applicable for EDS which is with clutch.

If any following conditions is met,the inverter state transit to Standby from Clutch Operation:

- HV main relay status ***HvSysRlySts***="Open"

Or

- Clutch actual state is open and ***SysSigModeReq***="Standby" and motor speed is less than 10rpm

If any following conditions is met,the inverter state transit to Standby from Torque Control mode:

- HV main relay status ***HvSysRlySts***="Open"

Or

- ***SysSigCluOperTypReq***="Engage" and ***SysSigModeReq***="Standby"

342961 v1 **Torque mode control with clutch [CS Released]**

Purpose: To define the conditions the inverter shall transit to torque mode when EDS is with clutch.
Verification Method: Analysis AND Test
Legacy ID:

The requirement is applicable for EDS which is with clutch.

If the following conditions are met,the inverter state transit to torque control mode from Clutch Operation:

- HV main relay status ***HvSysRlySts***="Closed"
- ***SysSigModeReq***="TqCtrl"
- Clutch actual state is closed

If the following condition is met,the inverter state transit to Clutch Operation from Torque Control mode:

- ***SysSigCluOperTypReq***="DisEngage"

6.3.3.13.5.1.11 PMA/SEA Launch Mode

676990 v1 **Launch Mode information feedback [CS Released]**

Purpose: Send the necessary information when HVED in launch mode
Verification Method: Analysis AND Test
Legacy ID:

In **Launch Mode**, HVED should feedback specific information to propulsion system:

- the maximum available torque of each motor on ***SysSigTqAvl***
- the actual applied torque of each motor on ***SysSigTqEst/SysSigTqAct***
- the actual speed of each motor on ***SysSigSpdAct/SysSigSpdActSgn***
- the system temperature of each motor on ***SysSigCoolT/SysSigMotT/SysSigInvrT***
- the LOS signal information of each motor on ***SysSigLimnIndcn***

1048103 v1 **Lanch Mode [CS Released]**

Purpose: By increasing the available torque to gain stronger accelerate performance.
Verification Method: Test
Legacy ID:

Usage signal	Description
LnchModSts	0 Disable 1 Initial 2 Standby 3 Ready 4 Active 5 Failed

Launch mode drive scene:

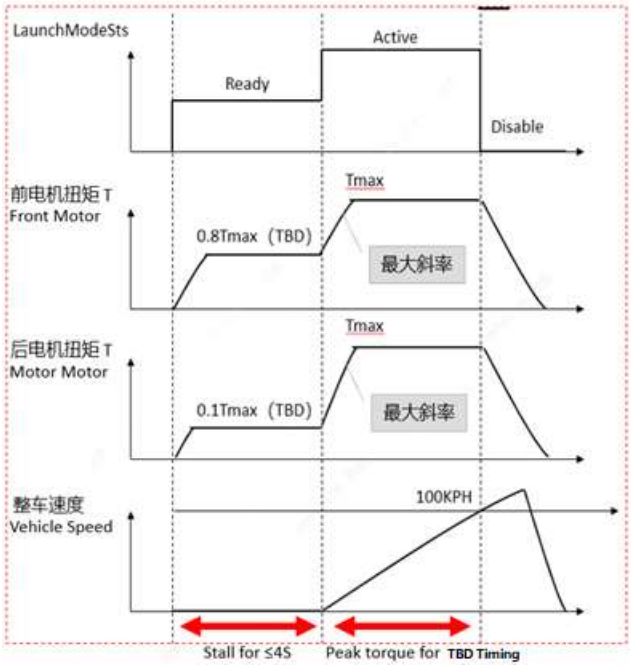
Pre-condition:

- Coolant Temperature＜40°C(TBD)；

- Motor Temperature＜90°C(TBD)；
- Inverter Temperature＜40°C(TBD)；
- HVDC voltage＜765V(TBD).

In Progress:

- The motor standstill no more than 4s;
- Extreme peak torque last for more than **3s**(TBD);
- Each motor extreme peak torque no less than (tbd) Nm;
- Torque response slope for each motor without gear ratio is 1931Nm/s(TBD)



HVED shall identify the Launch mode request to enter or exit Launch mode.

If LnchModSts=1/2/3/4 (Launch mode is active),

HVED shall enter to Launch mode:

- Request full coolant flow, ***SysSigCooltFlowMinReq*** = 10 L/min(TBD);
- If have oil pump to cool motor and haven't enter to Torque Mode, request low speed(1600rpm TBD,according to different supplier) ;
- If have oil pump to cool motor and has enter to Torque Mode, request full speed (3700rpm TBD);
- ***SysSigTqAvl:: MaxAvailTorq*** Peak positive available torque increase to extreme peak value.
  (For reference: peak positive available torque 4400Nm increase to 4800Nm. Final value defined by calibration).

If LnchModSts≠1/2/3/4 (Launch mode is Off),

HVED shall exit to Launch mode:

- Coolant flow request recover to common strategy;
- If have oil pump to cool motor, oil pump speed request recover to common strategy;

- ***SysSigTqAvl:: MaxAvailTorq*** Peak positive available torque recover to common strategy.

6.3.3.13.5.2 Torque Arbitration

250585 v1 Derating Control [CS Released]

Purpose: Protect inverter from taking damage (e.g. temperature derating etc.)
Verification Method: Test
Legacy ID:

The HV Electric Drive System shall protect itself from taking damage from e.g. over temperature, over voltage, over current, over speed etc. by having derating functionality in the SW.
Derating Control shall be performed when the HV Electric Drive System detects some values (e.g. high voltage battery over voltage or under voltage(***SysSigHvBattUDc***),
HV Electric Drive System over temperature, over speed, etc.) over threshold.
In Derating Control, the HV Electric Drive System need to decrease the Torque.

The HV Energy Store System will send the HV Electric Drive System the value of ***SysSigHvBattUDynMinLim*** and ***SysSigHvBattUDynMaxLim***, the HV Electric Drive System shall use these values as input for a derating strategy for Power Limit.

129580 v3 Torque prioritizations [CS Released]

Purpose: Setting priority between different torque affecting functionalities based on the time required to reach a non-negligable system or component consquence in case of uncompliancy.
Verification Method: Analysis AND Test
Legacy ID:

Whenever the inverter is in a state where torque production is expected the actual torque shall be controlled to fulfil the requirements of the various torque affecting functionalities according to the following prioritization order:

1. Any fault reaction specified in the diagnostic fault matrix
2. Anti-spin or traction control, when applicable
3. Fast limitations (Inverter temperature limitation/protection, DC-Voltage, DC-power and DC-current limitation (HVDC system limitations))
4. Anti-shuffle, when applicable
5. Slow limitations (EM temperature limitations, Mechanical power limitation and Stability limitation QM)
6. Following the requested set-point according to the control mode applicable at the time.

Note:

A requirement associated with one of these areas can be violated only if it is contradicted by a higher priority area.

E.g. A reduction of torque due to EM temperature limitation has lower priority than an addition of torque by anti-shuffle in the short term.

250588 v2 **Torque allowed [CS Released]**

Purpose: To ensure that torque production is stopped when the torque allowed signal goes false.
Verification Method: Test
Legacy ID:

Torque allowance by the propulsion master is recevied on port ***SysSigTqAllwd***. No torque shall be produced when the propulsion master is not allowing such.
The signal disables/enables torque output from the EM. It shall be interpreted as separate requests for torque in ISG rotating direction (propulsive torque) and torque working against the rotating direction of the ISG (generative/braking torque).
The allowed torque output is described as below:

Value	Torque allowed status
0x0	Propulsive and regenerative torque not allowed
0x1	Propulsive torque allowed
0x2	Regenerative torque allowed
0x3	Propulsive and regenerative torque allowed

The signal shall be E2E protected.

262956 v2 **Torque Inhibitation due to immobalizer [CS Released]**

Purpose: to ensure that torque is not lost due to immo while driving.
Verification Method: Test
Legacy ID:

When the HVElectricDrive system accomplished the activation of the electrical motor it shall disregard any attempt from Immobilizer system to disable the HVElectricDrive motor.

6.3.3.13.5.2.1 Anti-shuffle

250617 v1 **Active damping calibration [CS Released]**

Purpose: To enable the drivability aspect of anti-shuffle to be calibratable by Geely, either as a parameter or during operation from the propulsion subsystem, which would increase the flexibility in calibration of anti-shuffle functionality as well as making it possible to have calibration depending on driving situation
Verification Method: Test

Legacy ID:

The Anti-shuffle function behaviour shall be possible to calibrate with the signal ***SysSigActvDampgGain*** and with a dataset parameter.

There shall also be a parameter to set the maximum torque that can be damped out by the Anti-shuffle function which shall be respected by the function.

At least the following features shall be affected by the calibration of the anti-shuffle functionality:

- Default replacement value to be used whenever the gain signal(s) is not received or timed out.
- The maximum amplitude of the corrective torque which the anti-shuffle functionality is allowed to output.
- Maximum rate of change due to control input change (gain or corrective torque sign allowed/mode). This is only required during the transition time to the new operating conditions (e.g. if a step is introduced due to sudden change in gain).

250618 v1 **Active Damping [CS Released]**

Purpose: To reduce oscillations in the torque produced at the wheels to a level that cannot be felt by the driver or the passengers.
Verification Method: Analysis AND Test
Legacy ID:

The inverter shall adjust the torque request to the EM to reduce oscillations caused by delays in the drive train. It shall do this by a regulator that monitors the following signals: ***SysSigTqAct***, ***SysSigSpdAct***, ***SysSigTqReq*** and the esimated e*xternal wheel speed.* It shall not adjust the EM torque with more than the max corrective torque specified in req "Active damping calibration". The gain of the regulator shall be possible to calibrate and exchange with an external gain request as specified in req "Active damping calibration".
Active damping shall not cause the actual torque (***SysSigTqAct***) to change sign more than twice after the absolute value of the torque request to the EM has gone lower than the maximum corrective torque for active damping.

INFO: The drive train can be described with the image below. Typical resonances are in the range 5 Hz to 15 Hz



J_1 = Inertia of EM
k = Torsional spring representing elastic elements in drivetrain

250619 v2 **Active damping mode control [CS Released]**

Purpose: To make it possible for the propulsion to set different damping modes
Verification Method: Test
Legacy ID:

For minimize the shuffle in the vehicle when the motor drives the vehicle, the oscillation in the drive train should be damped by the Active Damping control function. The HV electric drive system shall perform a torque retard calculation based on a feedback from engine speed or vehicle speed etc.

The HV electric drive system shall perform motor active damping control when receive a request by signal ***SysSigActvDampgModReq*** which is defined as:

0 = No Damping
1 = Low Damping
2 = Medium Damping
3 = High damping

Motor active damping control shall be adjusted by active damping gain factor.

If the signal is lost for more than 1 s the mode shall be set to the default mode.

250620 v1 **Active damping reference speed [CS Released]**

Purpose: To limit the impact of the anti-shuffle functionality to the frequencies related to the drivetrain eigenfrequencies.
Verification Method: Test
Legacy ID:

The anti-shuffle output corrective torque shall only contain DC-components (frequencies lower than the eigenfrequencies of the drivetrain) small enough to not be felt or noticed in the torque estimate (shall not violate the torque control accuracy requirements).

A reference speed corresponding to the EM speed without the undesired oscillations is given by the signal ***SysSigEngSpdOfActvDampg***.

Note:

The output of the anti-shuffle is effectively cancelling out eigenfrequency oscillations in the drivetrain. The two most straight forward methods of achieving this is by either looking at the difference between the machine speed and a reference speed or by using only the machine speed and high-pass filtering the result.
If a reference speed is used this is supplied either directly from a sensor source (e.g. wheel speed) or from an estimate based on wheel speed and total effective speed ration at the moment.

6.3.3.13.5.2.1.1 Backlash Control

Backlash control

1045446 v1 **Backlash Control Interface [CS Released]**

Purpose: Backlash Control Interface
Verification Method: Test
Legacy ID:

控制器应该传送以下信号到BLC模块：

- 电机轴速Motor shaft speed：***SysSigSpdAct***
- 实际轮端扭矩Motor Torque Actual(Wheel)：***SysSigTqAct***
- 轮端请求扭矩Motor Torque Request：***SysSigTqReq***
- 轴速Axle speed：
  前轴SpdRotlForWhlsAtAxleFrnt（前单电机用轴速）
  后轴SpdRotlForWhlsAtAxleRe （后单电机用轴速）

- 轮速wheel speed:
  前轮WhlSpdCircumlFrnt（前双电机用轮速）
  后轮WhlSpdCircumlRe（后双电机用轮速）
- 纵向加速度Longitudinal acceleration (Ax)：ADataRawSafe::ALgt
- 挡位Drive Direction(PRND)：GearLvrIndcn
- 车速Vehicle Speed：VehSpdLgt
- 踏板扭矩请求（前）Pedal Torque Request-Frnt：PrpsnTqFrntReq（前单电机用）
- 踏板扭矩请求（后）Pedal Torque Request-Rear：PrpsnTqReReq（后单电机用）
- 踏板扭矩请求前左（前双电机用）Pedal Torque Request-Front-left：ZVCWhlTqFrntLeDrvReq
- 踏板扭矩请求前右（前双电机用）Pedal Torque Request- Front -right：ZVCWhlTqFrntRiDrvReq
- 踏板扭矩请求后左（后双电机用）Pedal Torque Request-Rear-left：ZVCWhlTqReLeDrvReq
- 踏板扭矩请求后右（后双电机用）Pedal Torque Request-Rear-right：ZVCWhlTqReRiDrvReq

控制器应该接收以下信号来自BLC模块：
- 请求扭矩方向Torque Request Sign
- BLC状态 BLC state
- BLC补偿扭矩状态BLC torque

1045447 v1 Backlash Control State [CS Released]

Purpose: Backlash Control State
Verification Method: Test
Legacy ID:

BLC状态分为以下状态
2 = Hold
3 = Recover
4 = Off_Exit

BLC模块应根据内部计算的扭转角度、扭矩请求方向 和 TipIn/TipOut工况切换BLC状态。

- 当BLC功能未激活时，BLC状态发送4 = Off_Exit；
- 当轮端请求扭矩穿越0Nm时（包括升扭穿越和降扭穿越）且扭转角度＞特定阈值1，说明正准备齿隙间的靠齿面切换，BLC功能激活，BLC状态发送2 = Hold；
- 当状态为2 = Hold且扭转角度＞特定阈值2，说明齿隙间的靠齿面切换已完成，恢复跟随轮端请求扭矩，BLC状态发送3 = Recover；
- 当状态为3 = Recover且实际轮端扭矩达到轮端请求扭矩时，退出BLC功能，BLC状态发送4 = Off_Exit；
- 当状态为3 = Recover时，再次轮端请求扭矩穿越0Nm时（包括升扭穿越和降扭穿越）且扭转角度＞特定阈值1时，BLC状态发送2 = Hold；

特定阈值1 和 特定阈值2 可标定。

使用信号：
- 实际轮端扭矩Motor Torque Actual(Wheel)：***SysSigTqAct***
- 轮端请求扭矩Motor Torque Request：***SysSigTqReq***

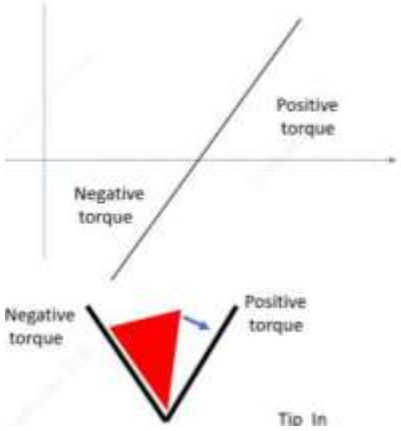
1045449 v1 Backlash Control Twist Angle [CS Released]

Purpose: Backlash Control Twist Angle
Verification Method: Test
Legacy ID:

当轮端请求扭矩穿越0Nm时（包括升扭穿越和降扭穿越）时，电机转速 对比 输出轴轴端转速/速比 产生转速差。可根据转速差积分计算扭矩过零时，齿隙间 齿的相对转动角度，称为扭矩角度。

使用信号

- 电机轴速Motor shaft speed：***SysSigSpdAct***
- 轴速Axle speed：
  前轴SpdRotlForWhlsAtAxleFrnt（前单电机用轴速）
  后轴SpdRotlForWhlsAtAxleRe （后单电机用轴速）
- 轮速wheel speed:
  前轮WhlSpdCircumlFrnt（前双电机用轮速）
  后轮WhlSpdCircumlRe（后双电机用轮速）
- 速比根据产品自身参数




Clunk Noise

1045450 v1 **Backlash Control Torque [CS Released]**

Purpose: Backlash Control Torque
Verification Method: Test
Legacy ID:

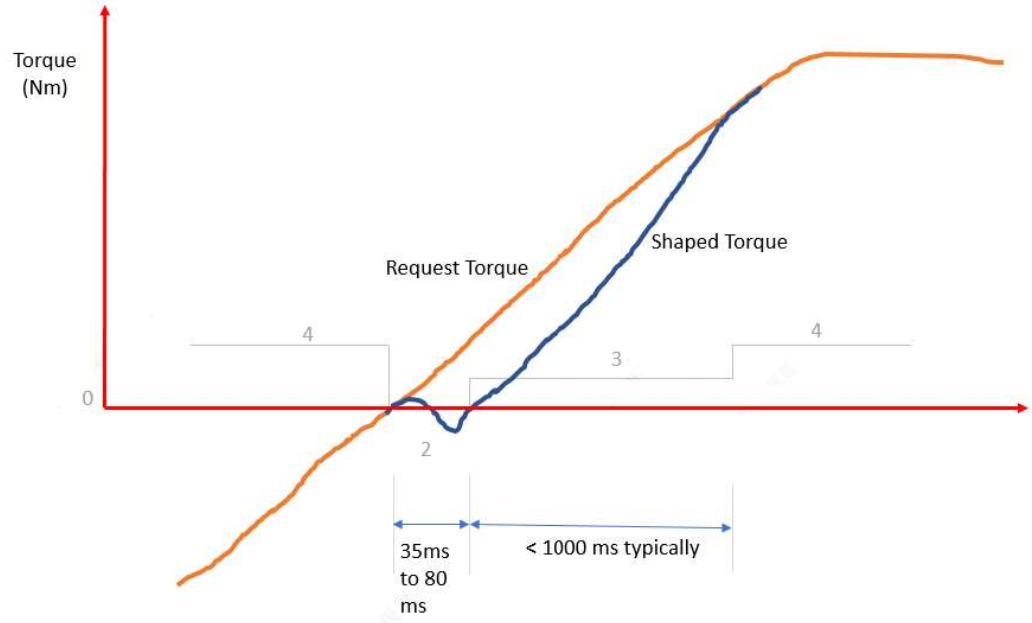
❖ 当状态为2 = Hold时，BLC功能激活，输出扭矩应加上BLC补偿扭矩；
如果ActiveDamping功能同时激活，应延迟补偿ActiveDamping扭矩，以免影响BLC功能效果。
控制器计算的BLC补偿扭矩，应综合考虑 抑制扭转角度 扭矩方向 和 扭矩请求 实际扭矩等参数。

❖ 当状态为3 = Recover时，BLC功能激活，输出扭矩应加上BLC补偿扭矩；
如果ActiveDamping功能同时激活，应延迟补偿ActiveDamping扭矩，以免影响BLC功能效果。
控制器计算的BLC补偿扭矩，应综合考虑 抑制扭转角度 扭矩方向 和 扭矩请求 实际扭矩等参数。
控制器计算的BLC补偿扭矩，按一定斜率减少，使输出扭矩达到请求扭矩。

❖ 当状态不为2 或者 3时，BLC功能关闭，输出扭矩不再加上BLC补偿扭矩；
恢复正常的扭矩响应策略。
控制器计算的BLC补偿扭矩为0。

❖ 当DTCS 或者 EDTC 触发时，BLC功能关闭，BLC 切换Off_Exit，输出扭矩不再加上BLC补偿扭矩。
DmcStsFrntToEsc::Dmssts =1OnDec/4OnInc(前单电机)
DmcStsRearToEsc::Dmssts =1OnDec/4OnInc(后单电机)
MotDynCtrlStsFrntLe::Dmssts =1OnDec/4OnInc/5YrLim(前双电机-左)
MotDynCtrlStsFrntRi::Dmssts =1OnDec/4OnInc/5YrLim(前双电机-右)
MotDynCtrlStsReLe::Dmssts =1OnDec/4OnInc/5YrLim(后双电机-左)
MotDynCtrlStsReRi::Dmssts =1OnDec/4OnInc/5YrLim(后双电机-右)

效果图如下：



6.3.3.13.5.2.2 Limitations

134154 v2 **Mechanical power limitation [CS Released]**

Purpose: Limit the power limits to ensure that the HW limitations are not violated nor that the HW is used in such a way causing stress and unnecessary wear
Verification Method: Analysis AND Test
Legacy ID:

Estimated available torque shall be limited such that the estimated available mechanical power, defined as the product between the estimated maximum and minimum available torque and actual EM speed, is limited to < *MechPwrLimMax* and > *MechPwrLimMin* *(which are HW dependant parameters)*.

250884 v1 **HVDC power limitation [CS Released]**

Purpose: Ensure that the power limits are not violated when the inverter controls the torque setpoint on its own (e.g. speed control) or when propulsion system underestimates the losses in the system
Verification Method: Analysis AND Test
Legacy ID:

Whenever the inverter torque set point is not controlled externally by ***SysSigTqReq**** and the HV-battery contactor status, ***SysSigHvRlySts***, indicates "closed", applied actual torque shall be limited such that the maximum and minimum DC power limits, ***SysSigPwrLimMin*** and ***SysSigPwrLimMax***, are not violated when defining the DC power as the product of the inverters estimated DC current and DC voltage.

The power limitation shall only act to reduce the absolute value of the applied torque, i.e. never add torque in the opposite direction (e.g. to compensate for losses at zero torque with a very low power limit).

Note:
* In a mode of operation where the inverter regulates the torque on its own in order to achieve a different target such as a speed or voltage set point.

Voltage control calibration target:

During voltage control initialization (before the HV-battery contactor is opened), the power limitations for the Electric Drive system must be sufficiently followed to ensure that an immediate contactor opening is avoided.

253446 v1 **Power limitation [CS Released]**

Purpose: To avoid violate the power limitation
Verification Method: Analysis AND Test
Legacy ID:

The Electric Drive System shall not violate the powerlimitations set by signals the ***SysSigPwrLimMin*** and ***SysSigPwrLimMax***

6.3.3.13.5.2.2.1 EM speed limitations

The maximum speed is limited by both the inverters capability to control the machine and the durability of the maximum machine itself.Exposing the EM for severe enough over speed scenarios will cause rotor deformation, due to fatigue. Apart from the impact on remaining EM life time a severe enough rotor deformation will cause the rotor to clash with other parts, e.g. the stator, which in turn can cause the machine axle to be locked ==> EM destroyed. Avoiding a locked rotor scenario is for some applications highly safety relevant.

Another aspect considered is the usage of safe state by the inverter/EM where it is desired to use a safe state which requires a limited speed.
Normal functionality (QM)

Subsystem HV Electric Drive:

EM speed limitation [QM] - Ramp down propulsive torque and enter standby (including disengagement of the EM where applicable) if exceeding the speed threshold, before exceeding the maximum operating speed of the EM.

EM speed limitation signaling [QM] - Speed limitation used to enable the usage of the preferred safe state.
This is only actively signaled for applications where the EM speed dependence of the vehicle speed (connection state under the control of the inverter). This value should by default go to its maximum value whenever the connection towards wheels is open.

Requirements on Subsystem propulsion:

EM usage speed limitation [QM] - Request the inverter to Standby before exceeding a speed threshold.

Vehicle speed limitation [QM] - Limit the vehicle speed to a maximum value when the EM cannot be disengaged from the wheels. For applications where this is under the control of the inverter this will be signaled by the EM speed limitation signaling from HV Electric Drive.

Safety functionality (ASIL)

Subsystem HV Electric Drive:

1. **Over speed safe state [ASIL X]** - If the speed exceeds maximum operating speed disable propulsive torque before exceeding the maximum operating speed + a control margin.

Requirements on Subsystem propulsion:

1. **EM over-speed prevention [ASIL X]** - Limit the vehicle speed so that the maximum mechanical speed of the EM is not reached. For applications with a transmission connection from EM to reference point, under the control of the inverter, this is handled with an ASIL C signal from the inverter.

For some applications in addition the following is needed from SubSystem : Brake Control, to prevent that the hazard occurs when the car is coasting in a downhill slope (This has a much lower ASIL level due to vastly lower exposure)

1. **EM over-speed prevention [ASIL A TBC]** - Limit the vehicle speed so that the maximum mechanical speed of the EM is not reached.

Note:

Maximum operating speed and maximum mechanical speed are fix limits from the machine/inverter design.

Speed thresholds need to be calibrated with SubSystem : Propulsion so that the limits are not violated.

129588 v1 **EM speed limitation [CS Released]**

Purpose: To limit the usability of the electric machine to the speed range for which it is specified.
Verification Method: Analysis AND Test
Legacy ID:

If the absolute value of the EM speed is above *EmSpeedLimitThresholdCalib* the inverter shall limit the available and applied EM accelerating torque towards zero Nm.

The calibration of *EmSpeedLimitThresholdCalib* shall be such that the EM accelerating torque has been reduced to zero before the absolute value of the EM speed becomes greater than *EmSpeedLimitMax*.

After the available and applied torque has been limited to zero Nm, due to EM speed limitation, and before the EM speed becomes greater than *EmSpeedLimitMax* the inverter shall be in Standby.

Once the inverter enters Standby due to EM speed limitation it shall remain in Standby until the absolute value of the EM speed is smaller than *EmSpeedLimitMax, h*ysteresis shall be applied to avoid toggling around *EmSpeedLimitMax*.

The limitation shall ensure that the full speed range of the EM is used. The EM speed is not allowed to overshoot the speed limit at all.

Note:

EmSpeedLimitMax is a fix limit set by the specification of the electrical machine, while *EmSpeedLimitThresholdCalib* need to be set during calibration for each application.

Smoothness of the torque limitation from this functionality has low priority since this is outside the normal (drivability relevant) operating range.

6.3.3.13.5.2.3 Power Dissipation Control

134140 v1 **Power dissipation operation general [CS Released]**

Purpose: Provide an option to reduce the efficiency of the electric drive train, either to effectively add a margin towards charing power limits or to inject heating power in differenct parts of the system.
Verification Method: Analysis AND Test
Legacy ID:

When the power dissipation functionality is active the motor control shall be adapted so that additional losses are introduced in inverter, electric machine or both compared to the losses associated with the normal control strategy (when power dissipation operation is not active).

While additional losses are produced in the system, via the power dissipation functionality, all normal control-, estimation-, limitation- and self-protection requirements are still applicable as if the additional power dissipation was not present.

Applicable for applications where *PowerDissipationApplicability* equals "yes".

Note: This implies that the DC-current signal shall show the actual DC-current, even while the additional power dissipation is active, accurately enough for the accuracy requirements of the HVDC current and voltage (and consequently the HVDC power limitation) to be fulfilled.

134141 v2 **Power dissipation in torque control [CS Released]**

Purpose: Provide correct information on available torque also when additional power dissipation is active
Verification Method: Analysis AND Test
Legacy ID:

If power dissipation is active and additional losses* are being produced while the inverter operates in torque control, the estimation of available torque, on IsgTqAvl, shall be adapted to account for the actual efficiency of the electric drive components.

Note:

*Additional losses actively and purposely injected compared to the losses present during normal torque control operation (without power dissipation).

Available torque due to HVDC power limits is not handled within HV Electric Drive, however if power dissipation is applied with the purpose to heat up different parts of the system during motoric drive the available torque will be reduced accordingly which need to be indicated to the Propulsion Vehicle Control ECU.

250919 v1 **Power dissipation internal activation [CS Released]**

Purpose: Specifying where power dissipation shall be available per application
Verification Method: Analysis AND Test
Legacy ID:

Internally activated power dissipation shall only be allowed if ***SysSigPwrLimMin***> *PowerDissipationGenPwrLim* AND ***SysSigModSts*** included in *PowerDissipationModeActive* AND *PowerDissipationApplicability* == "Yes"

PowerDissipationModeActive is bit mapped according to:

Bit 1: Transmission synchronization speed control

Bit 2: Torque control

Bit 3: Speed control

Bit 4: Clutch control

Note:

The extent (and amount) of additional losses appropriate to apply per situation is set by calibration for the relevant mode of operation and balanced with the impact on other areas, such as drivability, limitations and accuracy.

6.3.3.13.5.2.4 Distribute Traction Control_Lotus

Distributed Traction Control used for improving vehicle skidding perfermance

700095 v1 **dTCS library Integration Cybersecurity requirement [CS Released]**

Purpose: Integration software shall meet cyber security requirement
Verification Method: Analysis AND Test
Legacy ID:

Software integrated dTCS library shall meet sybersecurity requirment

1. Ensure the integrity and authenticity of software and avoid manipulation of software
2. Ensure the security of in-vehicle communication
3. Avoid DDOS (Distributed Denial Of Service) attack

700285 v1 **dTCS receive EDS Configuration [CS Released]**

Purpose: To set different parameter set according to vehicle configration
Verification Method: Test
Legacy ID:

Inverter shall send EDS position type to dTCS by port **VariantCodingConfigured** which Code define as bellows:

0x0: Undifned

0x1: EDSF mean this inverter is in front electric drive

0x2: EDSRL mean this inverter is in rear left electric drive

0x3: EDSRR mean this inverter is in rear right electric drive

- when the EDS is in front side, the inverter shall assign value 0x1(EDSF) to **VariantCodingConfigured** and send it to dTCS library;
- when the EDS is in rear side and with single motor,the inverter shall assign value 0x3(EDSRR) to **VariantCodingConfigured** and send it to dTCS library;
- when the EDS is in rear side and with double motor,the rear right inverter shall assign value 0x3(EDSRR) to **VariantCodingConfigured** and the rear left inverter shall assign value 0x2(EDSRL) to **VariantCodingConfigured** and send it to dTCS library;

700289 v1 **dTCS recieve operation command [CS Released]**

Purpose: The define the TCS contorl request flag
Verification Method: Test
Legacy ID:

EDS shall gateway **dTCSOperCmd** to dTCS library after receiving from VMCU

dTCSOperCmd will include three EDS Operate Command(dTCSOperCmdFrnt/dTCSOperCmdRL/ dTCSOperCmdRR),everyone state define as bellows

0x0: No Request

- which mean no any other request and dTCS library control priority is higher;

0x1:VMC TCS interventio

- dTCS library recieve this state, its torque limitation will exit and VMC will take over TCS function to command Inverter torque output;

0x2:VMC YAW intervention

- dTCS library recieve this state, its torque limitation will exit and VMC will take over YAW control function to command Inverter torque output;

0x3:Reserved

700290 v1 **dTCS receive each whee roadfric [CS Released]**

Purpose: dTCS Library calculate wheel slip rate based on wheel road friction
Verification Method: Test
Legacy ID:

EDS shall gateway **RoadFrictionFactor** to dTCS after receiving each wheel Road friction from VMCU;
dTCS Library will calculate the target slip rate based on wheel road friction which include four wheel road friction,define as bellows:
The signal RoadFric group:

RoadFricFL
RoadFricFLQf
RoadFricFR
RoadFricFRQf
RoadFricRL
RoadFricRLQf
RoadFricRR
RoadFricRRQf

700291 v1 **EDS torque state for dTCS [CS Released]**

Purpose: EDS state for dTCS
Verification Method: Analysis AND Test
Legacy ID:

EDS shall deliver its own state to dTCS via **TcsPssdState** to inform dTCS if It can response torque limitation;**TcsPssdState** define as bellow:

0x0:Normal:

- when Inverter have no fault and work in torque control that can response torque output normally.
- when Inverter is in torque control and derating, this state is normal too;

0x1:Shutdown:

- EDS have any fault or EDS can't enter torque mode that EDS cannot obey the torque limitation from dTCS.

0x2:Inhibit :Reserved

700292 v1 **dTCS control mode state feedback [CS Released]**

Purpose: The actual dTCS control mode feedback on CAN
Verification Method: Analysis AND Test
Legacy ID:

EDS shall send the state of dTCS function state on CAN by assembling dTCS libary intervention state and inverter state by Port **SysSigDtcsControlState;**

SysSigDtcsControlState define as bellows:

Bit 0 - dTCS_Init:

- dTCS is in initialization; this state is from dTCS library interface **TorqueControlState;**

Bit 1 - dTCS_Idle:

- dTCS is monitoring, but no intervention, the limit torque **dTCSIntvTqLimn** will be maximum which mean no dTCS torque limitation; this state is from dTCS library interface **TorqueControlState;**

Bit 2 - dTCS_On_Dec:

- dTCS intervention, dTCS will send the torque limit **dTCSIntvTqLimn** to inverter to reduce the wheel slip rate; this state is from dTCS library interface **TorqueControlState;**

Bit 3 - dTCS_On_Inc:

- dTCS is quitting the intervention, torque limitation **dTCSIntvTqLimn** will ramp to maximum torque until no torque limitaion; this state is from dTCS library interface **TorqueControlState;**

Bit 4 - dTCS_Fault:

- dTCS have internal fault in library; this state is from dTCS library interface **TorqueControlState;**

Bit 5 - EDS_Fault/dTCS Lib runtime

- this Code indication : Inverter monitor dTCS Library running stuck and **MonitorCounter** no update for 300ms ,then Dtcs Lib have something wrong and Inverter shall set bit 5 , else it shall set zero；

Bit 6 -Reserve

- this Code indication: Default set zero;

Note：

A calibration data shall be used to enable or disable **EDS Fault** detection (or dTCS Library task alive monitoring) and report and reaction on MCU Host SW.

only **EDS fault** and **dTCS_fault** can bypass dTCS torque limitations. dTCS Library can make sure the invalid value is 20000.

EDS_Fault priority is highest than dTCS fault state. when dTCS Library task alive monitoring time out, MCU Host can default dTCS torque limitations is 20000;

6.3.3.13.5.2.5 Torque Vector Control_Lotus

Torque Vector Control is applicable for axle double motor

697415 v1	**Torque deviation monitor each other for TVC [CS Released]**

Purpose: this item is used to monitor toque devatation between right and left emotor
Verification Method: Analysis AND Test
Legacy ID:

In order to avoid the vehicle lateral instability, double e-motor should receive right or left motor torque in port **SysSigTqAct** each other and calculate torque deviation between two emotor, If the deviation is larger than

500Nm(TBD) , It shall set indicator In port **SysSigLimnIndcn** and report DTC;

Note:

Torque deviation post reaction shall be aligned with OEM.the supplier can set LOS bit0 in port **SysSigLimnIndcn** and report DTC when torque deviation out-of-tolerance firstly

6.3.3.13.5.3 LOS

250940 v6	LOS Signalling [CS Released]

v6 change log:	增加IsgLimnIndcn2的定义

Purpose: Specifying format of limitation indication
Verification Method: Test
Legacy ID:

The LOS Signal shall use the below bits as following in the port ***SysSigLimnIndcn*** :

Bit 0 in Port ***SysSigLimnIndcn***: Service Request.
Shall indicate "Service Required". The mapping relation between Service Request with the DTC code refer to SSDS Part II
Bit 1 in Port ***SysSigLimnIndcn***: No Propulsion.
Shall indicate that the ECU or Electric Motor is currently not available, Shall send fault request to Propulsion and the vehicle should stop within a time series. Usually the inverter keeps active short circuit caused by severe faults.
Bit 2 in Port ***SysSigLimnIndcn***: Speed Limitation.
Shall indicate that EM is currently not available, but the vehicle can operate in limited speed if configured with other engine (ICE or EM). Usually the inverter keeps freewheel state or speed dependent Active Short Circuit/Freewheeling caused by not too severe faults
Bit 3 in Port ***SysSigLimnIndcn*** : Torque Limitation.
Shall indicate the Inverter is currently in limited torque operation, the inverter can drive in limp home or 0 Nm control.
Bit 4 in Port ***SysSigLimnIndcn***: Park Fault or Clutch Fault.
Shall indicate the integrated park lock control fault or disconnected clutch mechanisam fault.
Bit 5 and Bit6 in Port ***SysSigLimnIndcn***: IGBT status indication

bit6	bit5	IGBT status
0	0	PWM
0	1	Active Short Circuit
1	0	Freewheel
1	1	Reserved

Bit5 and bit6 in EEA4.0 architecture:
Bit 5 in Port ***SysSigLimnIndcn***: Speed Limitation (Torque Available).
Shall indicate that EM currently shall work in low speed, and the vehicle can operate in limited speed (30km/h) .
Applicable for oil pump fault or so on.
Bit 6 in Port ***SysSigLimnIndcn***: TBD

Bit 7~14 in Port ***SysSigLimnIndcn***: IEM DTC Mask number.
Mask all the DTCs with number 1, 2, 3...(no more than 511, 9 bits used for DTC mask)
If any fault occures, sending its mask number on the propulsion CAN by these bits. If two or more faults exist at one time stamp, sending the mask number cyclically each 2s.

The supplier should trace each DTC with one or more LOS bit in the column "Fault Effect in ECU" in the document SSDS Part II.

The inverter should send out the corresponding LOS bit value correctly and record the DTC, snapshot at the same time when fault occures.
The DTC mask number should have correct mapping relation with the DTC code.

Bit 15 in Port ***SysSigLimnIndcn***: LV supply abnormal indication.

- If HVED detect LV supply abnormal that will cause LOS Bit1/2/3 trigger (for example voltage too low), HVED shall trigger LOS Bit 15 in the same time.
- If HVED detect LV supply abnormal that will cause LOS Bit 0 trigger (for example voltage too high), HVED don't need to trigger LOS Bit 15.

The LOS Signal shall use the below bits as following in the port ***IsgLimnIndcn2*** :

Bit 2 in Port ***IsgLimnIndcn2*** : Fuel cut. (Only applicable for PCM/XCM2 which near the oil pipes)
When HVED have the serious fault/DTC which may possibly cause fire.
These fault/DTC when happen,shall indicate the propulsion system to stop fuel offering.
When HVED have the serious fault/DTC like Ac side input current or voltage error
These fault/DTC when happen,,shall indicate the propulsion system inhibit the P3 motor drive but allowed the P3 motor charging;

250948 v1 **Feedback from LOS Master [CS Released]**

Purpose: Specify reactions based on the feedback signal from the Master LOS manager.

Verification Method: Test
Legacy ID:

The feedback signal from the Master LOS manager (i.e. ECM) consist of “Failed =1” OR “Passed =0”

The Subsystem LOS SW-C, upon reception of ***SysSigFbLimnIndcn*** = 1, shall report FAILED for DEM_LimnIndcnFb

The Subsystem LOS SW-C, upon reception of ***SysSigFbLimnIndcn*** = 0, shall report PASSED for DEM_LimnIndcnFb

1045457 v1 **LOS Signalling special for coaxial dual motor [CS Released]**

Purpose: Additional limitation indication for Four Motor
Verification Method: Analysis AND Test
Legacy ID:

对于同轴双电机电驱：

如果单边电机出现Los Bit 1故障，应进行如下故障处理：

- 1秒（TBD）内，无故障的电机 应与 故障电机，进行一致的故障处理
- 左右两边电机的***SysSigTqavl*** = 0；
- 扭矩响应不再恢复。

如果单边电机出现Los Bit 2故障，应进行如下故障处理：

- 1秒（TBD）内，无故障的电机 应与 故障电机，进行一致的故障处理；

- 左右两边电机的***SysSigTqavl*** = 0；
- 但，无故障的电机不应报告Los Bit 2故障；
- 1秒（TBD）后，***SysSigTqavl*** 恢复，扭矩恢复跟随***SysSigTqReq***请求扭矩

HVED product integrate coaxial dual motors:

When one of the motors malfunctions and LOS Bit 1 occurs:

- Within 1 second (TBD), coaxial motor same processing shall be carried out;
- Both left and right available torque ***SysSigTqavl*** set to 0;
- Won't recover torque response.

When one of the motors malfunctions and LOS Bit 2 occurs:

- Within 1 second (TBD), coaxial motor same processing shall be carried out;
- Both left and right available torque ***SysSigTqavl*** set to 0;
- But normal motor shall not report LOS Bit 2;
- After 1 second (TBD), ***SysSigTqavl*** recover and responds to torque request ***SysSigTqReq***.

6.3.3.13.5.4 Measurements & Estimations

251172 v1 — Torque estimation accuracy levels [CS Released]

Purpose: Define the expectations on the torque estimations when it comes to accuracy
Verification Method: Test
Legacy ID:

The difference between the EM torque estimate, ***SysSigTqAct***, and the actual produced torque shall not exceed *TqEstActlErrStat* when steady state conditions apply.

The difference between the EM torque estimate, ***SysSigTqAct***, and the actual produced torque shall not exceed *TqEstActlErrDyn* when dynamic state conditions apply.

Steady state is defined as when the derivative of the requested torque, ***SysSigTqReq***, has been less than 20Nm/s for more than 500ms without significant interference from anti-shuffle and without any interference from torque limitations or anti-spin. All other states are considered dynamic state.

251173 v2 — Torque estimation [CS Released]

Purpose: This value should reflect how much torque the propulsive management system can utilize from the HVElectric Drive system. Time horizon aspect will be that it is until a new value is reported so that it is not a instant value.
Verification Method: Test
Legacy ID:

The actual electric motor torque on the reference point (crankshaft for SPA, even shaft of 7DCT for CMA, front axle for PMA) including all losses shall be estimated/measured and signalled on port ***SysSigTqAct***.

251174 v2 Available torque estimation [CS Released]

Purpose: Report the available torque that the EM can deliver.
Verification Method: Test
Legacy ID:

Taking all limiting factors in the inverter and the EM, the inverter shall estimate maximum and minimum available torque at the reference point (crankshaft for SPA, even shaft of 7DCT for CMA, front axle for PMA) at any given moment and report on port ***SysSigTqAvl***. The actual available torque reported shall be what the propulsive management can utilize until an updated value is reported. It shall not cause any problems to the component that this torque is applied.

In cases when the estimation cannot be done such in fault state, available torque shall be 0 Nm.

- *TqAvlIsgMax* - Maximum torque is defined as maximum torque in positive direction,
- *TqAvlIsgMin* - Minimum torque is defined as the maximum torque in the opposite direction and is always negative.

The torque shall be estimated for time window of TBD seconds.

The available torque shall be low pass filtered with a time constant of TBD ms (calibratable).

If the torque for any reason cannot be estimated the available torque shall be set to 0.

Note: The calculation shall not limit the torque calculation based on power limits from ECM.

251175 v3 EM Speed [CS Released]

Purpose: Measure and signal the speed of EM.
Verification Method: Test
Legacy ID:

The EM speed measured the reference point (crankshaft for SPA, even shaft of 7DCT for CMA, electric motor shaft for PMA/Lotus) shall be reported as in the following:

Actual speed of the EM (absolute value) shall be reported on port ***SysSigSpdAct***.
Actual speed of the EM with sign (forward/backwards) shall be reported on port ***SysSigSpdAct***.

The accuracy of the measured EM speed shall be Max(1% (of actual EM speed in rpm), 10 rpm)

267112 v2 EM Speed Safe [CS Released]

Purpose: Safe EM speed (ASIL B) is used for Chassis Ropp Function

Verification Method: Test
Legacy ID:

The safe EM speed measured the reference point (crankshaft for SPA, even shaft of 7DCT for CMA, electric motor shaft for PMA/Lotus) shall be reported as in the following:
Actual safe speed of the EM with sign (forward/backwards) shall be reported on port ***SysSigSpdActSafe***.
The accuracy of the measured EM speed shall be Max(1% (of actual EM speed in rpm), 10 rpm)

The Safe EM speed is used for ROPP solution, signal is used to identify vehicle standstill state when in specific failure mode. This signal should achive ASILB with quality factor and E2E protection.
Quality factor=3, if the rotation position sensor is no fault, and plausibility check of EM speed is ok
Quality factor=2, if the rotation position sensor is no fault, but plausibility check of EM speed is error.
Quality factor=1. initial value
Quality factor=0, if the rotation position sensor is erros, or ohter fault causing the speed is unreliability.

FTTI for safe EM speed sending, 200ms, include error dectection and sending out on network.

267239 v1 **Torque estimation steady state [CS Released]**

Purpose: Define the expectations on the torque estimations when it comes to accuracy
Verification Method: Test
Legacy ID:

The System shall, in steady state, control that torque control accuracy does not exceed ±4Nm when the torque is lower than100Nm; and the motor torque control accuracy shall not exceed ±4% when the torque is higher than 100Nm; The actual motor torque shall not exceed the minimum and maximum torque range.

6.3.3.13.5.4.1 PMA/SEA

685646 v4 **Torque estimation 800V Advanced [CS Released]**

Purpose: This value should reflect how much torque the propulsive management system can utilize from the HV Electric Drive system. Time horizon aspect will be that it is until a new value is reported so that it is not a instant value.
Advanced requirement aim to get more accurate data and enough torque during accelerating.

Verification Method: Test
Legacy ID:

Represented Signal in Requirement	Signal	Description
SysSigTqAct	IsgTqAct	Estimated Front Motor/Left Front Motor torque on the crankshaft
	IsgWhlTqAct	Estimated Front Motor/Left Front Motor torque on wheel
	IsgRiWhlTqAct	Estimated Right Front Motor torque on wheel
	WhlMotSysTqEst	Estimated Rear Motor torque on wheel
	WhlMotSysRiTqEst	Estimated Right Rear Motor torque on wheel
	ElecMot2TqAct	Estimated P1 Front Motor torque

The actual electric motor torque on the reference point (Wheel Axle for PMA) including all losses shall be

estimated/measured and signalled on port ***SysSigTqAct***.

PMA/SEA 800V :

- **When request output propulsive torque:**

Wheel Axle estimated torque
= Motor Axle estimated torque * Gear ratio * Torque accuracy Low limit * Transmission efficiency

- **When request output regenerate torque:**

Wheel Axle estimated torque
= Motor Axle estimated torque * Gear ratio / (Torque accuracy Low limit * Transmission efficiency)

For example:
If request output propulsive torque:
Motor Axle estimated torque = 400Nm , Gear ratio = 10.33 , Torque accuracy Low limit = 100% - 3% = 97% , Transmission efficiency = 97%.
So
Wheel Axle estimated torque = 400*10.33*97%*97% Nm = 3888 Nm.

For example:
If request output regenerate torque:
Motor Axle estimated torque = 400Nm , Gear ratio = 10.33 , Torque accuracy Low limit = 100% - 3% = 97% , Transmission efficiency = 97%.
So
Wheel Axle estimated torque = 400*10.33/(97%*97%) Nm = 4392 Nm.

685647 v4 **Available torque estimation 800V Advanced [CS Released]**

Purpose: Report the available torque that the EM can deliver for 800V.
Advanced requirement aim to get more accurate data and enough torque during accelerating.

Verification Method: Test

Legacy ID:

SysSigTqAvl		
Represented Signal in Requirement	Signal	Description
SysSigTqAvl	IsgTqAvl	Front Motor/Left Front Motor torque on motor shaft
	IsgWhlTqAvl	Front Motor/Left Front Motor torque on wheel
	IsgRiWhlTqAvl	Right Front Motor torque on wheel
	WhlMotSysTqAvl	Rear Motor torque on wheel
	WhlMotSysRiTqAvl	Right Rear Motor torque on wheel
	ElecMot2TqAvl	P1 Front torque Motor

Taking all limiting factors in the inverter and the EM, the HVED system shall estimate maximum and minimum available torque at the reference point (Axle for PMA) at any given moment and report on port ***SysSigTqAvl***.

The actual available torque reported shall be what the propulsive management can utilize until an updated value is reported. It shall not cause any problems to the component that this torque is applied.

TqAvlIsgMax **- is defined as the Maximum torque in Positive direction**
(Same as the absolute positive direction of EDS).

TqAvlIsgMin **- is defined as the Maximum torque in Opposite direction**

(Same as the absolute opposite direction of EDS , and it is always negative(With a minus sign)).

If the torque for any reason cannot be estimated the available torque shall be set to 0Nm.
(Fault state ,Derating and so on)

The torque shall be estimated for time window of TBD seconds.
The available torque shall be low pass filtered with a time constant of TBD ms (Calibratable).

Note: The calculation shall not limit the torque calculation based on power limits from Propulsion.

Maximum Axle Available torque
***SysSigTqAvl::MaxAvailTorq*= Motor Available torque * Gear ratio * Torque accuracy Low limit * Transmission efficiency**

Maximum Axle Available torque

- 如果电机可用负扭受减速器限制 且 无提升场景时，
  SysSigTqAvl::MinAvailTorq = 减速器反方向的承受的最大负扭矩
- 如果电机可用负扭不受减速器限制时，
  SysSigTqAvl::MinAvailTorq = 电机端最小可用扭矩 * 传动比 * 传动效率
- 如果电机可用负扭受减速器限制 但存在提升场景（如倒挡驱动）时，
  SysSigTqAvl::MinAvailTorq = 电机端最小可用扭矩 * 传动比 * 传动效率

685648 v1 EM Speed and EM Speed Safe Consistent PMA/SEA [CS Released]

Purpose: Consistent of motor speed signal processing
Verification Method: Analysis AND Test
Legacy ID:

SysSigSpdActSafe and ***SysSigSpdAct*** motor speed signal ,shall be processed by same method.

For example Low pass filtering is used,

SysSigSpdActSafe and ***SysSigSpdAct*** shall have same frequency low pass filtering then send out on CAN.

697338 v1 EM Speed Safe Zero [CS Released]

Purpose: Safe EM speed equal to Zero when it is very low.
Verification Method: Test
Legacy ID:

When Motor Speed ≤ 1 rpm， Send ***SysSigSpdActSafe*** = 0.

6.3.3.13.5.5 Safety

98307 v2 — Safe state [CS Released]

Purpose: For minimum impact on vehicle driveability when entering the safe state.
Verification Method: Test
Legacy ID: 357997

The inverter shall in safe state, in order of priority:

1. draw or produce no current on HVDC
2. produce no torque
3. minimize the impact on vehicle driveability

251236 v1 — Avoid wrong direction when drive-off [CS Released]

Purpose: To protect torque safety
Verification Method: Analysis AND Test
Legacy ID:

The Electric Drive System(EDS) shall not apply torque in a different direction than the driving direction unless there is a request for regeneration by the propulsion master
SysSigTqAllwd = 0x2 (Regenerative torque allowed) or 0x3(Propulsive and regeneraive torque allowed)
The torque monitoring mechanism shall compare the estimated torque with the request torque, if exceed TBD Nm, trigger the safe state by software (this value shall be calibratable).
The direction monitoring mechanism shall compare the torque request sign with the vehicle motion status ***SysSigVehMtnSts***, if inconformity, trigger the safe state by software(This value shall be possible to calibrate).

6.3.3.13.5.5.1 Active Discharge

194409 v2 — Active discharge diagnostics [CS Released]

Purpose: To be able to have the workshops repair cars where Active Discharge fails.

Verification Method: Test
Legacy ID: REQPROD:357911

The inverter shall have a diagnostic monitor that triggers when the voltage exceeds minimum voltage as specified in the DPR after Active Discharge has been requested for 900 ms.(CMA platform)

The inverter shall have a diagnostic monitor that triggers when the voltage exceeds minimum voltage as specified in the DPR after Active Discharge has been requested for 5 seconds.(MMA/PMA/SEA platform)

Other platform please select wanted one.

The fault reaction, treshold voltage and timing shall be specified in the Fault Matrix.

252644 v3 Active discharge [CS Released]

Purpose: To fulfil the requirements on safe voltage from the standard 6469
Verification Method: Test
Legacy ID:

If the active discharge process is initiated, when the HV-battery contactor is physically open, the voltage of the HVDC bus shall be reduced to below *ActiveDischargeSafeVoltageLevel (30V a.c. or 60V d.c.)* within *ActiveDischargeSafeVoltageTime (2s)*.
For applications where *ADSafeEnergyReqYesNo* equals "Yes" the voltage shall further be reduced to below *ActiveDischargeSafeEnergyLevel (<0.2J)* within *ActiveDischargeSafeEnergyTime (2s)*.

If the active discharge process is not finalized within the required time from the request, the system shall maintain the discharge process for at least *ActiveDischargeReTryTime (5s).*

For mode of operation during Active discharge refer to REQ;242474 Standy Pre-requisites

Parameter definition:
ActiveDischargeSafeVoltageLevel definition:
The voltage shall be reduced to less than 30V a.c. (rms) for a.c. circuits and 60V d.c. circuits

ActiveDischargeSafeEnergyLevel definition:

The total stored engergy of the circuit shall be <0.2J

134156 v2 Active discharge - Abort criterias [CS Released]

Purpose: To avoid that the active discharge will get stuck if there exist failures in the system that prevents it to be completed
Verification Method: Analysis AND Test
Legacy ID:

The active discharge shall remain active until either of below conditions are fulfilled:

- The HVDC voltage has been confirmed below the target voltage, *ActiveDischargeSafeVoltageLevel (30V a.c. or 60V d.c.)*.
- Active discharge has not been able to finish for *ActiveDischargeReTryTime (5s)*, from the time of the last active discharge request.
- A request for a functional safety safe state contradicting with the active discharge request

163127 v2 Active discharge safety [CS Released]

Purpose: To be sure active discharge is functional even without low voltage supply
Verification Method: Test
Legacy ID:

The active discharge shall be functional in the event that the low voltage supply of the corresponding high voltage components fails.

The active discharge should have path redundancy.

252654 v1 **Passive Discharge [CS Released]**

Purpose: Ensure that the HVDC voltage is harmless for humans when the vehicle is turned off or in case of a crash.
Verification Method: Test
Legacy ID:

The passive discharge shall be finished (that the DC voltage is reduced to less than 60V) within 2min when the high voltage contactor is open.

267636 v3 **Active discharge -Activation [CS Released]**

Purpose: Define activation criteria for when active discharge should be initiated and necessary actions to follow before activating active discharge
Verification Method: Test
Legacy ID:

The active discharge process shall be initiated when at least one of the following criteria are met:

- ***SysHvRlySts*** = "Open and request active discharge"
- ***SysSigCrashStsSafe***="Crash"
- CPSR = low for 50ms & BECM lost detected for 50ms.

The Active discharge process shall be activated every time the HV system is powered down.

In addition, all the following must be fulfilled to request active discharge:
-*Active discharge availability* indicates that at least one method of active discharge is available.
-*HVDC Voltage* is higher than parameter *ActvDchaUMin*.

6.3.3.13.5.6 Diagnostics

251404 v1 **E2E check fail diagnostic [CS Released]**

Purpose: To specify which signal to diagnose E2E
Verification Method: Test
Legacy ID:

The Electric Drive System shall diagnose the following received signal E2E status, including:
SysSigCrashStsSafe, ***SysSigHvRlySts***, ***SysSigTqAllwd***, ***SysSigVehMtnSts***

190632 v1 **Active discharge fail diagnostic [CS Released]**

Purpose: Inverter diagnostic
Verification Method: Test
Legacy ID:

The inverter shall monitor the active discharge status, the discharge status shall be in accordance with the active discharge command;
The inverter shall protect the active discharge resistor from thermal damage during discharge process;
The inverter shall set a DTC if discharge fails.

190634 v1 **Anti shuffle fail diagnostic [CS Released]**

Purpose: Inverter diagnostic
Verification Method: Test
Legacy ID:

The system shall set a DTC if severe oscillation occures.

190687 v1 **Torque monitor diagnostic [CS Released]**

Purpose: Inverter diagnostic
Verification Method: Test
Legacy ID:

The system shall compare the safety estimation torque with the non safety estimation torque, if exceed TBD Nm, the inverter should go to safe state.
The system shall compare the estimation torque with the requested torque, if exceed TBD Nm, the inverter should go to safe state.

6.3.3.13.5.7 P1 generator motor speed control

1054682 v1 **Speed Control mode - Performance [CS Released]**

Purpose: Speed Control mode - Performance
Verification Method: Test
Legacy ID:

电机处于转速模式时，应满足以下性能(电机端)：
1 空载：0-10000rpm/1s;
2 空载下误差范围：(<2000 rpm ±20rpm，>2000 rpm ±1%)。

1054683 v1 **Speed Control mode – Reduce available min torque when approaching the limit [CS Released]**

Purpose: Speed Control mode – Reduce available min torque when approaching the limit
Verification Method: Test
Legacy ID:

P1作用于曲轴端的可用正扭矩衰减：

- P1电机处于转速模式时，当 （P1电机扭矩 > 0Nm） 且 （P1可用电机正扭矩 - P1电机请求扭矩 < 10Nm TBD），即P1电机正扭矩调速能力已接近极限，此时应对CAN线上发送的P1作用于曲轴端的可用正扭矩 ElecMot2TqAvl***::MaxAvailTorq*** 进行线性衰减 以防止发动机扭矩超过P1扭矩导致飞转，最多不能衰减超过50%（TBD）。
  修正方法为：*ElecMot2TqAvlMax* + 衰减扭矩。(衰减扭矩为负值，不超过50%*(-1)**ElecMot2TqAvlMax*)
  如果衰减后的值＜0，则赋予0值。

P1作用于曲轴端的可用负扭矩衰减：

- P1电机处于转速模式时，当 （P1电机扭矩 < 0Nm） 且 （P1可用电机负扭矩 - P1电机请求扭矩 > -10Nm TBD），即P1电机负扭矩调速能力已接近极限，此时应对CAN线上发送的P1作用于曲轴端的可用负扭矩 ElecMot2TqAvl***::MinAvailTorq*** 进行线性衰减 以防止发动机扭矩超过P1扭矩导致飞转，最多不能衰减超过50%（TBD）。
  修正方法为：*ElecMot2TqAvlMin* - 衰减扭矩。(衰减扭矩为负值，不超过50%**ElecMot2TqAvlMin*)
  如果衰减后的值>0，则赋予0值。

1054684 v1 — Speed Control mode – Engine net torque [CS Released]

Purpose: Speed Control mode – Engine net torque
Verification Method: Test
Legacy ID:

电机处于转速模式时，应使用发动机净扭矩 EngTqReqFast 作为转速环的参考因数，提升转速模式的转速控制稳定性。

1054685 v2 — Power limitation in Speed Mode [CS Released]

Purpose: Power limitation in Speed Mode
Verification Method: Test
Legacy ID:

电驱动系统不得违反ElecMot2PwrLimMax和ElecMot2PwrLimMin信号设定的功率限制。

当P1处于**转速模式**时，功率为正功率（放电），高压侧正功率不能超过ElecMot2PwrLimMax放电功率限制。
（比如电机正转时，出正扭矩使转速提高，此时功率不应超过ElecMot2PwrLimMax）

当P1处于**转速模式**时，功率为负功率（回馈/发电），高压侧负功率不能超过ElecMot2PwrLimMin发电功率限制。
（比如电机正转时，出负扭矩使转速降低，此时功率不应超过ElecMot2PwrLimMin）

6.3.3.13.6 Signal Map

251605 v1 — Reference to Signals [CS Released]

Purpose: To show which interfaces that contains the signals
Verification Method: Documentation
Legacy ID:

Signal Map is found within LC: EM[Number]MotorControlInterface

EM1 (Main Front EM system)

EM2 (Auxillary Front EM system)
EM3 (Main Rear EM system)

Depending on System Voltage Signals are also mapped in

400V
LC: EM[Number]400VInterface

800V
LC: EM[Number]800VInterface

6.3.3.13.7 GHS2.0 Exclusive

361128 v1 **Inverter Mode feedback [CS Released]**

Purpose: status feedback to Propulsion system
Verification Method: Test
Legacy ID:

The HV e-drive system shall give the status feedback *WhlMotSysModSts* to propulsion system.

0 =EM initial mode;
1 = EM standby mode;
2 = EM torque control mode;
3 = EM SpdCtrl mode;
4 = EM ClutchOperation mode (clutch operation mode means clutch is disconnecting or connecting, when rear EDS operate in this mode, propulsion subsystem will not control the rear motor);
5 = EM PreChrg mode;
6 = EM PwrDwn mode;
7 = EM Flt
8 = WhlMotSysModStsTyp2_TcsCtrl

361145 v1 **Motor Stalling protection [CS Released]**

Purpose: Components protection
Verification Method: Test
Legacy ID:

Under the normal operation condition of the rear motor, and within a certain period of time (TBD, can be calibrated), the rear motor shall meet the stalling torque (TBD, calibration value);
When the rear motor is stalling, the motor shall be controlled by frequency reduction to make the NVH noise acceptable.

6.3.3.14 Logical component: Obd2Requirements

ID	Name	Version	Status	Last Changed
x0400000021F625C3	Obd2Requirements	(6)	CS Released	2025-11-24

On board diagnostics requirements for EDS

6.3.3.14.1 Included Virtual Function Clusters(VFCs)

6.3.3.14.2 Obd2Requirements: Receiver Ports

6.3.3.14.3 Obd2Requirements: Send Ports

6.3.3.14.4 Carconfig info:

Parameter number	Parameter name	Parameter value	Description
13	PROPULSION TYPE	03	Plugin hybrid vehicle
		02	Hybrid vehicle
46	DIAGNOSIS SYSTEM	03	JOBD
		02	EOBD
		01	OBD 2

6.3.3.14.5 Requirements

194360 v5 **Different parameter files for OBDII [CS Released]**

Purpose: To make the distinction between cars for OBDII and non-OBDII markets easier.
Verification Method: Test
Legacy ID: REQPROD:357937

To make the distinction between cars for OBDII and non-OBDII markets, there are two method.

Method 1 with different partnumber:
There shall be two different parameter files, one with the OBDII parameter set to ON, one with the parameter set to OFF.
Note:
The different parameter files must have unique part numbers.

Method 2 with Car Configuration parameter:

Propulsion Type CCP# 13 PROPULSION TYPE	OBDII ON/OFF
Plugin hybrid vehicle	ON
Hybrid vehicle	ON
other values	OFF

194365 v5 **Differentiate between OBDII relevant markets [CS Released]**

Purpose: To identifiy if the ECU is OBD II relevant or not.
Verification Method: Test
Legacy ID: REQPROD:357938

There must be an parameter where we can calibrate if the SW is OBD II relevant or not.
This parameter shall be accessed by relevant OBD II SWCs.

195130 v4 Scantool functionality [CS Released]

Purpose: DCM sends requests to Scantool for determening if the application should approve or reject the diagnostic request. Scantool can make the DCM abort the treatment of a request under certain circumstances.
Verification Method: Test
Legacy ID: REQPROD:431305

If a function call to Scantool is an OBD service (specified with a service ID in the range of OBD as specified in ISO regulations) sent from external tester and OBD is calibrated to be OFF, the Scantool shall set return value of the function call NOK.
Otherwise Scantool shall set the return value of the function call to OK.

Propulsion Type CCP#13 PROPULSION TYPE	OBDII ON/OFF
Plugin hybrid vehicle	ON
Hybrid vehicle	ON
other values	OFF

251406 v2 MIL Indication request [CS Released]

Purpose: To indicate need for service to driver as per OBD2 Legislation
Verification Method: Test
Legacy ID: REQPROD:431321

Req: The unit shall indicate the “MIL – Malfunction Indication Lamp” via interface Port:***SysSigErrIndcnReq***
Req: Port:***SysSigErrIndcnReq*** shall be set to “1”, when the unit detects ““MIL – Malfunction Indication Lamp” should be switched “ON”, and Port: IsgErrIndcnReq shall be set to “0”, when the unit detects ““MIL – Malfunction Indication Lamp” should be switched “OFF”
Req: It shall be possible to override the behaviour of the Port: ***SysSigErrIndcnReq*** via separate calibration switch, such that Port: ***SysSigErrIndcnReq*** shall indicate “0”.
Info: AUTOSAR provides standard API to get the current state of MIL request. This is called as Dem_GetIndicatorStatus()

251407 v2 Clear DTCs after a regrogramming event [CS Released]

Purpose: To fulfill legal requirements according to California Code of regulations 1968.2.
Verification Method: Test
Legacy ID: 428391

After reprogramming event shall readiness information be cleared (i.e., set all monitors to indicate “not complete”) for all monitored components and systems in all modules that report supported readiness for a readiness bit other than the comprehensive components readiness. Reprogrammed ECU shall do this by sending CAN message to ECM on port . Send this ***SysSigClrRdyReq*** message all the time during the first DCY after the programming event. Length of the DCY shall be at least 60 seconds. This shall be done only during the first DCY after reprogramming

event, but if the length of the DCY is shorter than 60 seconds then message shall be send also during the next DCY until DCY with the minimum length of 60 seconds is reached.

Reprogramming event shall be detected by the following method: When Driving Cycle has reached 60 seconds or more, value AAhex shall be stored in a KAM (Keep-alive memory) or a NVRAM (Non-volatile RAM) memory. If software is downloaded to the ECU then this area shall be cleared to value (=00hex or = FFhex).

Note: This requirement is valid for hybird vehicle only.

251408 v2 OBD snapshot signals [CS Released]

Purpose: Snapshot information by OBD
Verification Method: Inspection
Legacy ID:

If the DTC code is OBD relevant in SSDS Part II, then when it is triggered, the following signal should be recorded in snapshot.
SysSigEngNSafe, ***SysSigEngT***, ***SysSigLoadValCalcd***, ***SysSigThrPos,*** ***VehicleSpeed,*** ***TimeSinceEngineStart,*** ***VehBattUSysU,*** ***AmbTRaw***

251409 v1 OBD warm up cycle [CS Released]

Purpose: OBD warm up cycle definition
Verification Method: Inspection
Legacy ID:

OBD warm up cycle definition from the signal ***SysSigMonrCycCompl***.

251410 v1 IUPR calculation [CS Released]

Purpose: To fulfill on board diagnostics requirements
Verification Method: Test
Legacy ID:

Refer to GB 18352.6-2016
J.3.4.2 numerator counter
J.3.4.2.1 Definition：Refer to GB 18352.6。
J3.4.2.2 The increse of the numerator counter: Refer to GB 18352.6

J3.4.3 Denominator counter
Denominator is triggered by propulsion subsystem，and signal ***SysSigDenOfDiagRateGen*** jumps from 0 to 1 indicating the denominator increase。

J3.4.4 IUPR ratio
J3.4.4.1 Definition：IUPR ratio equals numerator/denominator。

251411 v1 Clear permanent fault [CS Released]

Purpose: Clear permanent fault

Verification Method: Test
Legacy ID:

All permanent DTCs (releated to the emission regulation) should be cleared when it's MIL off logic is fullfilled. (no fault during three continous drive cycle)

If the permanent DTC can not be cleared when the MIL off, it should be cleared by the indication of ***SysSigDrvCycToClrPrmntDtc*** when the test run passes.

251412 v2 MonrCycCmpl and DrvCycToClrPrmntDtc Reciever [CS Released]

Purpose: To clarify how to use the MonrCycCmpl signal and *DrvCycToClrPrmntDtc signal*
Verification Method: Test
Legacy ID: REQPROD:342890

The "**Erasing permanent fault code, driving cycle condition"** (CAN signal ***SysSigDrvCycToClrPrmntDtc***) shall be used.
The "**Erasing confirmed fault code, driving cycle condition**" (CAN signal ***SysSigMonrCycCompl***) shall be used.

California Code of Regulations 1968.2
(2.4) Erasing a confirmed fault code.
The OBD II system may erase a confirmed fault code if the identified malfunction has not been again detected in at least 40 engine "***Erasing confirmed fault code, driving cycle condition***", ***SysSigMonrCycCompl*** and the MIL is presently not illuminated for that malfunction.

The OBD definition of "***Erasing permanent fault code, driving cycle condition***" complete is transmitted over CAN from ECM via CAN signal ***SysSigDrvCycToClrPrmntDtc***

(d)(2.5.2)(B)
(iii)The following criteria are satisfied on any single driving cycle (which may be a different driving cycle than that in which the criteria of section (d)(2.5.2)(B)(i) are satisfied):
a. Except as provided in section (d)(2.5.2)(B)(iii)f. below, cumulative time since engine start is greater than or equal to 600 seconds;
b. Except as provided in section (d)(2.5.2)(B)(iii)e. below, cumulative vehicle operation at or above 25 miles per hour occurs for greater than or equal to 300 seconds (medium-duty vehicles with diesel engines certified on an engine dynamometer may use cumulative operation at or above 1150 rpm in lieu of at or above 25 miles per hour for purposes of this criteria);
c. Continuous vehicle operation at idle (i.e., accelerator pedal released by driver and either vehicle speed less than or equal to one mile per hour or engine speed less than or equal to 200 rpm above normal warmed-up idle (as determined in the drive position for vehicles equipped with an automatic transmission)) for greater than or equal to 30 seconds; and
….
f. For hybrid vehicles, manufacturers shall use "cumulative propulsion system active time" in lieu of "cumulative time since engine start" for the criterion in section (d)(2.5.2)(B)(iii)a.

Additional req for erasing permanent fault code according to **California Code of Regulations 1968.2**
(i) The monitor has run and made one or more determinations during a driving cycle that the malfunction of the component or the system is not present and has not made any determinations within the same driving cycle that the malfunction is present;
(ii) The monitor has not made any determinations that the malfunction is present subsequent to the most recent driving cycle in which the criteria of section (d)(2.5.2)(B)(i) are met;

251413 v1 High Voltage Battery State of Charge [CS Released]

Purpose: To be able to provide snap shot information about the system
Verification Method: Test
Legacy ID:

The system shall monitor the High Voltage Battery State of Charge ***SysSigHVBattSoc*** and make this information available in freeze frame data.

225918 v2 PID $1F [CS Released]

Purpose: PID $1F
Verification Method: Test
Legacy ID:

Time since engine starts or propulsion on with movement for PID $1F.
1. For component with KL15 connection, start counting when DrvgCyc=1, or PrpsnSysActv=1 (DrvgCyc is the same with PrpsnSysActv in hybrid vehicles), stop the counting when KL15=OFF.
2. For component without KL15 connection, start counting when DrvgCyc=1, or PrpsnSysActv=1 (DrvgCyc is the same with PrpsnSysActv in hybrid vehicles), stop the counting when UsageMode≠Driving (UsageMode defined in VehModMngtGlbSafe1 signal group).

3. After the PID $1F starting the counting,
3.1 If DrvgCyc signal missing, And UsageMode signal missing, then continue the timing counting till to ECU sleep
3.2 If DrvgCyc signal missing, And UsageMode signal live and valid, then continue the timing counting till to UsageMode≠Driving

251473 v1 Clear confirmed fault [CS Released]

Purpose: Clear confirmed DTC
Verification Method: Test
Legacy ID:

All confirmed DTCs should be cleared when the aging counter counts to the limit defined in SSDS.

297679 v1 PID $1C [CS Released]

Purpose: PID $1C

Verification Method: Test
Legacy ID:

PID $1C:OBD requirements to which vehicle or engine is certified.

Car Configuration Parameters DIAGNOSIS SYSTEM CCP# 46	OBDII different market
OBD 2	OBD2 for America and China
EOBD	OBD for Europe
JOBD	OBD for Japan

6.3.3.14.6 Signal Map

251607 v1 **Reference to Signals [CS Released]**

Purpose: To show which interfaces that contains the signals
Verification Method: Documentation
Legacy ID:

Signal Map is found within LC: EM[Number]Obd2Interface

EM1 (Main Front EM system)
EM2 (Auxillary Front EM system)
EM3 (Main Rear EM system)

6.3.3.15 Logical component: RmsComponentRequirements

ID	Name	Version	Status	Last Changed
x040000002228C6E9	RmsComponentRequirements	EEA3.525R3(5b3a1)	Released	2025-11-28

Remote Monitor System for Electric Drive System
Use for all Front Main EM ECU configurations

6.3.3.15.1 Included Virtual Function Clusters(VFCs)

6.3.3.15.2 RmsComponentRequirements: Receiver Ports

6.3.3.15.3 RmsComponentRequirements: Send Ports

6.3.3.15.4 Carconfig info:

Parameter number	Parameter name	Parameter value	Description
14	REMOTE MONITORING SYSTEM	02	RMS Activated
		01	RMS Disactivated
4	ELECTRIC DRIVELINE	08	Electric Motor RWD
		07	Electric Motor FWD
		06	Electric Motor AWD
695	Electric Motor Drive Type	03	Synchronous motor
		07	Front single synchronous motor + rear double synchronous motor

6.3.3.15.5 Requirements

242036 v5 **Remote Monitoring System data [CS Released]**

Purpose: Signal data from the electric drive system to fulfill Remote Monitoring System (RMS) legislations
Verification Method: Test
Legacy ID:

The HV Electric Drive system shall provide the information in the table below to be used by the remote monitoring system<GBT32960.3>.
The data is provided in port ***SysSigGeneric***. (If the signal sending type is "OnWrite", RMS Activated in CCP REMOTE MONITORING SYSTEM CCP#14 will be the trigger event, otherwise the signal can be sending periodically.)

Signal Name	Description	Requirement
EMQnty	Quantity of electrical machines	The quantity of electrical machines is 1,2,3..., according to Car Configuration VEHICLE (Parameter 1) and TRANSMISSION DRIVELINE(Parameter 3)for different project, MGM/IGM/IEM shall deal with the car configuration parameter during startup, and r the quantity of the electrical machine on this vehicle, encode the number into IgmGe signal and send out on bus.
EMSeqNr	Sequence Number of electrical machine	PMA: MGM=1, IEM=1 or 2 according to CarConfig VEHICLE TYPE (Parameter 1) and TRANSMISSION DRIVELINE(Parameter 3) SPA: IGM=1, IEM=1 or 2 according to CarConfig VEHICLE TYPE (Parameter 1) and TRANSMISSION DRIVELINE(Parameter 3) CMA1.5: IGM=1
ModStatusRms	State of electrical machine	0x00 invalid TURE Condition xModSts==0x04(Invalid) if voltage control mode is not applicable 0x01 PwrCns TURE Condition xModSts==0x02(TqCtrl) or xModSts==0x03(SpdCtrl) And xIDc*xUDc>0, hysteresis as a parameter, default value is TBD 0x02 PwrGen TURE Condition 1 xModSts==0x02(TqCtrl) or xModSts==0x03(SpdCtrl) And xIDc*xUDc<=0, hysteresis as a parameter, default value is TBD 0x02 PwrGen TURE Condition 2 xModSts==0x04(UDcCtrl) if voltage control mode exist for the vehicle 0x03 OffSts TURE Condition xModSts==0x0(Init) or xModSts==0x06(PwrDwn) or or xModSts==0x05(Precharge) 0x04 RdySts TURE Condition xModSts==0x01(std) 0x05 Abnormal TURE Condition xModSts==0x07(Flt) 0x06~0x0F Invalid TURE Condition xModSts==0x04(Invalid) if voltage control mode is not applicable Default 0x03(Offset)
InvrtTAlrmSt	Electrical machine controller temperature warning	If xInvrT > TBD, set flag=1, else set flag=0, warning level 2
MotTAlrmSt	Electrical machine temperature warning	If xMotT > TBD, set flag=1, else set flag=0, warning level 2
DTCLow	DTC low byte for UDS or OBD	Sending the inverter DTC with DTC status on propulsion CAN. If two or more DTC at the time series, sending them periodically every 2s

DTCMid	DTC middle byte for UDS or DTC low byte for OBD	Sending the inverter DTC with DTC status on propulsion CAN. If two or more DTC at the time series, sending them periodically every 2s
DTCHig	DTC high byte for UDS or OBD	Sending the inverter DTC with DTC status on propulsion CAN. If two or more DTC at the time series, sending them periodically every 2s
DTCSts	DTC status for UDS or OBD	Sending the inverter DTC with DTC status on propulsion CAN. If two or more DTC at the time series, sending them periodically every 2s
TypeInfo	Electric Drive System Type Information	Indicate different EDS suppliers. OEM defines this value. 0=not defined 1=PMA1 VseA, with ratio=10.134 2=PMA1 Nidec, with ratio=10.294 3=PMA1 VREMT, with ratio=11.56 4=SPAL Scheaffler, with ratio=9.071 5=SPA Simens, with ratio=10.008 6=PMA2 VREMT, with ratio=11.209 7=EPA ZF,with radio=9.187 8=EPA Polestar,with 2-gear ratio=7.155 and 13.645 9=PMA2 UAES,with ratio=11.067
FltAlrmSt	Fault alarm status	Any DTC will set this alarm
SpdAlrmSt	Overspeed alarm status	Over speed will set this alarm
IacAlrmSt	Phase Current Sensor alarm status	Any DTC related with phase current sensor will set this alarm
UDcAlrmSt	HVDC voltage sensor alarm status	Any DTC related with HVDC voltage sensor will set this alarm
RslAlrmSt	Resolver sensor alarm status	Any DTC related with resolver sensor will set this alarm
ADAlrmSt	Active discharge alarm status	Active discharge failure will set this alarm

1045056 v2 Remote Monitoring System data for 800V 900V [CS Released]

Purpose: Signal data from the electric drive system to fulfill Remote Monitoring System (RMS) legislations

Verification Method: Test

Legacy ID:

This RMS requirement is for 800V/900V vehicle.

❖ The HV Electric Drive system shall provide the information in the table below to be used by the Remote Monitoring System <GBT32960.3>. The data is provided in port ***SysSigGenericNew1RMS 和 SysSigGenericNew2RMS***.

❖ The used Car Configuration Parameter

Parameter number	Parameter name	Value (Hex)	Value Description
CCP#14	Remote Monitoring System	01	RMS Disactivated
		02	RMS Activated
		80	RMS Activated(2025)

❖ RMS activate

If ***CCP#14*** == "01", RMS function need to Disactivated.

If ***CCP#14*** == "80", RMS function need to Activate. HVED shall learn quantity and sequence number of electrical machines and so on. And send ***SysSigGeneric*** signal on CAN.

Signal Name	Description	Requirement
EMQnty	Quantity of electrical machines	The quantity of electrical machines is 1,2,3..., According to the motor configuration the EDS have , HVED shall encode the number into ***SysSigGeneric*** signal and send out on CAN. If **XCM2/MGM** (Front Axle have One Controller with One Motor): EMQnty(IgmGenericNew1RMS) = "1"; If **IEM3/XCUP/XCUP3** (Rear Axle have One Controller with One Motor): EMQnty(IemGenericNew1RMS) = "1"; If **IEM2/XCUP2**(Rear Axle have One Controller with Two Motor distributed in Left and Right): EMQnty(IemGenericNew1RMS) = "1",EMQnty(Iem2GenericNew1RMS) = "1"; Total motor quantity will be calculate by Telematics Applications.
EMSeqNr	Sequence Number of electrical machine	The Sequence Number of electrical machines is 1,2,3..., According to the motor configuration the EDS have , HVED shall encode the number into ***SysSigGeneric*** signal and send out on CAN. If **XCM2/MGM** (Front Axle have One Controller with One Motor): EMSeqNr(IgmGenericNew1RMS) = "1"; If **IEM3/XCUP/XCUP3** (Rear Axle have One Controller with One Motor): EMSeqNr(IemGenericNew1RMS)= "1"; If **IEM2/XCUP2**(Rear Axle have One Controller with Two Motor distributed in Left and Right): EMSeqNr(IemGenericNew1RMS) = "1",EMSeqNr(Iem2GenericNew1RMS) = "2"; Sequence Number will be calculate by Telematics Applications.

Signal Name	Description	Requirement
ModStatusRms	State of electrical machine	0x00 invalid TURE Condition(One of below fulfill): • if Speed mode not applicable，but ***SysSigModSts*** send 0x03(SpdCtrl); • if UDC mode not applicable，but ***SysSigModSts*** send 0x04(UdcCtrl); • if Sync mode not applicable，but ***SysSigModSts*** send 0x05(PreChrg); • if E-cluch not applicable，but ***SysSigModSts*** send 0x09(CluOpe); 0x01 PwrCns TURE Condition (One of below fulfill): • ***SysSigModSts***==0x02(TqCtrl) and ***SysSigIDc*******SysSigUDc*** > 0; (hysteresis as a parameter, default value is TBD) • ***SysSigModSts***==0x03(SpdCtrl), if Speed mode applicable And ***SysSigIDc*******SysSigUDc*** > 0(hysteresis as a parameter, default value is TBD); • ***SysSigModSts***==0x08(TcsCtrl), if TCS mode applicable And ***SysSigIDc*******SysSigUDc*** > 0(hysteresis as a parameter, default value is TBD); 0x02 PwrGen TURE Condition (One of below fulfill): • ***SysSigModSts***==0x02(TqCtrl) and ***SysSigIDc*******SysSigUDc*** <=0 (hysteresis as a parameter, default value is TBD); • ***SysSigModSts***==0x03(SpdCtrl), if Speed mode applicable And ***SysSigIDc*******SysSigUDc*** <= 0(hysteresis as a parameter, default value is TBD);

		• ***SysSigModSts***==0x04(UdcCtrl), if UDC mode applicable And ***SysSigIDc*******SysSigUDc*** <= 0(hysteresis as a parameter, default value is TBD); • ***SysSigModSts***==0x08(TcsCtrl), if TCS mode applicable And ***SysSigIDc*******SysSigUDc*** <= 0(hysteresis as a parameter, default value is TBD); 0x03 OffSts TURE Condition (One of below fulfill): • When HVED initialize, send Default value • ***SysSigModSts***==0x00(Init); • ***SysSigModSts***==0x06(PwrDwn); 0x04 RdySts TURE Condition(One of below fulfill): • ***SysSigModSts***==0x01(Std); • ***SysSigModSts***==0x09(CluOpe),if Clutch function applicable; 0x05 Abnormal TURE Condition(One of below fulfill): • ***SysSigModSts***==0x07(Flt); 0x06~0x0F Reserve.

Signal Name	Description	Requirement
InvrT	Electrical machine controller temperature	***SysSigGenericInvrT*** = ***SysSigInvrT***
MotT	Electrical machine temperature	***SysSigGenericMotT*** = ***SysSigMotT***
EMSpdAct	Electrical machine Actual speed	**Speed of motor shaft** ***SysSigGenericEMSpdAct*** =***SysSigSpdAct***
EMTqAct	Electrical machine Actual Torque	**Torque of motor shaft** ***SysSigGenericEMTqAct*** =***SysSigTqAct***/gear ratio
Udc	DC voltage	***SysSigGenericUdc*** = ***SysSigUDc***
Idc	DC current	***SysSigGenericIdc*** = ***SysSigIDc*** * 0.95 （因电机母线电流存在上偏差，会导致电耗达到超100%的效率值。所以为避免此问题，RMS上传的母线电流应取下偏差值：补偿95%.）

Signal Name	Description	Requirement
InvrtTAlrmSt	Electrical machine controller temperature warning	If ***SysSigInvrT*** > TBD,not affecting normal driving of vehicle, Set flag=1 (fault Level 1); If ***SysSigInvrT*** > TBD, affecting vehicle performance and requiring the driver to limit driving,Set flag=2 (fault Level 2); Else set flag=0(No fault).
MotTAlrmSt	Electrical machine temperature warning	If ***SysSigMotT*** > TBD,not affecting normal driving of vehicle, Set flag=1 (fault Level 1); If ***SysSigMotT*** > TBD, affecting vehicle performance and requiring the driver to limit driving,Set flag=2 (fault Level 2); Else set flag=0(No fault).
FltAlrmSt	Fault alarm status	Any DTC will set this alarm
SpdAlrmSt	Over speed alarm status	Over speed will set this alarm
IacAlrmSt	Phase Current Sensor alarm status	Any DTC related with phase current sensor will set this alarm
UDcAlrmSt	HVDC voltage sensor alarm status	Any DTC related with HVDC voltage sensor will set this alarm
RslAlrmSt	Resolver sensor alarm	Any DTC related with resolver sensor will set this alarm

	status	
ADAlrmSt	Active discharge alarm status	Active discharge failure will set this alarm

6.3.3.15.6 Signal Map

251608 v1 **Reference to Signals [CS Released]**

Purpose: To show which interfaces that contains the signals
Verification Method: Documentation
Legacy ID:

Signal Map is found within LC: EM[Number]Obd2Interface

EM1 (Main Front EM system)
EM2 (Auxillary Front EM system)
EM3 (Main Rear EM system)

Note: If Interface is not allocated then shall all requirements related to said interface be disregarded.

6.3.3.16 Logical component: EngineStartPriorityRequirements

ID	Name	Version	Status	Last Changed
x0400000007DBE257	EngineStartPriorityRequirements	(2)	CS Released	2021-03-31

LC only for Hybrid Systems

6.3.3.16.1 Included Virtual Function Clusters(VFCs)

6.3.3.16.2 EngineStartPriorityRequirements: Receiver Ports

6.3.3.16.3 EngineStartPriorityRequirements: Send Ports

6.3.3.16.4 Requirements

252580 v1 **Prioritization of motors for engine start [CS Released]**

Purpose: Too specify the general order of motors to start en engine with
Verification Method: Test
Legacy ID:

In setups where there are multiple Electrical Motors(EM) in combination with a combustion engine there shall be prioritization of which EM shall be used to start the Combustion engine, unless the combustion engine has it's own start motor.

1. P0 -BSG
2. P1 -On Engine Crank Shaft
3. P2 -On Transmission Input Shaft
4. P3 -On Transmission Output Shaft

If an EM with a higher priority as a start motor has a fault but can still be turned, it shall be possible to start the combustion engine with a lower priority EM.

253286 v1 **Maximize motor ability to start engine [CS Released]**

Purpose: Ensure successful engine start, engine stop and immediate engine start (i.e. at sudden driver change-of-mind). Avoid engine stall due to too low available motor torque.
Verification Method: Test
Legacy ID:

When the signal ***SysSigEngSt1WdSts*** **=** StartingInProgress(Value = 4) OR ***SysSigEngSt1WdSts*** = Running_StartingInProgress(Value = 7), then the motor drive system shall maximize its motoring torque ability such that full motoring torque is available for at least 3 seconds. The motor drive shall increase its torque ability by easening up its slow thermal motor limits.

After 3 seconds, the motor drive may use its standard torque limitations.

6.3.3.16.5 Signal Map

253303 v1 **Reference to Signals [CS Released]**

Purpose: To show which interfaces that contains the signals
Verification Method: Documentation
Legacy ID:

Signal Map is found within LC: EM[Number]MotorControlInterface

EM1 (Main Front EM system)
EM2 (Auxillary Front EM system)
EM3 (Main Rear EM system)

Depending on System Voltage Signals are also mapped in

400V
LC: EM[Number]400VInterface

800V
LC: EM[Number]800VInterface

6.3.3.17 Logical component: TerminatingECUsonCANorCANFD

ID	Name	Version	Status	Last Changed
x0400000022E8F6B0	TerminatingECUsonCANorCANFD	(3)	CS Released	2026-01-09

Terminal node definition according to the topology.

6.3.3.17.1 Included Virtual Function Clusters(VFCs)

6.3.3.17.2 TerminatingECUsonCANorCANFD: Receiver Ports

6.3.3.17.3 TerminatingECUsonCANorCANFD: Send Ports

6.3.3.18 Logical component: EM2GeneralInterface

ID	Name	Version	Status	Last Changed
x04000000224AE8DC	EM2GeneralInterface	EEA 25R3(5b1a4)	CS Released	2025-12-09

High Voltage Electric Drive - Front Part
Auxillary Front Electric Drive System
Configuration From:

1. Front RIght Hub Motor
2. DHT(PCM)(P1)

6.3.3.18.1 Included Virtual Function Clusters(VFCs)

6.3.3.18.2 EM2GeneralInterface: Receiver Ports

Port	Interface	DataElement	DataType	E2E ID
CooltFlowForDtElec/	CooltFlowForDtElec/;5	CooltFlowForDtElec/;	CooltFlow2/;2	
Description				
Predicted coolant low for Electric Drivetrain (ED) coolant circuit.				

Port	Interface	DataElement	DataType	E2E ID
CooltTSigForDtElec/	CooltTSigForDtElec/;5	CooltTSigForDtElec/;	CooltTHiResl/;2	
Description				
Coolant temperature measured by sensors installed in emotor cooling citcuit.				

Port	Interface	DataElement	DataType	E2E ID
DrvgCyc/	DrvgCyc/;3	DrvgCyc/;	OnOff1/;2	
Description				
Transition from 0 to 1 indicates a start of a new Driving Cycle.				

Port	Interface	DataElement	DataType	E2E ID
ElecMot2ActvDampgModReq/	ElecMot2ActvDampgModReq/;1	ElecMot2ActvDampgModReq/;	WhlMotSysActvDampgModCodSts/;2	
Description				
Generator motor active damping mode request 0 = No damping 1 = Low damping 2 = Medium damping 3 = High damping				

Port	Interface	DataElement	DataType	E2E ID
ElecMot2EngSpdOfActvDampg/	ElecMot2EngSpdOfActvDampg/;1	ElecMot2EngSpdOfActvDampg/;	N1/;2	

Description

Generator motor reference speed of active damping

Port	Interface	DataElement	DataType	E2E ID
ElecMot2ModReq/	ElecMot2ModReq/;1	ElecMot2ModReq/;	IsgModTyp1/;4	

Description

Generator motor operation mode request
1: standby
2: torque control
3: speed control
4: UDC control
5: Speed Control Dead Zone

Port	Interface	DataElement	DataType	E2E ID
ElecMot2PwrLimMax/	ElecMot2PwrLimMax/;1	ElecMot2PwrLimMax/;	PrpsnPwrLimMax/;2	

Description

Maximum generator motor drive system powerlimit on DC side. Positive power means power flowing into inverter.

Port	Interface	DataElement	DataType	E2E ID
ElecMot2PwrLimMin/	ElecMot2PwrLimMin/;1	ElecMot2PwrLimMin/;	PrpsnPwrLimMin/;2	

Description

Minimum generator motor drive system powerlimit on DC side. Positive power means power flowing into inverter.

Port	Interface	DataElement	DataType	E2E ID
ElecMot2SpdReq/	ElecMot2SpdReq/;1	ElecMot2SpdReq/;	IsgSpdWSgnTyp/;2	

Description

Generator motor speed request

Port	Interface	DataElement	DataType	E2E ID
ElecMot2TqReq/	ElecMot2TqReq/;1	ElecMot2TqReq/;	Tq34/;1	

Description

Generator motor torque request

Port	Interface	DataElement	DataType	E2E ID
EngCooltInhbFlow/	EngCooltInhbFlow/;4	EngCooltInhbFlow/;	CooltInhb/;2	

Description

Inhibitation of non-crucial coolant flow requests. Sent out to inform that the eninge is in heat up phase and that all additional coolant flow will postphone the time when the engine has reached normal operating temperature.

Port	Interface	DataElement	DataType	E2E ID
EngSt1WdSts/	EngSt1WdSts/;4	EngSt1WdSts/;	EngSt1WdSts/;1	137

Description

Signaling the state of the powerpack

Port	Interface	DataElement	DataType	E2E ID
HvBattUDc/	HvBattUDc/;2	HvBattUDc/;	BCLRqrdU/;1	

Description

Actual Hv Battery pack voltage (measuered before contactors, i.e. always giving pack voltage).

Port	Interface	DataElement	DataType	E2E ID
HvBattUDynMaxLim/	HvBattUDynMaxLim/;2	HvBattUDynMaxLim/;	NewBattVolt/;1	

Description

The pack voltage high limit of the Hv Battery, dynamically based on cell voltage distribution.

Port	Interface	DataElement	DataType	E2E ID
HvBattUDynMinLim/	HvBattUDynMinLim/;2	HvBattUDynMinLim/;	NewBattVolt/;1	

Description

The pack voltage low limit of the Hv Battery, dynamically based on cell voltage distribution.

Port	Interface	DataElement	DataType	E2E ID
HvSysSts/	HvSysSts/;3	HvSysSts/;	HvSysSts/;2	

Description

Hv System Status during integrity test. Informs vehicle that longer Hv System start up time is necessary to perform Extended Integrity Test due to Hv System fault or safety issue.

Port	Interface	DataElement	DataType	E2E ID
PrpsnFbLimnIndcn/	PrpsnFbLimnIndcn/;2	PrpsnFbLimnIndcn/;	UInt4/;2	

Description

Feedback signal that the present limitation has lead to a recofiguraton that has affected the driver.

Port	Interface	DataElement	DataType	E2E ID
PtEngTqAct/	PtEngTqAct/;1	PtEngTqAct/;	PtTqType/;2	

Description

Engine Net torque actual without P1

Port	Interface	DataElement	DataType	E2E ID
VehModMngtGlbSafe1/	VehModMngtGlbSafe1/;7	VehModMngtGlbSafe1/;	VehModMngtGlbSafe1/;2	116

Description

VehModMngtGlbSafe1 is a system signal group sent globally in an end to end protected record that consists of CarModSts1, CarModSubtypWdCarModSubtyp, UsgModSts, EgyLvlElecMai, EgyLvlElecSubtyp, PwrLvlElecMai, PwrLvlElecSubtyp and FltEngyCnsWdSts

CarModSts1 : Factory, Transport, Dyno, Normal. Crash

UsgModSts : Abandoned, Inactive, Convenience, Active, Driving

CarModSubtypWdCarModSubtyp: examples for CarModSubtyp: Factory Driving, Transport Driving

6.3.3.18.3 EM2GeneralInterface: Send Ports

Port	Interface	DataElement	DataType	Generation Type	E2E ID
ElecMot2IDc/	ElecMot2IDc/;1	ElecMot2IDc/;	Ampr1/;1	Periodic	

Description

DC current measured in the auxiliary motor.

Port	Interface	DataElement	DataType	Generation Type	E2E ID
ElecMot2UDc/	ElecMot2UDc/;3	ElecMot2UDc/;	U52/;1	Periodic	

Description

The voltage level on the inverter HV DC link

6.3.3.18.4 Signal Map

252586 v2a1 **Signal Map [CS Released]**

Purpose: To map SystemSignal to actual signals used on CAN
Verification Method: Documentation
Legacy ID:

Signal Map

System Signal	Actual Signal	Description
SysSigIDc	ElecMot2IDc	Measured current on the DC-link
SysSigUDc	ElecMot2UDc	Measured voltage on the DC-link
SysSigVehModMngt	VehModMngtGlbSafe1	Vehicle Mode Management signal
SysSigHvBattUDynMinLim	HvBattUDynMinLim	Dynamic minimum voltage limit from HV battery
SysSigHvBattUDynMaxLim	HvBattUDynMaxLim	Dynamic maximum voltage limit from HV battery
SysSigCooltTSigForDtElec	CooltTSigForDtElec	Temperature of the Coolant for Electric Drivetrains
SysSigCooltFlowForDtElec	CooltFlowForDtElec	Coolant flow for Electric Drivetrains
SysSigEngCooltInhbFlow	EngCooltInhbFlow	Coolant flow inhibitation for Propulsion Engine
SysSigEngSt1WdSts	EngSt1WdSts	Engine status
SysSigHvBattUdc	HvBattUDc	HV battery Voltage DC
SysSigHvBattuDynMinLim	HvBattUDynMinLim	Dynamic Voltage Limit of HV battery
SysSigHvBattuDynMaxLim	HvBattUDynMaxLim	Dynamic Voltage Limit of HV battery
SysSigActvDampgModReq	ElecMot2ActvDampgModReq	Active dampening mode request
SysSigEngSpdOfActvDmpg	ElecMot2EngSpdOfActvDampg	Engine speed target for active dampening
SysSigModReq	ElecMot2ModReq	Drive mode request to the Electric Drive System
SysSigPwrLimMin	ElecMot2PwrLimMin	Power limitaion to the Electric Drive System
SysSigPwrLimMax	ElecMot2PwrLimMax	Power limitaion to the Electric Drive System
SysSigSpdReq	ElecMot2SpdReq	Speed request to the Electric Drive System
SysSigTqReq	ElecMot2TqReq	Torque request to the Electric Drive System
SysSigFbLimnIndcn	PrpsnFbLimnIndcn	Feedback signal of Limitation Indication

SysSigDrvCyc	DrvgCyc	Driving Cycle indicator

6.3.3.19 Logical component: SecOCSecMngFVSIGM

ID	Name	Version	Status	Last Changed
x040000002784CD3E	SecOCSecMngFVSIGM	(3)	CS Released	2026-07-08

The FVM salve node receives TripCnt&RestCnt in synchronization packets for MAC authentication

6.3.3.19.1 Included Virtual Function Clusters(VFCs)

6.3.3.19.2 SecOCSecMngFVSIGM: Receiver Ports

Port	Interface	DataElement	DataType	E2E ID
TripResetSyncMesage/	TripResetSyncMesage/;4a1	TripResetSyncMesage/;	TripResetSync/;1	

Description

FVM Master发送的报文，用于整车新鲜度值同步
周期性发送，每次发送3桢，间隔50ms。

The FVM Master cycle sends messages for synchronizing the freshness value of ECU that support SecOC

Description
feature. The FVM Master send 3 frames each time and an interval of 50ms in each frame.

6.3.3.19.3 SecOCSecMngFVSIGM: Send Ports

Port	Interface	DataElement	DataType	Generation Type	E2E ID
TripResetSyncReqMsgIGM/	TripResetSyncReqMsgIGM/;1	TripResetSyncReqMsgIGM/;	NoYes1/;2	Sporadic	

Description
当FVM slave向FVM master获取同步报文TripCnt&RestCnt的信息 The FVM slave obtains information about the synchronization packet TripCnt&RestCnt from the FVM master 请求一次，发送3桢，每帧间隔50ms

6.3.3.19.4 SecOC FVM SLAVE IGM

957048 v4 **FVM initialize [CS Released]**

Purpose: the requirement of FVM initialize
Verification Method: Test
Legacy ID: 747246

- SALVE FVM 所在的ECU在启动/唤醒/复位时涉及到FVM模块的初始化（如ECU方涉及到其他FVM初始化的场景请提出），FVM执行以下过程。
  1、获取存储在非易失性存储器中的行程计数器值；
  2、在第一次启动时，将行程计数器的最新值设置为初始值0x00 00 00，
  3、无法从非易失性存储器读取同步计数器值时使用0x00 00 00。
  4、将重置计数器的最新值设置为初始值0。
  5、每个FV中，先前的同步计数器设置为最新的同步计数器值，先前的重置计数器设置和先前的消息计数器均设置为初始值。

The ECU where SALVE FVM is located involves the initialization of the FVM module during startup/wake-up/reset (if the ECU involves other FVM initialization scenarios, please provide them). The FVM performs the following process.
1. Obtain the trip counter value stored in non volatile memory;
2. At the first startup, set the latest value of the travel counter to the initial value of 0x00 00 00.
3. When the synchronization counter value cannot be read from non-volatile memory, and set it as value 0x00 00 00.
4. Set the latest value of the reset counter to the initial value of 0.
5. In each FV, the previous trip counter is set to the latest trip counter value, the previous reset counter and previous message counter are set to the initial value.

957049 v2 TripResetSynReqMsg Format [CS Released]

Purpose: define the format of TripResetSynReqMsg
Verification Method: Test
Legacy ID: 747244

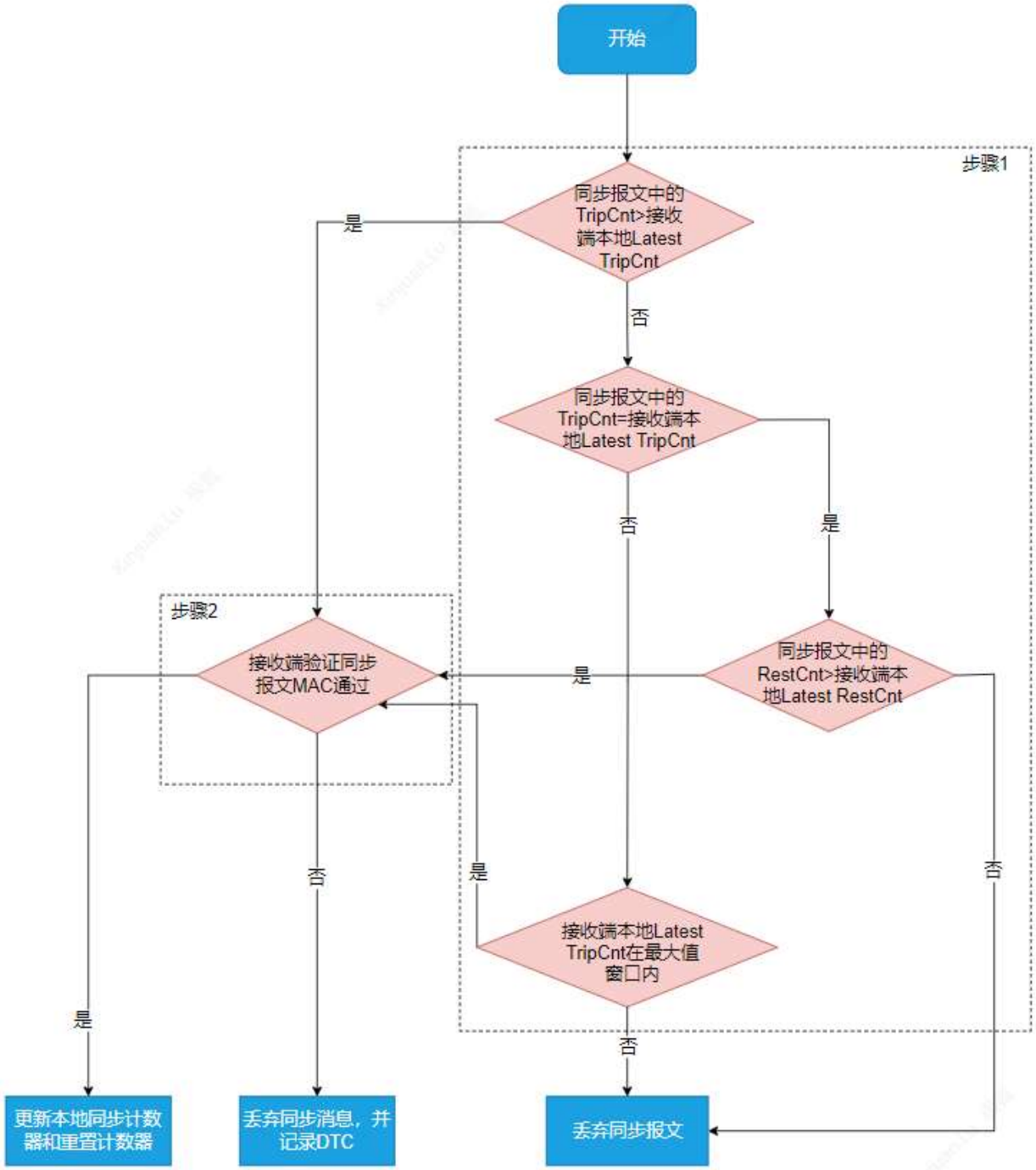
ID	DataField
Messgae ID	ReqSyn
	1bit

ReqSyn值:
1：同步请求
0：忽略
注：非SecOC报文，不需要截断FV及MAC。

ReqSyn value:
1: synchronization request
0: ignore
Note: Non-SECOCPDU does not need to truncate FV and MAC.

957051 v6 After Receiving TripResetSyncMsg [CS Released]

Purpose: the reqirement for FVM slave after receving TripResetSyncMsg from FVM master
Verification Method: Test
Legacy ID: 747247

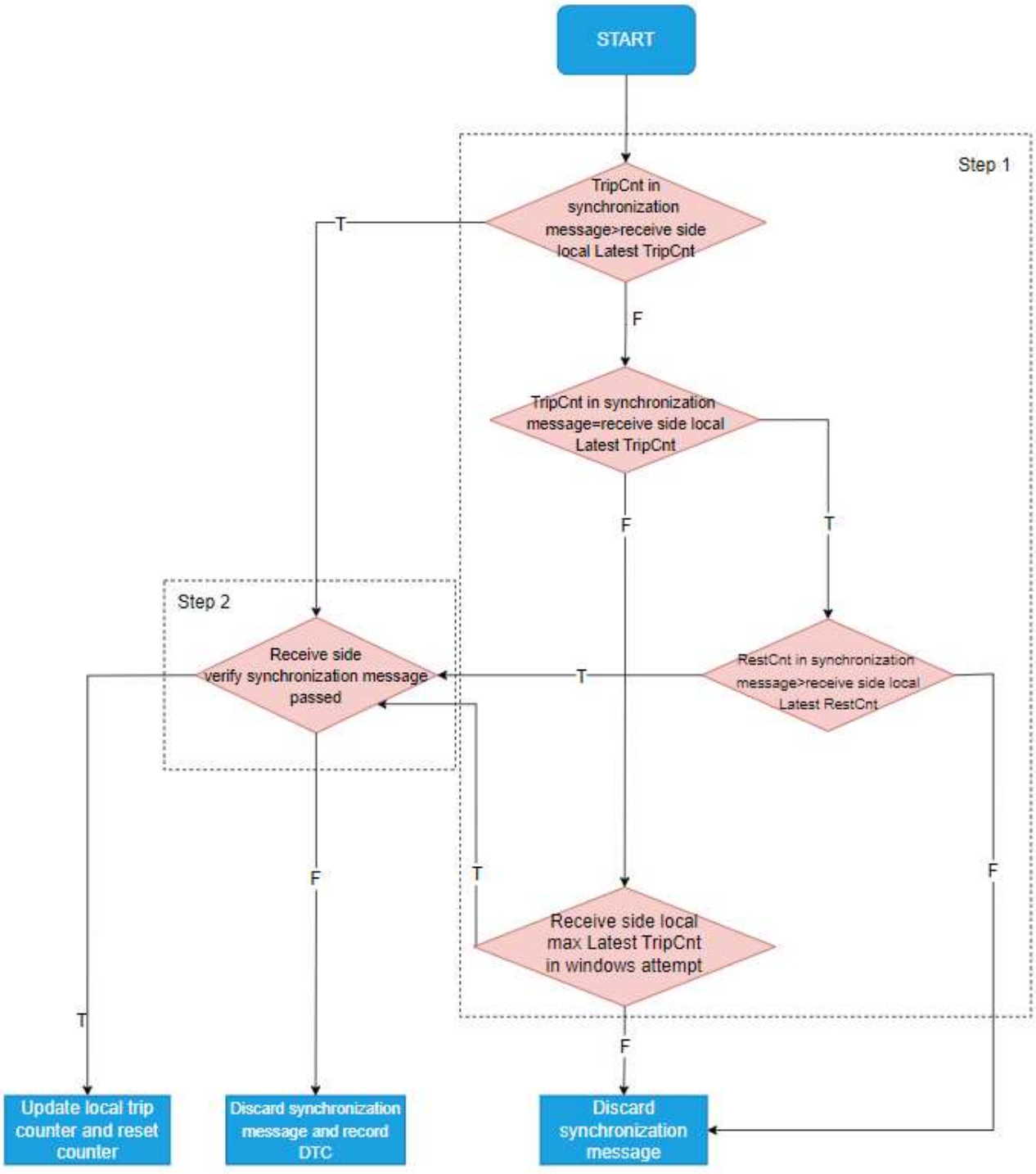


1. FVM SLAVE ECU获取当前最新的行程计数器、最新的重置计数器值，与新鲜度值同步报文TripResetSyncMesage中的TripCnt、RestCnt进行对比：
   - 当新鲜度值同步报文中的TripCnt>ECU当前最新的行程计数器成立，则进入步骤2
   - 当新鲜度值同步报文中的TripCnt=ECU当前最新的行程计数器，判断新鲜度值同步报文中的RestCnt>当前最新的重置计数器值是否成立，如果不成立，丢弃新鲜度值同步报文；如果成立，则进入步骤2
   - 当新鲜度值同步报文中的TripCnt<ECU当前最新的行程计数器，判断当前最新的行程计数器是否在最大值窗口内（按照条件1&条件2判断），如不在最大值窗口内，丢弃新鲜度值同步报文；如在最大值窗口内，则进入步骤2

     条件 1：

同步计数器的最大值 - 同步计数器接收窗口
<= FVM 中保存的最新同步计数器
<= 同步计数器的最大值
条件 2：
同步计数器的初始值
<= 新鲜度值同步消息中的同步计数器值
<= 同步计数器的初始值 + 同步计数器接收窗口
备注：同步计数器接收窗口=3
Condition 1:
“Maximum value of the trip counter” – “ClearAcceptanceWindow” ≤ “Latest value of the trip counter maintained by the slave ECU” ≤ “Maximum value of the trip counter”
Condition 2: “Initial value of the trip counter” ≤ “Trip counter value in the synchronization message” ≤ “Initial value of the trip counter” + “ClearAcceptanceWindow”

2. 校验新鲜度值同步报文TripResetSyncMesage的MAC值，具体校验方式参考CSRS
   - 如校验通过，FVM更新本地最新的行程计数器、最新的重置计数器的值，并将行程计数器的值存储到NVM中；
   - 如校验不通过，则丢弃新鲜度值同步报文TripResetSyncMesage，并记录DTC（DTC参数详见CSRS）。



1. FVM SLAVE ECU obtains the latest travel counter and reset counter values, and compares them with the TripCnt and RestCnt in the freshness value synchronization message TripResetSyncMesage:

When TripCnt in the synchronization message is greater than the latest trip counter of the ECU, proceed to step 2

When TripCnt in the synchronization message=the latest trip counter of the ECU, determine whether the RestCnt in the synchronization message>the latest reset counter value is valid. If not, discard the synchronization message; If true, proceed to step 2

When the TripCnt in the synchronization message is less than the latest trip counter of the ECU, determine whether the latest trip counter is within the maximum value window (according to conditions 1 and 2). If it is not

within the maximum value window, discard the freshness value synchronization message; If within the maximum value window, proceed to step 2

condition 1:

"Maximum value of the trip counter" – "ClearAcceptanceWindow" ≤ "Latest value of the trip counter maintained by the slave ECU" ≤ "Maximum value of the trip counter"

Condition 2:

"Initial value of the trip counter" ≤ "Trip counter value in the synchronization message" ≤ "Initial value of the trip counter" + "ClearAcceptanceWindow"

2. Verify the MAC value of the synchronization message TripResetSyncMesage, and refer to CSRS for specific verification methods

If the verification is successful, FVM updates the latest local trip counter and the latest reset counter values, and stores the trip counter values in NvM;
If the verification fails, the synchronization message TripResetSyncMesage will be discarded and a DTC will be recorded (refer CSRS for DTC parameters).

1072361 v1 TripResetSyncReqMsg Send Condition [CS Released]

Purpose: Condition for sending TripResetSynReqMsg
Verification Method: Test
Legacy ID:

- SALVE FVM ECU启动/唤醒/重启后：

  SALVE FVM 初始化后，在网络管理报文发出之后；如果150ms 内没有收到新鲜度值同步消息，应发送新鲜度值同步请求消息。需要连续发送3次，间隔50ms。如果在发送期间收到新鲜度值同步消息，并且验证通过后，则停止发送新鲜度值同步请求消息。

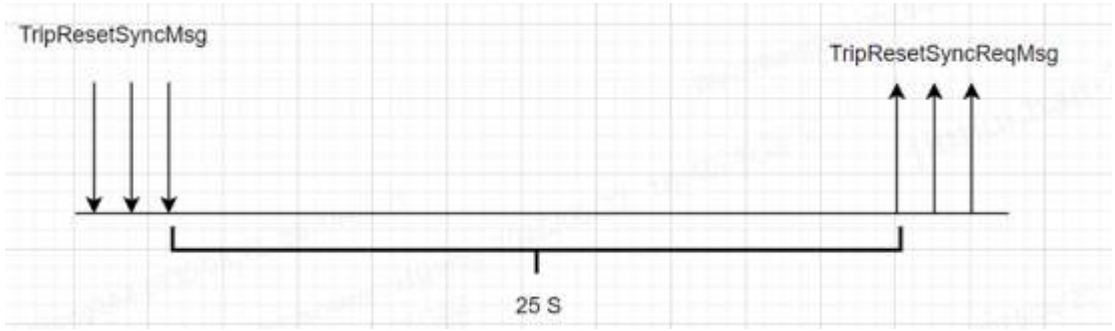
- SALVE FVM ECU在Busoff恢复时，如果150ms 内没有收到新鲜度值同步消息，应发送新鲜度值同步请求消息TripResetSyncReqMsgIGM。
- 注意：如果在发送新鲜度值同步请求消息期间，收到了的新鲜度值同步消息并验证通过，则停止发送后续的新鲜度值同步请求消息。 例如，在发送了1 次新鲜度值同步请求消息后，收到了新鲜度值同步消息并验证通过，则不再发送后续的两次新鲜度值同步请求消息。
- 从节点在最后一次成功接收到同步消息后，如果在25秒内没有收到同步消息，则发送同步请求消息。

- After SALVE FVM ECU starts/wakes up/restarts:

  After SALVE FVM initialization, after the network management message is sent; If no freshness value synchronization message is received within 150ms, a freshness value synchronization request message should be sent. It needs to be sent 3 consecutive times with an interval of 50ms. If a freshness value synchronization message is received during transmission and the verification is passed, the freshness value synchronization request message will be stopped.

- If the SALVE FVM ECU does not receive a freshness value synchronization message within 150ms during Busoff recovery, it should send a freshness value synchronization request message .
- The slave node shall send a synchronization request message if no synchronization message is received within 25 seconds after the last successful reception.

Note: If the freshness value synchronization request message is received and verified during the sending of the freshness value synchronization request message, the subsequent freshness value synchronization request message will be stopped. For example, after sending a freshness value synchronization request message once, if a freshness value synchronization message is received and verified, the subsequent two freshness value synchronization request messages will not be sent.



6.3.4 Data Types

Record Data Type: **AmbTWithQly (2)**

Description:

Group Signal	Data Type
AmbTVal	AmbT
Qly	GenQf1

Primitive Data Type: AmbT"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 11
Physical Lower Lim interval type: CLOSED
Physical lower limit: -70
Physical Upper Lim interval type: CLOSED
Physical upper limit: 134.7

UUID: e98be107-cd30-48bb-aee9-43cb8c40a2c1
BaseType UUID: 6FF78AA4-086F-4F05-BEE0-41EB475EFA83
Unit: DegCgrd
CompuCategory: LINEAR
CompuMethod Table:

Min	Max	Interval type	Numerators	Denominators	Constant
0	2047	[ ]	-70.0; 0.1	1	

Primitive Data Type: GenQf1"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 2
Lower limit internal range: 0
lower limit interval type internal range: CLOSED
upper limit internal range: 3
upper limit interval type internal range: CLOSED

UUID: 68330404-f471-4bbd-8fc8-271f1473d230
BaseType UUID: 26DD2B41-A428-4C18-8753-9DD5279354A2
Unit: NoUnit
CompuCategory: TEXTTABLE

CompuMethod Table:

Min	Max	Interval type	Numerators	Denominators	Constant
0	0	[ ]			GenQf1_UndefindDataAccur
1	1	[ ]			GenQf1_TmpUndefdData
2	2	[ ]			GenQf1_DataAccurNotWithinSpcn
3	3	[ ]			GenQf1_AccurData

Primitive Data Type: **Ampr1 (1)**

Description:
Primitive Data Type: Ampr1"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 14
Physical Lower Lim interval type: CLOSED
Physical lower limit: -818.8
Physical Upper Lim interval type: CLOSED
Physical upper limit: 819.2

UUID: C00AE1C9-0BBE-4243-9650-D258774E0E2F
BaseType UUID: 0483C040-F857-409E-A24B-0437456B212F
Unit: Ampr
CompuCategory: LINEAR
CompuMethod Table:

Min	Max	Interval type	Numerators	Denominators	Constant
0	16380	[ ]	-818.8; 0.1	1	

Primitive Data Type: **BCLRqrdU (1)**

Description:
Primitive Data Type: BCLRqrdU"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 16
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 6553.5

UUID: 0811C7CE-053A-46B9-A2E3-88892C50EAD7
BaseType UUID: D7C8FA74-0CAB-4BAB-BACA-8712DF0209A7
Unit: Volt
CompuCategory: LINEAR
CompuMethod Table:

Min	Max	Interval type	Numerators	Denominators	Constant
0	65535	[ ]	0; 0.1	1	

Primitive Data Type: Boolean (2)

Description:

Primitive Data Type: Boolean"

System Signal Encoding: BOOLEAN
category: BOOLEAN
SystemSignalLength: 1
Lower limit internal range: 0
lower limit interval type internal range: CLOSED
upper limit internal range: 1
upper limit interval type internal range: CLOSED

UUID: 70F22060-192B-AC2F-E040-901367011E69
BaseType UUID: 1F2A41AA-1EAF-408A-A5C7-3D1CD7D41AB2
Unit: NoUnit
CompuCategory: TEXTTABLE
CompuMethod Table:

Min	Max	Interval type	Numerators	Denominators	Constant
0	0	[ ]			Boolean_FALSE
1	1	[ ]			Boolean_TRUE

Primitive Data Type: CarTiGlb (2)

Description:

Primitive Data Type: CarTiGlb"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 32
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 4.294967294E8

UUID: 166efcb0-f994-4f2e-a5d8-bb588ad047a3
BaseType UUID: 09F0B1CA-AAEB-47CD-A09E-2EC7DC3DFF5D
Unit: Sec
CompuCategory: LINEAR
CompuMethod Table:

Min	Max	Interval type	Numerators	Denominators	Constant
0	4294967295	[ ]	0.0; 0.1	1	

Primitive Data Type: CooltFlow2 (2)

Description:
Primitive Data Type: CooltFlow2"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 9
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 51.1

UUID: ccc216c3-c546-4624-801d-b45ec89f523d
BaseType UUID: EAE2BBF2-C6FE-441D-9B58-8DA346A26AC8
Unit: LPerMins
CompuCategory: LINEAR
CompuMethod Table:

Min	Max	Interval type	Numerators	Denominators	Constant
0	511	[ ]	0.0; 0.1	1	

Primitive Data Type: **CooltFlow6 (1)**

Description:
Primitive Data Type: CooltFlow6"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 10
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 20.46

UUID: A42B1D98-1CA2-4C43-94D0-D54E0212E484
BaseType UUID: 0D1E990B-49F1-4246-904B-0EC39B8A26BB
Unit: LPerMins
CompuCategory: LINEAR
CompuMethod Table:

Min	Max	Interval type	Numerators	Denominators	Constant
0	1023	[ ]	0; 0.02	1	

Primitive Data Type: **CooltInhb (2)**

Description:
Primitive Data Type: CooltInhb"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 1
Lower limit internal range: 0
lower limit interval type internal range: CLOSED
upper limit internal range: 1

upper limit interval type internal range: CLOSED

UUID: a3b70fc1-c37e-48c4-b879-7310898ad2e6
BaseType UUID: D2776960-2FBA-41EC-A25F-DC69E891288D
Unit: NoUnit
CompuCategory: TEXTTABLE
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	0	[ ]			CooltInhb_CoolntFlowReqToEngAllwd
1	1	[ ]			CooltInhb_CoolntFlowReqToEngNotAllwd

Primitive Data Type: **CooltTHiResl (2)**

Description:

Primitive Data Type: CooltTHiResl"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 11
Physical Lower Lim interval type: CLOSED
Physical lower limit: -40
Physical Upper Lim interval type: CLOSED
Physical upper limit: 140

UUID: 162ccdfa-e4c2-44d1-ba9a-8d8f327e7486
BaseType UUID: 0627891E-E8DA-46B3-B3D7-5A1F149A29C3
Unit: DegCgrd
CompuCategory: LINEAR
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	1800	[ ]	-40.0; 0.1	1	

Record Data Type: **CrashStsSafe (1)**

Description:

Group Signal	**Data Type**
Chks	UInt8
Cntr	UInt4
Sts	CrashSts2

Primitive Data Type: CrashSts2"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 1
Lower limit internal range: 0
lower limit interval type internal range: CLOSED
upper limit internal range: 1

upper limit interval type internal range: CLOSED

UUID: CB3D510C-8832-4474-AAF2-8800078F07B8
BaseType UUID: 0AF76B65-498E-4D65-BE9A-EF355819AF9E
Unit: NoUnit
CompuCategory: TEXTTABLE
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	0	[ ]			CrashSts2_NoCrash
1	1	[ ]			CrashSts2_Crash

Primitive Data Type: UInt4"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 4
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 15

UUID: 70F22060-1935-AC2F-E040-901367011E69
BaseType UUID: D6A89C6F-2F3D-4C8E-8DC9-191B048C4E5D
Unit: NoUnit
CompuCategory: IDENTICAL

Primitive Data Type: UInt8"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 8
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 255

UUID: 70F22060-1937-AC2F-E040-901367011E69
BaseType UUID: 6E2EFCA8-43A6-4F1E-9530-99AE1843F3D5
Unit: NoUnit
CompuCategory: IDENTICAL

Record Data Type: **DateTi30 (2)**

Description:

Group Signal	**Data Type**
DataValid	NoYes1
Day	Day1
Hr1	Hr1
Mins1	Mins1
Mth1	Mth1
Sec1	Sec1
Yr1	Yr1

Primitive Data Type: Day1"

System Signal Encoding:	NONE
category:	VALUE
SystemSignalLength:	5
Physical Lower Lim interval type:	CLOSED
Physical lower limit:	1
Physical Upper Lim interval type:	CLOSED
Physical upper limit:	31
UUID:	cabbf8a2-83d3-45a4-802d-7791d847bd2b
BaseType UUID:	B788D709-1D00-4D4D-8A87-998EF3C1F582

Unit: Day
CompuCategory: IDENTICAL

Primitive Data Type: Hr1"

System Signal Encoding:	NONE
category:	VALUE
SystemSignalLength:	5
Physical Lower Lim interval type:	CLOSED
Physical lower limit:	0
Physical Upper Lim interval type:	CLOSED
Physical upper limit:	23
UUID:	7aa83ff1-927e-48a4-9889-c092b7bb1f53
BaseType UUID:	69B00CAD-33A6-4BF8-9BAE-F851947A8D19

Unit: Hr
CompuCategory: IDENTICAL

Primitive Data Type: Mins1"

System Signal Encoding:	NONE
category:	VALUE
SystemSignalLength:	6
Physical Lower Lim interval type:	CLOSED
Physical lower limit:	0
Physical Upper Lim interval type:	CLOSED
Physical upper limit:	59
UUID:	659f1b05-62d1-4cdf-b39f-ccb9839bd1a2
BaseType UUID:	812E9F61-79DA-4A29-9973-983AF879BECB

Unit: Mins
CompuCategory: IDENTICAL

Primitive Data Type: Mth1"

System Signal Encoding:	NONE
category:	VALUE
SystemSignalLength:	4
Physical Lower Lim interval type:	CLOSED
Physical lower limit:	1

Physical Upper Lim interval type: CLOSED
Physical upper limit: 12

UUID: 0f04673d-52b1-4aae-adef-703100e86cb6
BaseType UUID: 23E15304-0D6A-4724-ABEE-DD2165B5D702
Unit: Mth
CompuCategory: IDENTICAL

Primitive Data Type: NoYes1"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 1
Lower limit internal range: 0
lower limit interval type internal range: CLOSED
upper limit internal range: 1
upper limit interval type internal range: CLOSED

UUID: 565f4a56-9a7f-452b-8d9b-8084a6038e19
BaseType UUID: 07B2CC10-2C39-4414-B773-20E7313390A2
Unit: NoUnit
CompuCategory: TEXTTABLE
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	0	[ ]			NoYes1_No
1	1	[ ]			NoYes1_Yes

Primitive Data Type: Sec1"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 6
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 59

UUID: 8546e855-740d-4476-9ea6-5a96e0146734
BaseType UUID: 1407FA6D-54F1-4D5C-AAFE-2F09B409A085
Unit: Sec
CompuCategory: IDENTICAL

Primitive Data Type: Yr1"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 7
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 99

UUID: 611ca10b-5b74-49de-a9fe-1d492223a357
BaseType UUID: 3491C66B-BD7E-46B2-94F2-F1BAD71D5D5C
Unit: Yr
CompuCategory: IDENTICAL

Record Data Type: **EEMReqOilSpd (1)**

Description:

Group Signal	Data Type
EEMReqOilSpdSts	OnOff1
EEMReqOilSpdVal	IsgSpd1

Primitive Data Type: IsgSpd1"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 13
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 8191

UUID: 3c68d137-d345-4812-afb5-4edb366ad191
BaseType UUID: 0D83BE15-E221-4403-92E1-780A7E86AFCF
Unit: Rpm
CompuCategory: IDENTICAL

Primitive Data Type: OnOff1"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 1
Lower limit internal range: 0
lower limit interval type internal range: CLOSED
upper limit internal range: 1
upper limit interval type internal range: CLOSED

UUID: cd8171e6-5792-4296-bd41-91423cb48b93
BaseType UUID: E3808B2F-A4A4-46DA-B6A3-3729862ACE60
Unit: NoUnit
CompuCategory: TEXTTABLE
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	0	[ ]			OnOff1_Off
1	1	[ ]			OnOff1_On

Record Data Type: **EngNSafe (2)**

Description:

Group Signal	Data Type
EngN	N30
EngNChks	UInt8
EngNCntr	UInt4

Group Signal	Data Type
EngNGrdt	NGrdt1

Primitive Data Type: N30"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 15
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 16383

UUID: 513058ca-175d-45ba-8e34-d63246da7cd2
BaseType UUID: EAB05279-E673-42E6-B397-54D83B11A106
Unit: Rpm
CompuCategory: LINEAR
CompuMethod Table:

Min	Max	Interval type	Numerators	Denominators	Constant
0	32767	[ ]	0.0; 0.5	1	

Primitive Data Type: NGrdt1"

System Signal Encoding: 2C
category: VALUE
SystemSignalLength: 16
Physical Lower Lim interval type: CLOSED
Physical lower limit: -30000
Physical Upper Lim interval type: CLOSED
Physical upper limit: 30000

UUID: 28039716-8a25-400d-82c9-f5d2039bfea7
BaseType UUID: 6EADA2BB-1BA0-46A9-9FA7-5714AB3EBEF6
Unit: RpmPerSec
CompuCategory: LINEAR
CompuMethod Table:

Min	Max	Interval type	Numerators	Denominators	Constant
-30000	32767	[ ]	0.0; 1.0	1	

Primitive Data Type: UInt4"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 4
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 15

UUID: 70F22060-1935-AC2F-E040-901367011E69
BaseType UUID: D6A89C6F-2F3D-4C8E-8DC9-191B048C4E5D
Unit: NoUnit
CompuCategory: IDENTICAL

Primitive Data Type: UInt8"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 8
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 255

UUID: 70F22060-1937-AC2F-E040-901367011E69
BaseType UUID: 6E2EFCA8-43A6-4F1E-9530-99AE1843F3D5
Unit: NoUnit
CompuCategory: IDENTICAL

Record Data Type: **EngSt1WdSts (1)**

Description:

Group Signal	Data Type
Chks	UInt8
Cntr	UInt4
EngSt1WdSts	EngSt1

Primitive Data Type: EngSt1"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 4
Lower limit internal range: 0
lower limit interval type internal range: CLOSED
upper limit internal range: 9
upper limit interval type internal range: CLOSED

UUID: 29d82241-8b2b-46c3-b99e-31c188de550b
BaseType UUID: C0057E7A-F7CB-4AB3-9180-A6E5521E1759
Unit: NoUnit
CompuCategory: TEXTTABLE
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	0	[ ]			EngSt1_Ini
1	1	[ ]			EngSt1_Awake
2	2	[ ]			EngSt1_Rdy
3	3	[ ]			EngSt1_PreStrtg
4	4	[ ]			EngSt1_StrtgInProgs
5	5	[ ]			EngSt1_RunngRunng

6	6	[ ]			EngSt1_RunngStb
7	7	[ ]			EngSt1_RunngStrtgInProgs
8	8	[ ]			EngSt1_RunngRemStrtd
9	9	[ ]			EngSt1_AftRun

Primitive Data Type: UInt4"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 4
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 15

UUID: 70F22060-1935-AC2F-E040-901367011E69
BaseType UUID: D6A89C6F-2F3D-4C8E-8DC9-191B048C4E5D
Unit: NoUnit
CompuCategory: IDENTICAL

Primitive Data Type: UInt8"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 8
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 255

UUID: 70F22060-1937-AC2F-E040-901367011E69
BaseType UUID: 6E2EFCA8-43A6-4F1E-9530-99AE1843F3D5
Unit: NoUnit
CompuCategory: IDENTICAL

Record Data Type: **EngT (2)**

Description:

Group Signal	Data Type
EngT	T3
EngTQf	Qf1

Primitive Data Type: Qf1"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 2
Lower limit internal range: 0
lower limit interval type internal range: CLOSED

upper limit internal range: 3
upper limit interval type internal range: CLOSED

UUID: c0219f56-ad19-4bad-9bc6-ccfea3507592
BaseType UUID: 14DBA2F6-858C-4B3D-AD1E-A7B8DB8B68EF
Unit: NoUnit
CompuCategory: TEXTTABLE
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	0	[ ]			Qf1_DevOfDataUndefd
1	1	[ ]			Qf1_DataTmpUndefdAndEvlnInProgs
2	2	[ ]			Qf1_DevOfDataNotWithinRngAllwd
3	3	[ ]			Qf1_DataCalcdWithDevDefd

Primitive Data Type: T3"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 8
Physical Lower Lim interval type: CLOSED
Physical lower limit: -50
Physical Upper Lim interval type: CLOSED
Physical upper limit: 200

UUID: a98be047-536f-4c0f-bdfc-1e160d93ce0e
BaseType UUID: 212FDAAF-8ACC-4CA0-991E-36929AF441A2
Unit: DegCgrd
CompuCategory: LINEAR
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	250	[ ]	-50.0; 1.0	1	

Primitive Data Type: **GainTypActvDampg8 (2)**

Description:
Primitive Data Type: GainTypActvDampg8"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 8
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 0.051

UUID: 06dd8020-a6e3-45d1-82ca-7a6ca0af3506
BaseType UUID: 572BC4FC-6119-495A-A644-501ECFB9BFFC
Unit: NoUnit
CompuCategory: LINEAR
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	255	[ ]	0.0; 2.0E-4	1	

Record Data Type: **HvSysRlySts (1)**

Description:

Group Signal	**Data Type**
Chks	UInt8
Cntr	UInt4
HvSysRlySts	HvActvnSts

Primitive Data Type: HvActvnSts"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 2
Lower limit internal range: 0
lower limit interval type internal range: CLOSED
upper limit internal range: 3
upper limit interval type internal range: CLOSED

UUID: 560a6675-8511-409c-99a9-678b23d4ba9a
BaseType UUID: 1105BF71-773D-4B5C-9765-35D3E5BBA3DA
Unit: NoUnit
CompuCategory: TEXTTABLE
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	0	[ ]			HvActvnSts_Open
1	1	[ ]			HvActvnSts_Clsd
2	2	[ ]			HvActvnSts_KeepSt
3	3	[ ]			HvActvnSts_OpenAndReqActvDcha

Primitive Data Type: UInt4"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 4
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 15

UUID: 70F22060-1935-AC2F-E040-901367011E69
BaseType UUID: D6A89C6F-2F3D-4C8E-8DC9-191B048C4E5D
Unit: NoUnit
CompuCategory: IDENTICAL

Primitive Data Type: UInt8"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 8
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 255

UUID: 70F22060-1937-AC2F-E040-901367011E69
BaseType UUID: 6E2EFCA8-43A6-4F1E-9530-99AE1843F3D5
Unit: NoUnit
CompuCategory: IDENTICAL

Primitive Data Type: HvSysSts (2)

Description:

Primitive Data Type: HvSysSts"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 2
Lower limit internal range: 0
lower limit interval type internal range: CLOSED
upper limit internal range: 2
upper limit interval type internal range: CLOSED

UUID: b63ea3fb-9522-49e1-ba1a-6cb0963c5b4a
BaseType UUID: 2E7FE9AE-C72F-440E-9800-3D59AE58186E
Unit: NoUnit
CompuCategory: TEXTTABLE
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	0	[ ]			HvSysSts_Inin
1	1	[ ]			HvSysSts_Test
2	2	[ ]			HvSysSts_Rdy

Primitive Data Type: InhibitSts (1)

Description:

Primitive Data Type: InhibitSts"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 1
Lower limit internal range: 0

lower limit interval type internal range: CLOSED
upper limit internal range: 1
upper limit interval type internal range: CLOSED

UUID: 396351FB-FC8E-4A82-9124-EE868BAEF160
BaseType UUID: 95B6DFBF-558A-40C5-AE30-875B52231D35
Unit: NoUnit
CompuCategory: TEXTTABLE
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	0	[ ]			InhibitSts_NotInhibited
1	1	[ ]			InhibitSts_Inhibited

Primitive Data Type: IsgModTyp1 (4)

Description:
Primitive Data Type: IsgModTyp1"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 4
Lower limit internal range: 0
lower limit interval type internal range: CLOSED
upper limit internal range: 15
upper limit interval type internal range: CLOSED

UUID: 9fa68994-b45f-417d-b5ca-f0a30d78dbb7
BaseType UUID: 33E7CE2C-31AE-4E31-A71C-ED49F9B8374F
Unit: NoUnit
CompuCategory: TEXTTABLE
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	0	[ ]			IsgModTyp1_Inin
1	1	[ ]			IsgModTyp1_Stb
2	2	[ ]			IsgModTyp1_TqCtrl
3	3	[ ]			IsgModTyp1_SpdCtrl
4	4	[ ]			IsgModTyp1_UDcCtrl
5	5	[ ]			IsgModTyp1_PreChrg
6	6	[ ]			IsgModTyp1_PwrDwn
7	7	[ ]			IsgModTyp1_Flt
8	8	[ ]			IsgModTyp1_TcsCtrl

9	9	[ ]			IsgModTyp1_Reserved1
10	10	[ ]			IsgModTyp1_Reserved2
11	11	[ ]			IsgModTyp1_Reserved3
12	12	[ ]			IsgModTyp1_Reserved4
13	13	[ ]			IsgModTyp1_Reserved5
14	14	[ ]			IsgModTyp1_Reserved6
15	15	[ ]			IsgModTyp1_Reserved7

Primitive Data Type: **IsgModTyp2 (3)**

Description:

Primitive Data Type: IsgModTyp2"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 4
Lower limit internal range: 0
lower limit interval type internal range: CLOSED
upper limit internal range: 9
upper limit interval type internal range: CLOSED

UUID: 9F4AAEBF-A2AC-4015-B161-6C4311B4E14F
BaseType UUID: DEF42E5A-F055-40AC-97D0-51E6888AFA9A
Unit: NoUnit
CompuCategory: TEXTTABLE
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	0	[ ]			IsgModTyp2_Inin
1	1	[ ]			IsgModTyp2_Stb
2	2	[ ]			IsgModTyp2_TqCtrl
3	3	[ ]			IsgModTyp2_SpdCtrl
4	4	[ ]			IsgModTyp2_UDcCtrl
5	5	[ ]			IsgModTyp2_PreChrg
6	6	[ ]			IsgModTyp2_PwrDwn
7	7	[ ]			IsgModTyp2_Flt
8	8	[ ]			IsgModTyp2_TcsCtrl
9	9	[ ]			IsgModTyp2_C

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Primitive Data Type: **IsgSpdWSgnTyp (2)**

Description:

Primitive Data Type: IsgSpdWSgnTyp"

System Signal Encoding: 2C
category: VALUE
SystemSignalLength: 15
Physical Lower Lim interval type: CLOSED
Physical lower limit: -16384
Physical Upper Lim interval type: CLOSED
Physical upper limit: 16383

UUID: cf06397e-dd77-4b73-a3df-e2f27afee0a1
BaseType UUID: C19762F9-91F0-4ED2-8EA0-060D3C7152D8
Unit: Rpm
CompuCategory: LINEAR
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
-16384	16383	[ ]	0.0; 1.0	1	

Record Data Type: **IsgTqActSafe2 (4)**

Description:

Group Signal	**Data Type**
Chks	UInt8
Cntr	UInt4
IsgTqAct	Tq34
QualityFactor	QualityFactor

Primitive Data Type: QualityFactor"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 2
Lower limit internal range: 0
lower limit interval type internal range: CLOSED
upper limit internal range: 3
upper limit interval type internal range: CLOSED

UUID: 70f22060-19dd-ac2f-e040-901367011e69
BaseType UUID: 04D3A5FB-0F8E-49D9-A17F-DF32C9077415
Unit: NoUnit
CompuCategory: TEXTTABLE
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	0	[ ]			QualityFactor_QfUndefd
1	1	[ ]			QualityFactor_QfInProgs

2	2	[ ]			QualityFactor_QfNotSpc
3	3	[ ]			QualityFactor_QfSnsrDataOk

Primitive Data Type: Tq34"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 14
Physical Lower Lim interval type: CLOSED
Physical lower limit: -8188
Physical Upper Lim interval type: CLOSED
Physical upper limit: 8192

UUID: D55D4E0C-1A9E-4164-B722-5B17A555319D
BaseType UUID: 153FBD12-431E-4D74-BC9D-AEDB8EE476EC
Unit: Nm
CompuCategory: LINEAR
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	16380	[ ]	-8188; 1	1	

Primitive Data Type: UInt4"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 4
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 15

UUID: 70F22060-1935-AC2F-E040-901367011E69
BaseType UUID: D6A89C6F-2F3D-4C8E-8DC9-191B048C4E5D
Unit: NoUnit
CompuCategory: IDENTICAL

Primitive Data Type: UInt8"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 8
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 255

UUID: 70F22060-1937-AC2F-E040-901367011E69
BaseType UUID: 6E2EFCA8-43A6-4F1E-9530-99AE1843F3D5
Unit: NoUnit

CompuCategory: IDENTICAL

Record Data Type: **IsgTqAllwd1 (3)**

Description:

Group Signal	Data Type
Chks	UInt8
Cntr	UInt4
IsgTqAllwd	Nr37

Primitive Data Type: Nr37"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 4
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 15

UUID: 6750a0f0-386c-46a7-a978-7ba5cdd93aeb
BaseType UUID: A9DAEDF8-ADEA-4250-B882-82EDEF6DF8B3
Unit: NoUnit
CompuCategory: IDENTICAL

Primitive Data Type: UInt4"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 4
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 15

UUID: 70F22060-1935-AC2F-E040-901367011E69
BaseType UUID: D6A89C6F-2F3D-4C8E-8DC9-191B048C4E5D
Unit: NoUnit
CompuCategory: IDENTICAL

Primitive Data Type: UInt8"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 8
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 255

UUID: 70F22060-1937-AC2F-E040-901367011E69
BaseType UUID: 6E2EFCA8-43A6-4F1E-9530-99AE1843F3D5
Unit: NoUnit
CompuCategory: IDENTICAL

Record Data Type: **IsgTqAvl (4)**

Description:

Group Signal	Data Type
IsgTqAvlMax	Tq34
IsgTqAvlMin	Tq34

Primitive Data Type: Tq34"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 14
Physical Lower Lim interval type: CLOSED
Physical lower limit: -8188
Physical Upper Lim interval type: CLOSED
Physical upper limit: 8192

UUID: D55D4E0C-1A9E-4164-B722-5B17A555319D
BaseType UUID: 153FBD12-431E-4D74-BC9D-AEDB8EE476EC
Unit: Nm
CompuCategory: LINEAR
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	16380	[ ]	-8188; 1	1	

Primitive Data Type: **Len31 (2)**

Description:

Primitive Data Type: Len31"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 21
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 2000000

UUID: 6d6f4f0d-9bee-4f7c-bd93-d3c4c619cba0
BaseType UUID: 44C73339-661B-42CC-BE36-317F669E93A2
Unit: km
CompuCategory: LINEAR
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	2000000	[ ]	0.0; 1.0	1	

Primitive Data Type: **N1 (2)**

Description:

Primitive Data Type: N1"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 15
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 16383.5

UUID: 7b155427-508f-4f6d-92f2-95cd11161e04
BaseType UUID: FCD972F0-2BAE-4F4E-BC37-C629601FDE1E
Unit: Rpm
CompuCategory: LINEAR
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	32767	[ ]	0.0; 0.5	1	

Primitive Data Type: **NewBattSOC (1)**

Description:
Primitive Data Type: NewBattSOC"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 14
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 163.83

UUID: 9B7D7602-F71B-45A9-ABDB-A7EE4B20D122
BaseType UUID: ECFA50BD-760D-4346-B688-3163B1786664
Unit: Perc
CompuCategory: LINEAR
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	16383	[ ]	0; 0.01	1	

Primitive Data Type: **NewBattVolt (1)**

Description:
Primitive Data Type: NewBattVolt"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 13
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 2047.75

UUID: 5B310E62-33AE-410F-AF67-D345AE136CC8

BaseType UUID: FCA83C9E-6378-4B73-97AE-7AD92C665EDD
Unit: Volt
CompuCategory: LINEAR
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	8191	[ ]	0; 0.25	1	

Primitive Data Type: NoYes1 (2)

Description:

Primitive Data Type: NoYes1"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 1
Lower limit internal range: 0
lower limit interval type internal range: CLOSED
upper limit internal range: 1
upper limit interval type internal range: CLOSED

UUID: 565f4a56-9a7f-452b-8d9b-8084a6038e19
BaseType UUID: 07B2CC10-2C39-4414-B773-20E7313390A2
Unit: NoUnit
CompuCategory: TEXTTABLE
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	0	[ ]			NoYes1_No
1	1	[ ]			NoYes1_Yes

Primitive Data Type: NotCmpl1 (2)

Description:

Primitive Data Type: NotCmpl1"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 1
Lower limit internal range: 0
lower limit interval type internal range: CLOSED
upper limit internal range: 1
upper limit interval type internal range: CLOSED

UUID: 7c135382-e066-43d6-b712-11a88fb3b216
BaseType UUID: F482AAB4-F5B5-4825-8C73-A8BE05FEC537
Unit: NoUnit
CompuCategory: TEXTTABLE
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	0	[ ]			NotCmpl1_NotCmpl

1	1	[ ]			NotCmpl1_Cmpl

Primitive Data Type: **OnOff1 (2)**

Description:
Primitive Data Type: OnOff1"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 1
Lower limit internal range: 0
lower limit interval type internal range: CLOSED
upper limit internal range: 1
upper limit interval type internal range: CLOSED

UUID: cd8171e6-5792-4296-bd41-91423cb48b93
BaseType UUID: E3808B2F-A4A4-46DA-B6A3-3729862ACE60
Unit: NoUnit
CompuCategory: TEXTTABLE
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	0	[ ]			OnOff1_Off
1	1	[ ]			OnOff1_On

Primitive Data Type: **Perc2 (2)**

Description:
Primitive Data Type: Perc2"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 8
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 100

UUID: f2a5ea18-125a-4e8c-93b4-9d52ed3fb440
BaseType UUID: 70B0D3AC-6B10-4E47-AA40-20CB13F6DADF
Unit: Perc
CompuCategory: LINEAR
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	200	[ ]	0.0; 0.5	1	

Primitive Data Type: **Perc5 (2)**

Description:
Primitive Data Type: Perc5"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 8
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 100

UUID: 2fe18442-8ba1-4bb3-82f3-e536d2b4f5f5
BaseType UUID: ED0EE45E-5A96-48EE-A20A-089292520767
Unit: Perc
CompuCategory: LINEAR
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	255	[ ]	0.0; 0.3921568627 4509803	1	

Record Data Type: **PrpsnDataEvt (1)**

Description:

Group Signal	Data Type
Cnt	UInt7
Data1	UInt8
Data2	UInt8
Data3	UInt8
Data4	UInt8
Data5	UInt8
Data6	UInt8
ID	UInt8

Primitive Data Type: UInt7"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 7
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 127

UUID: 832723BE-03FA-40BC-963C-53BEC98975D7
BaseType UUID: 98E00118-3873-4CAA-9FA7-0EB316C8FBDF
Unit: NoUnit
CompuCategory: IDENTICAL

Primitive Data Type: UInt8"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 8
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0

Physical Upper Lim interval type: CLOSED
Physical upper limit: 255

UUID: 70F22060-1937-AC2F-E040-901367011E69
BaseType UUID: 6E2EFCA8-43A6-4F1E-9530-99AE1843F3D5
Unit: NoUnit
CompuCategory: IDENTICAL

Record Data Type: PrpsnDataPrd (1)

Description:

Group Signal	Data Type
Cnt	UInt7
Data1	UInt8
Data2	UInt8
Data3	UInt8
Data4	UInt8
Data5	UInt8
Data6	UInt8
Data7	UInt8

Primitive Data Type: UInt7"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 7
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 127

UUID: 832723BE-03FA-40BC-963C-53BEC98975D7
BaseType UUID: 98E00118-3873-4CAA-9FA7-0EB316C8FBDF
Unit: NoUnit
CompuCategory: IDENTICAL

Primitive Data Type: UInt8"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 8
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 255

UUID: 70F22060-1937-AC2F-E040-901367011E69
BaseType UUID: 6E2EFCA8-43A6-4F1E-9530-99AE1843F3D5
Unit: NoUnit
CompuCategory: IDENTICAL

Primitive Data Type: PrpsnPwrLimMax (2)

Description:

Primitive Data Type: PrpsnPwrLimMax"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 10
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 511

UUID: 07b1707d-6566-4155-9159-1bf36ec0c5ca
BaseType UUID: 9BE19EE8-7BE6-4CBB-A779-72DA5C27F5A6
Unit: KiloWatt
CompuCategory: LINEAR
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	1022	[ ]	0.0; 0.5	1	

Primitive Data Type: **PrpsnPwrLimMin (2)**

Description:

Primitive Data Type: PrpsnPwrLimMin"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 10
Physical Lower Lim interval type: CLOSED
Physical lower limit: -511
Physical Upper Lim interval type: CLOSED
Physical upper limit: 0

UUID: 391d09d6-7dfc-42a9-89a6-f417f9705cbc
BaseType UUID: 0A5E5015-F3D2-45AF-98AE-5656279318E6
Unit: KiloWatt
CompuCategory: LINEAR
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	1022	[ ]	-511.0; 0.5	1	

Record Data Type: **PtTqAtAxleReq (1)**

Description:

Group Signal	Data Type
Chks	UInt8
Cntr	UInt4
PtTqAtAxleFrntReq	Tq2
PtTqAtAxleReReq	Tq2

Primitive Data Type: Tq2"

System Signal Encoding: 2C
category: VALUE
SystemSignalLength: 16

Physical Lower Lim interval type: CLOSED
Physical lower limit: -20000
Physical Upper Lim interval type: CLOSED
Physical upper limit: 20000

UUID: c454eb3c-a49e-44fe-aa75-a2eca11ee072
BaseType UUID: BC2C6AD8-BE13-4C0A-B482-356536EF3356
Unit: Nm
CompuCategory: LINEAR
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
-20000	20000	[ ]	0.0; 1.0	1	

Primitive Data Type: UInt4"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 4
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 15

UUID: 70F22060-1935-AC2F-E040-901367011E69
BaseType UUID: D6A89C6F-2F3D-4C8E-8DC9-191B048C4E5D
Unit: NoUnit
CompuCategory: IDENTICAL

Primitive Data Type: UInt8"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 8
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 255

UUID: 70F22060-1937-AC2F-E040-901367011E69
BaseType UUID: 6E2EFCA8-43A6-4F1E-9530-99AE1843F3D5
Unit: NoUnit
CompuCategory: IDENTICAL

Primitive Data Type: **PtTqType (2)**

Description:

Primitive Data Type: PtTqType"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 11
Physical Lower Lim interval type: CLOSED

Physical lower limit: -1024
Physical Upper Lim interval type: CLOSED
Physical upper limit: 1023

UUID: da5b9484-6916-4b38-a2d6-b89c81a83560
BaseType UUID: BD076CCF-6F7E-4BB3-BF95-BEC2EC79F65A
Unit: Nm
CompuCategory: LINEAR
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	2047	[ ]	-1024.0; 1.0	1	

Record Data Type: **SysU (4)**

Description:

Group Signal	**Data Type**
SysU	U1
SysUQf	GenQf1

Primitive Data Type: GenQf1"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 2
Lower limit internal range: 0
lower limit interval type internal range: CLOSED
upper limit internal range: 3
upper limit interval type internal range: CLOSED

UUID: 68330404-f471-4bbd-8fc8-271f1473d230
BaseType UUID: 26DD2B41-A428-4C18-8753-9DD5279354A2
Unit: NoUnit
CompuCategory: TEXTTABLE
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	0	[ ]			GenQf1_UndefindDataAccur
1	1	[ ]			GenQf1_TmpUndefdData
2	2	[ ]			GenQf1_DataAccurNotWithinSpcn
3	3	[ ]			GenQf1_AccurData

Primitive Data Type: U1"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 8
Physical Lower Lim interval type: CLOSED

Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 25

UUID: 7bc41522-00dd-4966-985f-8c5e73b5a08d
BaseType UUID: 580B6CD1-9069-4CDA-89C2-C8ECAE8EEB3A
Unit: Volt
CompuCategory: LINEAR
CompuMethod Table:

Min	Max	Interval type	Numerators	Denominators	Constant
0	250	[ ]	0.0; 0.1	1	

Primitive Data Type: T3 (2)

Description:
Primitive Data Type: T3"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 8
Physical Lower Lim interval type: CLOSED
Physical lower limit: -50
Physical Upper Lim interval type: CLOSED
Physical upper limit: 200

UUID: a98be047-536f-4c0f-bdfc-1e160d93ce0e
BaseType UUID: 212FDAAF-8ACC-4CA0-991E-36929AF441A2
Unit: DegCgrd
CompuCategory: LINEAR
CompuMethod Table:

Min	Max	Interval type	Numerators	Denominators	Constant
0	250	[ ]	-50.0; 1.0	1	

Primitive Data Type: TimeDesForSWDL (2)

Description:
Primitive Data Type: TimeDesForSWDL"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 16
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 65535

UUID: 8FEB9E8D-ABF8-4C43-A22C-243D56EC12FF
BaseType UUID: AF7444E0-7297-4609-AF10-DBF2C45528A8
Unit: Sec
CompuCategory: LINEAR
CompuMethod Table:

Min	Max	Interval type	Numerators	Denominators	Constant
0	65535	[ ]	0.0; 1.0	1	

Primitive Data Type: Tq34 (1)

Description:

Primitive Data Type: Tq34"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 14
Physical Lower Lim interval type: CLOSED
Physical lower limit: -8188
Physical Upper Lim interval type: CLOSED
Physical upper limit: 8192

UUID: D55D4E0C-1A9E-4164-B722-5B17A555319D
BaseType UUID: 153FBD12-431E-4D74-BC9D-AEDB8EE476EC
Unit: Nm
CompuCategory: LINEAR
CompuMethod Table:

Min	Max	Interval type	Numerators	Denominators	Constant
0	16380	[ ]	-8188; 1	1	

Primitive Data Type: TrCoolgReq (2)

Description:

Primitive Data Type: TrCoolgReq"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 3
Lower limit internal range: 0
lower limit interval type internal range: CLOSED
upper limit internal range: 7
upper limit interval type internal range: CLOSED

UUID: CA1A690D-726F-4402-9896-08CADEEE01C6
BaseType UUID: 5FAC52EE-F146-4492-A205-DA5603E71DB7
Unit: NoUnit
CompuCategory: TEXTTABLE
CompuMethod Table:

Min	Max	Interval type	Numerators	Denominators	Constant
0	0	[ ]			TrCoolgReq_NoRequestForMoreCoolantPower
1	1	[ ]			TrCoolgReq_IncreaseLevel1ForMoreCoolantPower
2	2	[ ]			TrCoolgReq_In

					creaseLevel2ForMoreCoolantPower
3	3	[ ]			TrCoolgReq_MaxCoolingPower
4	4	[ ]			TrCoolgReq_NotDefined

Record Data Type: **TripResetSync (1)**

Description:

Group Signal	Data Type
Resetcounter	UInt16
Tripcounter	Uint24

Primitive Data Type: UInt16"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 16
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 65535

UUID: 70F22060-193B-AC2F-E040-901367011E69
BaseType UUID: DB278DB6-49F4-4C78-9B69-7CDDB0E0DE4C
Unit: NoUnit
CompuCategory: IDENTICAL

Primitive Data Type: Uint24"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 24
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 16777215

UUID: 1878981b-ce30-4558-9676-70edd2458f66
BaseType UUID: 1FFDC6FC-905E-4657-A4E7-4748C8CFED2F
Unit: NoUnit
CompuCategory: IDENTICAL

Primitive Data Type: **U52 (1)**

Description:

Primitive Data Type: U52"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 15
Physical Lower Lim interval type: CLOSED

Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 1638.4

UUID: 7F4F7E20-B068-44F3-9012-6DCA8BBEB898
BaseType UUID: 8FBF8AF8-1A37-4E82-BCE1-2464F51C9286
Unit: Volt
CompuCategory: LINEAR
CompuMethod Table:

Min	Max	Interval type	Numerators	Denominators	Constant
0	16384	[ ]	0; 0.1	1	

Primitive Data Type: UInt16 (2)

Description:
Primitive Data Type: UInt16"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 16
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 65535

UUID: 70F22060-193B-AC2F-E040-901367011E69
BaseType UUID: DB278DB6-49F4-4C78-9B69-7CDDB0E0DE4C
Unit: NoUnit
CompuCategory: IDENTICAL

Primitive Data Type: UInt4 (2)

Description:
Primitive Data Type: UInt4"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 4
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 15

UUID: 70F22060-1935-AC2F-E040-901367011E69
BaseType UUID: D6A89C6F-2F3D-4C8E-8DC9-191B048C4E5D
Unit: NoUnit
CompuCategory: IDENTICAL

Record Data Type: VehCfgPrm (3)

Description:

Group Signal	Data Type
BlkIDBytePosn1	Nr2
CCPBytePosn2	Nr2
CCPBytePosn3	Nr2

Group Signal	Data Type
CCPBytePosn4	Nr2
CCPBytePosn5	Nr2
CCPBytePosn6	Nr2
CCPBytePosn7	Nr2
CCPBytePosn8	Nr2

Primitive Data Type: Nr2"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 8
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 255

UUID: dab46ccf-30f0-4c1a-aea9-a986607e22a7
BaseType UUID: 1F3A60AE-10C1-4F7D-901A-8A36205AD28C
Unit: NoUnit
CompuCategory: LINEAR
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	255	[ ]	0.0; 1.0	1	

Record Data Type: **VehModMngtGlbSafe1 (2)**

Description:

Group Signal	Data Type
CarModSts1	CarModSts1
CarModSubtypWdCarModSubtyp	Nr5
Chks	UInt8
Cntr	UInt4
EgyLvlElecMai	UInt4
EgyLvlElecSubtyp	UInt4
FltEgyCnsWdSts	FltEgyCns1
PwrLvlElecMai	UInt4
PwrLvlElecSubtyp	UInt4
UsgModSts	UsgModSts1

Primitive Data Type: CarModSts1"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 3
Lower limit internal range: 0
lower limit interval type internal range: CLOSED
upper limit internal range: 5
upper limit interval type internal range: CLOSED

UUID: d3b1fb00-db0d-4795-98fd-3460a7feb3ff
BaseType UUID: 8F2AC7D8-D465-4CF2-88F0-62C934B938AE
Unit: NoUnit

CompuCategory: TEXTTABLE
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	0	[ ]			CarModSts1_CarModNorm
1	1	[ ]			CarModSts1_CarModTrnsp
2	2	[ ]			CarModSts1_CarModFcy
3	3	[ ]			CarModSts1_CarModCrash
5	5	[ ]			CarModSts1_CarModDyno

Primitive Data Type: FltEgyCns1"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 1
Lower limit internal range: 0
lower limit interval type internal range: CLOSED
upper limit internal range: 1
upper limit interval type internal range: CLOSED

UUID: a3ddcbfe-e406-4bea-a27e-968c4249dc2b
BaseType UUID: 65800EB1-B2BA-4096-920E-7A28D404CE3D
Unit: NoUnit
CompuCategory: TEXTTABLE
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	0	[ ]			FltEgyCns1_NoFlt
1	1	[ ]			FltEgyCns1_Flt

Primitive Data Type: Nr5"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 3
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 7

UUID: ed06a498-e79f-44ef-b4b4-5f16af8acbd3
BaseType UUID: 31352803-B10F-46D0-8B6E-04D14ED080A4
Unit: NoUnit
CompuCategory: IDENTICAL

Primitive Data Type: UInt4"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 4
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 15

UUID: 70F22060-1935-AC2F-E040-901367011E69
BaseType UUID: D6A89C6F-2F3D-4C8E-8DC9-191B048C4E5D
Unit: NoUnit
CompuCategory: IDENTICAL

Primitive Data Type: UInt8"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 8
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 255

UUID: 70F22060-1937-AC2F-E040-901367011E69
BaseType UUID: 6E2EFCA8-43A6-4F1E-9530-99AE1843F3D5
Unit: NoUnit
CompuCategory: IDENTICAL

Primitive Data Type: UsgModSts1"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 4
Lower limit internal range: 0
lower limit interval type internal range: CLOSED
upper limit internal range: 13
upper limit interval type internal range: CLOSED

UUID: 6c410900-dd7c-46ca-b22f-5f4b53b714c3
BaseType UUID: E72751B2-CDD0-40F0-B2FA-00045E361543
Unit: NoUnit
CompuCategory: TEXTTABLE
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	0	[ ]			UsgModSts1_ UsgModAbdnd
1	1	[ ]			UsgModSts1_ UsgModInActv
2	2	[ ]			UsgModSts1_ UsgModCnvinc

11	11	[ ]			UsgModSts1_UsgModActv
13	13	[ ]			UsgModSts1_UsgModDrvg

Record Data Type: **VehMtnStSafe1 (2)**

Description:

Group Signal	Data Type
VehMtnSt	VehMtnSt2
VehMtnStChks	UInt8
VehMtnStCntr	UInt4

Primitive Data Type: UInt4"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 4
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 15

UUID: 70F22060-1935-AC2F-E040-901367011E69
BaseType UUID: D6A89C6F-2F3D-4C8E-8DC9-191B048C4E5D
Unit: NoUnit
CompuCategory: IDENTICAL

Primitive Data Type: UInt8"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 8
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 255

UUID: 70F22060-1937-AC2F-E040-901367011E69
BaseType UUID: 6E2EFCA8-43A6-4F1E-9530-99AE1843F3D5
Unit: NoUnit
CompuCategory: IDENTICAL

Primitive Data Type: VehMtnSt2"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 3
Lower limit internal range: 0
lower limit interval type internal range: CLOSED
upper limit internal range: 7
upper limit interval type internal range: CLOSED

UUID: 3177ddbc-f3ca-47a2-be16-3a83bb785209

BaseType UUID: 836EDB9C-2E2E-456A-8ECC-2513244B9336
Unit: NoUnit
CompuCategory: TEXTTABLE
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	0	[ ]			VehMtnSt2_Ukwn
1	1	[ ]			VehMtnSt2_StandStillVal1
2	2	[ ]			VehMtnSt2_StandStillVal2
3	3	[ ]			VehMtnSt2_StandStillVal3
4	4	[ ]			VehMtnSt2_RollgFwdVal1
5	5	[ ]			VehMtnSt2_RollgFwdVal2
6	6	[ ]			VehMtnSt2_RollgBackwVal1
7	7	[ ]			VehMtnSt2_RollgBackwVal2

Record Data Type: **VehSpdLgtSafe1 (3)**

Description:

Group Signal	Data Type
VehSpdLgtA	Spd1
VehSpdLgtChks	UInt8
VehSpdLgtCntr	UInt4
VehSpdLgtQf	GenQf1

Primitive Data Type: GenQf1"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 2
Lower limit internal range: 0
lower limit interval type internal range: CLOSED
upper limit internal range: 3
upper limit interval type internal range: CLOSED

UUID: 68330404-f471-4bbd-8fc8-271f1473d230
BaseType UUID: 26DD2B41-A428-4C18-8753-9DD5279354A2
Unit: NoUnit
CompuCategory: TEXTTABLE
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	0	[ ]			GenQf1_UndefindDataAccur
1	1	[ ]			GenQf1_TmpUndefdData
2	2	[ ]			GenQf1_Data

					AccurNotWithinSpcn
3	3	[ ]			GenQf1_AccurData

Primitive Data Type: Spd1"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 15
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 125

UUID: 3a742b77-5971-4074-a752-87a6699ecb51
BaseType UUID: F60D578D-E124-4A92-9BA1-92E7D8332CBD
Unit: MtrPerSec
CompuCategory: LINEAR
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	31970	[ ]	0.0; 0.00391	1	

Primitive Data Type: UInt4"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 4
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 15

UUID: 70F22060-1935-AC2F-E040-901367011E69
BaseType UUID: D6A89C6F-2F3D-4C8E-8DC9-191B048C4E5D
Unit: NoUnit
CompuCategory: IDENTICAL

Primitive Data Type: UInt8"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 8
Physical Lower Lim interval type: CLOSED
Physical lower limit: 0
Physical Upper Lim interval type: CLOSED
Physical upper limit: 255

UUID: 70F22060-1937-AC2F-E040-901367011E69
BaseType UUID: 6E2EFCA8-43A6-4F1E-9530-99AE1843F3D5
Unit: NoUnit

CompuCategory: IDENTICAL

Primitive Data Type: **WhlMotSysActvDampgModCodSts (2)**

Description:

Primitive Data Type: WhlMotSysActvDampgModCodSts"

System Signal Encoding: NONE
category: VALUE
SystemSignalLength: 2
Lower limit internal range: 0
lower limit interval type internal range: CLOSED
upper limit internal range: 3
upper limit interval type internal range: CLOSED

UUID: 6fb9d2c9-4552-4d0b-841f-0551c6b8a87e
BaseType UUID: 8D3F4D58-ED68-457A-9F81-8472E5184A58
Unit: NoUnit
CompuCategory: TEXTTABLE
CompuMethod Table:

Min	*Max*	*Interval type*	*Numerators*	*Denominators*	*Constant*
0	0	[ ]			WhlMotSysActvDampgModCodSts_NoDampg
1	1	[ ]			WhlMotSysActvDampgModCodSts_LoDampg
2	2	[ ]			WhlMotSysActvDampgModCodSts_MedDampg
3	3	[ ]			WhlMotSysActvDampgModCodSts_HiDampg

6.4 External interfaces requirements

6.5 Internal interfaces requirements

6.6 Internal data requirements

6.7 Adaptation requirements

6.8 Safety requirements

6.9 Security and privacy requirements

6.10 Computer resource requirements

6.11 Software quality factors

6.12 Design and implementation constraints

6.13 Personnel-related requirements

6.14 Delivery requirements

6.15 General requirements

7 SOFTWARE PROJECTS REQUIREMENTS

8 APPENDIX

8.1 Car Config Parameters used in this SWRS

Parameter ID	Parameter Name	Parameter Value ID (hex)	Parameter Value Name
4	ELECTRIC DRIVELINE	02	Combustion Engine FWD and Electric Motor FWD
		03	Combustion Engine FWD and Electric Motor RWD
		04	Combustion Engine RWD and Electric Motor FWD
		05	Combustion Engine RWD and Electric Motor RWD
		06	Electric Motor AWD
		07	Electric Motor FWD
		08	Electric Motor RWD

Parameter ID	Parameter Name	Parameter Value ID (hex)	Parameter Value Name
		80	Combustion Engine FWD and Electric Motor FWD2
9	GEARBOX	81	8ATEAWD
		A4	FSSR+R2SR
		A5	LFWD+R2SR
13	PROPULSION TYPE	02	Hybrid vehicle
		03	Plugin hybrid vehicle
14	REMOTE MONITORING SYSTEM	01	RMS Disactivated
		02	RMS Activated
46	DIAGNOSIS SYSTEM	01	OBD 2
		02	EOBD
		03	JOBD
695	Electric Motor Drive Type	03	Synchronous motor
		07	Front single synchronous motor + rear double synchronous motor